

# **Health Data Science: AI and Knowledge Translation**

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# The Project

Welcome to the course notes for **Health Data Science: AI and Knowledge Translation**.

This platform bridges the gap between traditional epidemiology and modern data science. Unlike standard textbooks that focus on a single language, this resource adopts a **“Polyglot”** approach—teaching R and Python side-by-side—and integrates **Artificial Intelligence** directly into the workflow.

Whether you are a health researcher looking to modernize your skills or a data scientist seeking to understand epidemiological rigor, you are in the right place.

Here, we offer:

- **“Zero to AI Co-Pilot” Training:** Learn to use Large Language Models (LLMs) to generate, debug, and translate code while critically auditing them for bias and hallucinations.
- **Real-World Data:** Guides on navigating “messy” open data sources such as NHANES and WHO, rather than sanitized “toy” datasets.
- **Knowledge Translation:** Step-by-step walkthroughs on building interactive dashboards (Shiny/Dash) and dynamic reports to communicate findings to policymakers.

## What We Aim to Achieve

We are on a mission to:

- **Democratize Compute Access:** Eliminate hardware barriers by using cloud-native environments (Google Colab, Kaggle) that run on any device.

- **Bridge the “Polyglot” Gap:** Provide “Rosetta Stone” tutorials where concepts are implemented in both R and Python, allowing learners to be fluent in both.
- **Teach “AI-Resilience”:** Shift the focus from manual syntax generation to **rigorous code auditing**, security validation, and reproducibility.
- **Enable Professional Portfolios:** Help students transform static analyses into public-facing data science portfolios.

## Dive into Our Modules

Embark on a journey through our structured modules, designed to take you from data literacy to advanced AI workflows.

### Note

The tutorials are designed with a consistent structure to provide a cohesive learning experience. Here is what you can expect in each chapter:

1. **Overview:** A concise summary of learning objectives and topics.
2. **Concepts:** Core theoretical materials (slides, video lessons).
3. **Polyglot Tutorials:** In-depth walkthroughs with tabbed code blocks allowing you to toggle between **R** and **Python** solutions.
4. **AI Auditing:** Specific sections dedicated to verifying and “stress-testing” AI-generated code for the chapter’s topic.
5. **Knowledge Check:** Ungraded self-assessments to reinforce key concepts.
6. **Practice Exercises:** Hands-on tasks using real-world health data to build your portfolio.

## How Our Content is Presented

All our resources are hosted on an easy-to-access GitHub page. The format? Engaging text, reproducible code, clear outputs, and crisp videos that distill complex topics.

This document is created using [Quarto](#), allowing us to weave analysis, code, and interpretation into a single reproducible document.

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## How to Cite

The BibTeX format can be downloaded from [here](#).

**Part I**

**Course Planning**

# 1 Syllabus vs HDSx Inventory Completion Matrix

Status: baseline audit created before the normalized folder migration. The normalized structure described below has now been applied to the source tree.

Source syllabus: `syllabus/SPPH-381H-Course-Outline-v4.qmd` and `syllabus/SPPH-381H-Course-Outline-v4.pdf`.

Inventory checked: `root` `.qmd` source files, `_quarto.yml`, `assets/`, `examples/`, `images/`, and `syllabus/`. Generated `docs/`, `.git/`, and `.Rproj.user/` were excluded from the inventory assessment.

## 1.1 Extracted Schedule

| Week        | Topic                    | Activity / deliverable                                                                              |
|-------------|--------------------------|-----------------------------------------------------------------------------------------------------|
| 0 (Async)   | Onboarding               | GitHub account, orientation video, pre-course survey. Due before first class.                       |
| 1 (Sept 10) | Health Data, KT & Ethics | Open data portals, Data Intake Card, provenance, stewardship, responsible AI. No graded submission. |



| Week        | Topic                            | Activity / deliverable                                                                                                    |
|-------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 2 (Sept 17) | Modern Workflows                 | Fork template, launch Codespace, edit/render Quarto, commit/sync. Assignment 1 due Mon 4 PM.                              |
| 3 (Sept 24) | R with AI                        | R basics, tidyverse, functions, debugging with LLMs, <code>renv</code> , Quarto documentation. Assignment 2 due Mon 4 PM. |
| 4 (Oct 1)   | Git & Collaboration              | Commits, branches, pull requests, repo hygiene, group repository setup. Assignment 3 and M0 due Mon 4 PM.                 |
| 5 (Oct 8)   | Polyglot Awareness & R Deepening | Python/pandas comparison, R-to-Python translation, AI parity checks, R practice. Assignment 4 due Mon 4 PM.               |

| Week       | Topic                                | Activity / deliverable                                                                                                 |
|------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 6 (Oct 15) | Data Visualization                   | AI-assisted ggplot2/plotly, visual clarity/bias/misrepresentation checks.<br>Assignment 5 and M1 due Mon 4 PM.         |
| 7 (Oct 22) | EDA, Table 1 & AI Auditing           | EDA, Table 1, planted-error AI audit, provenance/stewardship revision.<br>Assignment 6 and M2 due Mon 4 PM.            |
| 8 (Oct 29) | Dashboard Prototypes for KT          | Minimal dashboard-style KT product, Quarto/Shiny concepts, deployment pathway choice.<br>Assignment 7 due Mon 4 PM.    |
| 9 (Nov 5)  | Communication of Scientific Findings | Audience-specific KT, plain-language summaries, revealjs slides, citations, peer review.<br>Assignment 8 due Mon 4 PM. |

| Week        | Topic                                            | Activity / deliverable                                                                                                                             |
|-------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 (Nov 12) | Writing & Publishing Reports                     | Quarto reports, citations, interactive elements, GitHub Pages, portfolio publishing. Assignment 9 and M3 due Mon Nov 16, 4 PM.                     |
| 11 (Nov 19) | Portfolio Surgery & Optional Advanced Deployment | Fresh Codespace reproducibility test, path/dependency/README fixes, optional deployment/Python pathway. Assignment 10 and M4 due Mon Nov 23, 4 PM. |
| 12 (Nov 26) | Term Project Presentations, Part 1               | M5 presentations, live Q&A, peer review. Slides due before class.                                                                                  |
| 13 (Dec 3)  | Term Project Presentations, Part 2               | M5 presentations, live Q&A, peer review. Slides due before class.                                                                                  |
| Final       | Final Portfolio                                  | Final portfolio due Dec 11, Fri 4 PM.                                                                                                              |

## 1.2 Folder Inventory Snapshot

| Area               | Current contents                                                                                                      | Notes                                                                |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Root Quarto source | <code>index.qmd</code> , <code>_quarto.yml</code> , 11 module part files, many numbered child <code>.qmd</code> files | Source is mostly flat, making week/module ownership hard to scan.    |
| syllabus/          | PDF and QMD syllabus source                                                                                           | Good source of truth for schedule and assessment logic.              |
| assets/data/       | NHANES_Demo_Converter                                                                                                 | Only one CSV data asset found under shared assets.                   |
| assets/images/     | Image subfolders for data1-data12, workflow*, git*, python*, r*, plus week3                                           | Assets are topic-fragment based, not normalized by course week.      |
| examples/          | bmi39.R, demo5.R, NHANES .rds files                                                                                   | Useful for R/EDA examples, but not tied to weeks or assignments.     |
| images/            | License/support images                                                                                                | Small global image folder separate from <code>assets/images</code> . |
| docs/              | Rendered Quarto output                                                                                                | Treat as generated output, not source curriculum inventory.          |

## 1.3 Completion Matrix

Status key: **Ready** = source is substantial and surfaced in navigation; **Partial** = source exists but has gaps, placeholders, or navigation mismatch; **Missing** = no clear source content found.

| Week | Syllabus need                             | Current matching inventory                                                                                                                        | Status  | Gap / action                                                                                                                                                                |
|------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | Async onboarding module                   | Syllabus-only; no dedicated onboarding .qmd found                                                                                                 | Missing | Add <code>week00-onboarding/</code> or an onboarding chapter with GitHub setup, orientation link placeholder, and survey instructions.                                      |
| 1    | Health data, ethics, KT, Data Intake Card | <code>data.qmd</code> ; <code>data1.qmd</code> ; extra <code>data2.qmd</code> - <code>data12.qmd</code> ; <code>assets/images/data1-data12</code> | Partial | Strong content exists, but only <code>data1.qmd</code> is in <code>_quarto.yml</code> ; decide whether data-source pages are Week 1 references or a separate data appendix. |

| Week | Syllabus need                                                 | Current matching inventory                                                                                                           | Status | Gap / action                                                                                        |
|------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------|
| 2    | Codespaces, Quarto basics, reproducible structure, commits    | <code>workflows.qmd</code> ; <code>workflows1.qmd</code> - <code>workflows11.qmd</code> ; <code>assets/images/workflows</code>       | Ready  | Good coverage. Normalize the naming and keep the parent/child relationship explicit.                |
| 3    | R with AI, tidyverse, functions, debugging, <code>renv</code> | <code>r.qmd</code> ; <code>r1.qmd</code> - <code>r10.qmd</code> ; <code>assets/images/r*</code> ; <code>examples/NHANES</code> files | Ready  | Add explicit <code>renv</code> activity/checkpoint if not already embedded in assignment materials. |
| 4    | Git collaboration, PRs, repo hygiene, group setup             | <code>git.qmd</code> ; <code>git1.qmd</code> - <code>git10.qmd</code> ; <code>assets/images/git*</code>                              | Ready  | Looks aligned; M0 group setup should be made visibly tied to Week 4.                                |

| Week | Syllabus need                                              | Current matching inventory                                           | Status  | Gap / action                                                                                                                                |
|------|------------------------------------------------------------|----------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | Python/pandas awareness and R deepening                    | python.qmd; Partial python1.qmd- python14.qmd; assets/images/python* |         | python.qmd is substantial, but only exposes python1.qmd; surface or consolidate the rest. R-deepening assignment materials are not obvious. |
| 6    | Data visualization with ggplot2/plotly and AI visual audit | viz.qmd; viz1.qmd                                                    | Missing | Current files are placeholders. Add real tutorial/activity content and assets.                                                              |

| Week | Syllabus need                                            | Current matching inventory                                                 | Status  | Gap / action                                                                                                        |
|------|----------------------------------------------------------|----------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| 7    | EDA, Table 1, planted-error AI audit, prove-nance update | <code>eda.qmd</code> ; <code>eda1.qmd</code> ; NHANES example data/scripts | Missing | Current EDA files are placeholders. Build the activity around the existing NHANES examples and add audit materials. |
| 8    | Dashboard-style KT prototype, Quarto/Shiny concepts      | <code>dashboards.qmd</code> ; <code>dashboards1.qmd</code>                 | Missing | Current files are placeholders. Need a minimal dashboard path, starter template, and testing instructions.          |



| Week | Syllabus<br>need                                                                           | Current<br>matching<br>inventory                                    | Status  | Gap /<br>action                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9    | Scientific<br>communi-<br>cation,<br>revealjs<br>slides,<br>peer<br>feedback               | <code>comms.qmd</code> ;<br><code>comms1.qmd</code>                 | Missing | Current<br>files are<br>placeholders. Add<br>revealjs<br>tutorial,<br>plain-<br>language<br>summary<br>activity,<br>and peer<br>review<br>template.            |
| 10   | Reproduciblereports.qmd<br>reporting,<br>publish-<br>ing,<br>citations,<br>GitHub<br>Pages | <code>reports1.qmd</code> ;<br><code>references.bib</code><br>empty | Missing | Current<br>files are<br>placeholders. Need<br>report<br>template,<br>citation<br>workflow,<br>bibliogra-<br>phy<br>example,<br>and<br>publishing<br>checklist. |

| Week | Syllabus need                                              | Current matching inventory                                                                | Status  | Gap / action                                                                                                     |
|------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------|
| 11   | Portfolio reproducibility stress test, optional deployment | <del>advanced.qmd</del><br><b>advanced1.qmd</b> ; maybe <b>reports/dashboards</b> overlap | Missing | Current advanced files are placeholders. Create a portfolio surgery checklist and fresh Codespace test protocol. |
| 12   | Presentations Part 1                                       | No dedicated presentation week materials beyond <b>comms*</b> placeholders                | Missing | Add presentation rubric/checklist and peer feedback form.                                                        |
| 13   | Presentations Part 2                                       | No dedicated presentation week materials beyond <b>comms*</b> placeholders                | Missing | Add repeat presentation logistics or combine with Week 12 materials.                                             |

| Week  | Syllabus need   | Current matching inventory                                                             | Status  | Gap / action                                                                                             |
|-------|-----------------|----------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------|
| Final | Final portfolio | Syllabus has reproducibility contract; no dedicated portfolio submission package found | Partial | Convert the syllabus reproducibility contract into a student-facing final portfolio check-list/template. |

## 1.4 Key Mismatches

1. The source tree is organized as topic families (`data`, `workflows`, `r`, `git`, `python`, etc.), while the syllabus is organized as weeks.
2. The first five technical weeks are substantially developed; Weeks 6-13 are mostly placeholders.
3. Several developed files are not surfaced in `_quarto.yml`: `data2.qmd`-`data12.qmd` and most `python*.qmd` files.
4. Parent files such as `data.qmd`, `workflows.qmd`, `r.qmd`, `git.qmd`, and `python.qmd` contain substantial chapter content but are used as Quarto `part:` pages, while many child pages are short fragments. This creates duplication and discoverability friction.
5. Assignment and milestone deliverables are present in the syllabus, but there is no obvious normalized assignment template folder.

## 1.5 Proposed Normalized Structure

Recommended target: keep the Quarto book, but normalize source around syllabus weeks and move reusable references into appendices.

```
1 /
2 _quarto.yml
3 index.qmd
4 syllabus/
5   SPPH-381H-Course-Outline-v4.qmd
6   SPPH-381H-Course-Outline-v4.pdf
7   syllabus-inventory-completion-matrix.md
8 weeks/
9   week00-onboarding/
10     index.qmd
11     assets/
12   week01-health-data-ethics/
13     index.qmd
14     activity-data-intake-card.qmd
15     assets/
16   week02-modern-workflows/
17     index.qmd
18     tutorials/
19     assets/
20   week03-r-with-ai/
21     index.qmd
22     tutorials/
23     assets/
24   week04-git-collaboration/
25     index.qmd
26     tutorials/
27     assets/
28   week05-polyglot-r-deepening/
29     index.qmd
30     tutorials/
31     assets/
32   week06-visualization/
33     index.qmd
34     activity-ai-visual-audit.qmd
```

```
35     assets/
36 week07-eda-ai-audit/
37     index.qmd
38     activity-table1-audit.qmd
39     assets/
40 week08-dashboard-kt/
41     index.qmd
42     starter-dashboard/
43     assets/
44 week09-science-communication/
45     index.qmd
46     peer-feedback-template.qmd
47     assets/
48 week10-reporting-publishing/
49     index.qmd
50     report-template.qmd
51     assets/
52 week11-portfolio-surgery/
53     index.qmd
54     reproducibility-test.qmd
55     assets/
56 week12-presentations/
57     index.qmd
58     rubric.qmd
59 week13-presentations/
60     index.qmd
61 assignments/
62     assignment01-workflows/
63     assignment02-r-with-ai/
64     assignment03-git/
65     assignment04-polyglot/
66     assignment05-visualization/
67     assignment06-eda-ai-audit/
68     assignment07-dashboard/
69     assignment08-communication/
70     assignment09-reporting/
71     assignment10-portfolio-surgery/
72 milestones/
73     m0-group-formation/
74     m1-proposal/
```

```

75     m2-preliminary-analysis/
76     m3-project-update/
77     m4-peer-review/
78     m5-presentation/
79     final-portfolio/
80 references/
81     data-sources/
82         nhis.qmd
83         nhanes.qmd
84         brfss.qmd
85         ccs.qmd
86         cads.qmd
87         ctns.qmd
88         csads.qmd
89         phac-infobase.qmd
90         ccdss.qmd
91         gho.qmd
92         open-portal.qmd
93         bc-data-catalogue.qmd
94     cheatsheets/
95     troubleshooting/
96 examples/
97     r/
98     python/
99     data/
100 assets/
101     shared/

```

## 1.6 Migration Notes

| Current source                                       | Proposed home                                    |
|------------------------------------------------------|--------------------------------------------------|
| data.qmd                                             | weeks/week01-health-data-ethics/index.qmd        |
| data1.qmd-data12.qmd and<br>assets/images/data*      | references/data-sources/<br>plus matching assets |
| workflows.qmd,<br>workflows1.qmd-<br>workflows11.qmd | weeks/week02-modern-workflows/                   |
| r.qmd, r1.qmd-r10.qmd                                | weeks/week03-r-with-ai/                          |

| Current source                                                                                                                                                  | Proposed home                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <code>git.qmd</code> ,<br><code>git1.qmd-git10.qmd</code>                                                                                                       | <code>weeks/week04-git-collaboration/</code>                                    |
| <code>python.qmd</code> ,<br><code>python1.qmd-python14.qmd</code>                                                                                              | <code>weeks/week05-polyglot-r-deepening/</code>                                 |
| <code>viz.qmd</code> , <code>eda.qmd</code> ,<br><code>dashboards.qmd</code> , <code>comms.qmd</code> ,<br><code>reports.qmd</code> , <code>advanced.qmd</code> | Expand into Weeks 6-11<br>modules                                               |
| <code>examples/bmi39.R</code> ,<br><code>examples/demo5.R</code> ,<br><code>examples/data/*.rds</code>                                                          | Keep under <code>examples/</code> , but<br>add week references in<br>activities |

## 1.7 Immediate Build Priorities

1. Decide whether the Quarto book should be navigated by week or by topic. For syllabus alignment, week-based navigation is the cleaner primary structure.
2. Surface existing developed content currently missing from `_quarto.yml`, especially `data2.qmd-data12.qmd` and `python2.qmd-python14.qmd`, or move it into a reference appendix.
3. Build Weeks 6-11, because those are the largest content gaps.
4. Add assignment and milestone folders so deliverables are first-class artifacts rather than only syllabus text.
5. Create a final portfolio checklist from the syllabus reproducibility contract.

## **Part II**

### **Week 0: Asynchronous Onboarding**



# Week 0: Asynchronous Onboarding

## Purpose

Students complete the basic account and orientation work before the first class so Week 1 can focus on health data, ethics, and course expectations.

## Required Student Tasks

- Create a GitHub account using a professional username.
- Watch the course workflow orientation video posted on Canvas.
- Complete the pre-course survey.
- Confirm access to Canvas and the course GitHub entry point.

## TODO Before Release

### Content TODO

- Add the final Canvas orientation video link.
- Add the pre-course survey link or instructions.
- Add a screenshot walkthrough for creating a GitHub account with a professional username.
- Add a brief troubleshooting note for students who already have a GitHub account.

## Definition of Done

- The page gives students one clear path to complete onboarding without installing local software.
- All external links have been checked in a fresh browser session.
- The GitHub username guidance is privacy-aware and professional.
- The page states what students should bring to Week 1.

## **Part III**

### **Week 1: Health Data, KT & Ethics**

# Health Data & Ethics

## Overview

This chapter sets the stage by defining the core philosophy of the course: data science does not end with a p-value, but with Knowledge Translation (KT). Students are introduced to the “Last Mile” of research—communicating findings effectively—and the ethical landscape of modern health data, including Indigenous Data Sovereignty (OCAP® principles) and the risks of AI hallucinations. You will explore open data sources like NHANES and establish the ground rules for privacy and security before touching a single line of code.

## Learning Objectives

- Explore the landscape of **open health data** (e.g., NHANES, WHO, World Bank, BC Data Catalogue) and learn where high-quality public datasets live.
- Practice finding a dataset, skimming its description/metadata, and recording basic provenance.
- Review **privacy and ethics** for public data and **responsible GenAI use**.
- Learn the meaning and importance of **Indigenous Data Sovereignty** (specifically OCAP® for First Nations) and CARE principles.
- Identify common **file formats** and **metadata** to support reproducibility.

**Note:** Week 1 is **light on code** and **heavy on awareness**. There is **no graded assignment submission** this week. Instead, you will complete

a “Data Intake Card” to prepare for your future portfolio.

## Connection

This first week starts with provenance, ethics, and KT framing before any dashboard work. Next week you will place the same dataset inside a reproducible project structure.

## Case Study Data Spine

The **NHANES Health Equity data spine** is the recurring dataset thread for this course. In the early weeks, use it only as a documented public-health data artifact: provenance, ethics, file organization, and reproducible paths.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The cached files are provided so early work can happen without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_spine <- read_csv(nhanes_path, show_col_types = FALSE)
15
```

```

16 dim(nhanes_spine)
17 #> [1] 105626      18
18 glimpse(nhanes_spine)
19 #> Rows: 105,626
20 #> Columns: 18
21 #> $ Cycle      <chr> "1999-2000", "1999-2000", "1999-2000", "1999-2000", "19~
22 #> $ BMI        <dbl> 14.90, 24.90, 17.63, NA, 29.10, 22.56, 29.39, 15.51, 18~
23 #> $ Weight     <dbl> 12.5, 75.4, 32.9, 13.3, 92.5, 59.2, 78.0, 40.7, 45.5, 1~
24 #> $ Height     <dbl> 91.6, 174.0, 136.6, NA, 178.3, 162.0, 162.9, 162.0, 156~
25 #> $ Waist      <dbl> 45.7, 98.0, 64.7, NA, 99.9, 81.6, 90.7, 64.1, 64.6, 108~
26 #> $ WeightMEC  <dbl> 10982.899, 28325.385, 46192.257, 10251.260, 99445.066, ~
27 #> $ Strata     <dbl> 5, 1, 7, 2, 8, 2, 4, 6, 9, 7, 1, 6, 13, 12, 11, 11, 5, ~
28 #> $ PSU        <dbl> 1, 3, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, 2, 1, 1, 1, 2~
29 #> $ Gender     <chr> "Female", "Male", "Female", "Male", "Male", "Female", "~
30 #> $ Race       <chr> "Non-Hispanic Black", "Non-Hispanic White", "Non-Hispan~
31 #> $ Education  <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
32 #> $ Marital    <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
33 #> $ Age        <dbl> 2, 77, 10, 1, 49, 19, 59, 13, 11, 43, 15, 37, 70, 81, 3~
34 #> $ PIR        <dbl> 0.86, 5.00, 1.47, 0.57, 5.00, 1.21, NA, 0.53, NA, NA, 1~
35 #> $ EducationClean <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
36 #> $ MaritalClean <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
37 #> $ IncomeGroup <chr> "Low Income (<1.3)", "High Income (>3.5)", "Middle Inco~
38 #> $ WHtR       <dbl> 0.4989083, 0.5632184, 0.4736457, NA, 0.5602916, 0.50370~

```

## In-Class Activity

- Use the case-study dataset as a practice entry for the Data Intake Card.
- Identify the public source, local cached file, and the README that describes the case study.
- Write one KT framing sentence: who might use this dashboard, and what decision could it support?
- Name one ethics or stewardship question to verify before publishing a dashboard based on public health survey data.

Student output: a completed Data Intake Card entry for either the NHANES case-study dataset or another approved public-health dataset.

---

## The Open Data Landscape

Open data can accelerate health research—if we understand **what** we’re downloading and **how** to use it responsibly. This week, we locate credible sources and map what “good” looks like.

### Typical Sources (Examples)

- **National health surveys** (e.g., *NHANES* style): De-identified microdata with documentation and code-books.
- **International agencies** (e.g., *WHO Global Health Observatory* style): Indicators aggregated by country/region over time.
- **Government open data portals** (e.g., *BC Data Catalogue* style): Administrative or program data with varying schemas.
- **Multilateral / development banks** (e.g., *World Bank Data* style): Socio-economic indicators for cross-country comparisons.

**Instructor Note:** In class, we will share specific links via Canvas. The goal today is *pattern recognition*—you should be able to recognize a trustworthy portal, find the **About/Documentation**, and locate a **download** link.

### What to Look For on a Portal

1. A human-readable **About** page for the dataset.
2. Clear **licensing/terms of use**.
3. **Metadata** (definitions of variables, units, time coverage).
4. A **download** button (CSV/JSON, etc.) and/or API.
5. A **citation** suggestion.

## Mini-Activity (10 minutes)

Pick any reputable portal (from Canvas links). Find a dataset you personally care about (e.g., smoking prevalence, hypertension, BMI). Write down:

- Dataset name + link (URL)
  - Who publishes it
  - Time coverage
  - One variable you understand immediately
  - One variable whose meaning is unclear
- 

## How to Find, Evaluate, and Cite a Dataset

### A. Finding & Quick Evaluation

When you land on a dataset page, scan for:

- **Scope:** Population, geography, years.
- **Data Level:** Individual/micro vs. aggregated indicators.
- **Update Cadence:** Last updated date.
- **Documentation:** Codebook, methodology PDF.

If any of these are missing, reconsider using it or document the gap.

### B. The Data Intake Card (Copy this block into your notes)

Use this template every time you bring a dataset into your project.

```
1 dataset_name: ""
2 publisher_org: ""
3 permalink_url: ""
4 publisher_contact_or_about: ""
5 license_terms: ""           # e.g., CC BY 4.0; "public domain"; "restricted"
```



```

6 geography: "" # e.g., Canada, global, BC, health authority
7 time_coverage: "" # e.g., 2005-2024, point-in-time
8 data_level: "" # microdata, aggregate indicators
9 variables_of_interest:
10   - name: ""
11     definition: ""
12     unit: ""
13     caveats: ""
14 potential_risks:
15   - privacy: "low/med/high (why?)"
16   - misinterpretation: "what could be misunderstood?"
17 indigenous_data_relevance: "yes/no/unknown (explain if yes)"
18 how_to_cite: "" # copy the suggested citation if provided
19 date_accessed: "YYYY-MM-DD"

```

---

## Privacy, Ethics, and Responsible GenAI Use

Even open data can be used irresponsibly. This course expects **ethical use**: - **No Personal Identifiers**: We only use public, de-identified data in Week 1. - **Attribution**: Acknowledge sources and licenses. - **Context**: Avoid overstating what aggregate indicators can support.

### Using GenAI Responsibly This Week

GenAI (e.g., a coding co-pilot or chat assistant) may be used to:

- **Summarize** a dataset description into plain language.
- **Draft** a first pass of a data dictionary from a codebook.
- **Suggest** variable interpretations you will verify.

You must:

- **Disclose** in your notes when AI is used (what prompt, what output).
- **Verify** with the official documentation.

- **Accept Accountability** for errors (AI is not a source of truth).

We will return to GenAI auditing later in the term; this week is limited to **summaries** and **notes**—**no data uploads** to AI tools.

---

## Indigenous Data Sovereignty (OCAP® & CARE) — Awareness

Before downloading or using data about Indigenous Peoples, be aware that **Indigenous Data Sovereignty** affirms rights and interests in data that pertain to Indigenous Peoples. Two widely cited frameworks you should **know exist**:

- **OCAP® Principles:** specifically referring to **First Nations** ownership, control, access, and possession of data.
- **CARE Principles:** Collective benefit, Authority to control, Responsibility, Ethics.

**What this means for Week 1:** - If your chosen dataset includes Indigenous content, flag it on your **Data Intake Card**.  
- Read the publisher's guidance and respect any use conditions.  
- When in doubt, **ask the instructor** before using or publishing analyses.

We acknowledge these frameworks now and revisit concrete applications later in the course.

---

## File Formats & Metadata (What You Need to Know Now)

Common formats you'll meet:

- **CSV:** Rows/columns; simplest to start.
- **JSON:** Nested; common for APIs.
- **XLSX:** Spreadsheets; watch for merged cells and headers.
- **GeoJSON / Shapefile:** Spatial data; used for maps (later weeks).

#### Best practices this week:

- **Download** the smallest useful file (e.g., one year) to explore the structure. - Keep the raw file **unchanged** in a **data/** folder (we will create this next week).
- Create a short **README section** in your notes describing:
  - Where you got it.
  - What the columns mean (link to codebook).
  - Any obvious pitfalls (missing values, units).

---

## Looking Ahead: The Lab Bench

Next week, we leave the browser and enter the lab. We will introduce **GitHub Codespaces**, your cloud-based “Public Computer” where we will load this data.

To prepare for next week:

1. Ensure you have a **GitHub Account**.
2. Ensure you can access the course repository link on Canvas.

## Portfolio Prep (Ungraded)

### Your Week 1 task (Complete/Incomplete):

1. Choose **one** dataset page from a reputable portal (links on Canvas).
2. Fill out the **Data Intake Card** (copy/paste the YAML block and complete it).
3. Save this card; you will use this dataset for your first real project in Week 5.

### I understand:

- ☐ The difference between open **aggregate indicators** and microdata.
- ☐ How to find documentation/metadata for a dataset.
- ☐ My responsibility when using GenAI (disclose, verify, be accountable).
- ☐ That datasets involving Indigenous Peoples require extra care and may need instructor guidance before use.

# Activity: Data Intake Card

## Goal

Students practice evaluating whether a health dataset is usable, ethical, documented, and appropriate for reproducible analysis.

## Approved Starter Portals

Choose **one** dataset or indicator page from one of the portals below.

1. **Survey-style microdata:** NHANES or NHIS from CDC/NCHS.
2. **Aggregated indicators:** WHO Global Health Observatory.
3. **Canadian open data:** BC Data Catalogue or the Open Government Portal.

The NHANES Health Equity data spine can be used as the shared practice example when students need a common dataset.

## Student Template

```
1 dataset_name: ""
2 publisher_org: ""
3 permalink_url: ""
4 date_accessed: "YYYY-MM-DD"
5 license_terms: ""
```

```

6 geography: ""
7 time_coverage: ""
8 data_level: "" # microdata / aggregated / unknown
9 variables_of_interest:
10   - name: ""
11     definition: ""
12 known_limitations:
13   - ""
14 stewardship_notes: ""
15 ai_use_note: ""

```

## Completed Example: NHANES

```

1 dataset_name: "NHANES Health Equity classroom dataset"
2 publisher_org: "CDC / National Center for Health Statistics"
3 permalink_url: "https://wwwn.cdc.gov/nchs/nhanes/"
4 date_accessed: "YYYY-MM-DD"
5 license_terms: "Public-use NHANES files; cite CDC/NCHS documentation"
6 geography: "United States"
7 time_coverage: "Multiple NHANES cycles represented in the prepared class file"
8 data_level: "Microdata-derived classroom dataset"
9 variables_of_interest:
10   - name: "BMI"
11     definition: "Body mass index from NHANES body measures data"
12 known_limitations:
13   - "Prepared class dataset is descriptive and should not be treated as a causal analysis"
14   - "Population inference requires attention to NHANES survey design and weights"
15 stewardship_notes: "Use public-use data responsibly, avoid small-cell disclosure, and state limitations"
16 ai_use_note: "AI may help draft plain-language summaries, but source documentation must be checked"

```

## Completion Checklist

To be marked **Complete**, your card must include:

- ☐ A stable link or access date.
- ☐ Publisher or organization.

- ☐ License or terms of use.
- ☐ Data level: microdata, aggregated indicator, or unknown with explanation.
- ☐ At least one variable definition grounded in documentation.
- ☐ One limitation or stewardship risk.
- ☐ AI-use note explaining what was drafted and how it was checked.

## Submission

Submit your completed card in Canvas under the Week 1 module. Starting in Week 2, keep a copy in your course repository as `data-intake-card.md`.

## Scoring Guide

- **Complete:** Stable source, publisher, terms, data level, documented variable definition, limitation/stewardship note, and AI-use note are present.
- **Needs work:** Source documentation is missing, license/terms are unclear, variable definitions are guessed, or limitations are absent.

## Reproducibility Check

- The dataset link is stable or includes an access date.
- The publisher and license are named.
- Variable definitions come from source documentation, not AI guesses.
- The AI-use note states how the student checked AI-assisted text.

## **Part IV**

### **Week 2: Modern Workflows**



# Modern Workflows

## Overview

**Summary:** This week you set up the computational environment you will use for the rest of the course. By the end, the technology should fade into the background so you can focus on health data science, not installation errors. We adopt a **“Zero to AI Co-Pilot”** approach: you do not need to install R, Python, or Git on your personal laptop. Everything runs in a cloud-native environment called GitHub Codespaces.

### Learning Objectives:

- Understand the “public computer vs cloud drive” analogy for cloud computing.
- Create a GitHub account, join the course GitHub Classroom, and accept the starter assignment.
- Launch a Codespace and identify the three key areas of the VS Code interface.
- Distinguish between an IDE, a notebook, and a script, and explain why this course uses Quarto notebooks.
- Explain the concept of literate programming and why it matters for reproducible health research.
- Edit and render a Quarto (`.qmd`) document: YAML header, Markdown text, and code chunks.
- Maintain a reproducible project structure using the `data/` and `output/` folder convention.
- Stage, commit, and sync changes using the VS Code Source Control panel.
- Stop your Codespace to conserve your free monthly hours.

## Connection

Week 1 focused on whether a dataset is trustworthy enough to use. This week focuses on where that dataset lives in a project and how relative paths make the same work portable across machines. Next week you will start inspecting the data with R.

## Case Study Data Spine

The **NHANES Health Equity data spine** is the recurring dataset thread for this course. In the early weeks, use it only as a documented public-health data artifact: provenance, ethics, file organization, and reproducible paths.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The cached files are provided so early work can happen without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_spine <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 dim(nhanes_spine)
17 #> [1] 105626      18
```

```

18 glimpse(nhanes_spine)
19 #> Rows: 105,626
20 #> Columns: 18
21 #> $ Cycle      <chr> "1999-2000", "1999-2000", "1999-2000", "1999-2000", "19~
22 #> $ BMI        <dbl> 14.90, 24.90, 17.63, NA, 29.10, 22.56, 29.39, 15.51, 18~
23 #> $ Weight     <dbl> 12.5, 75.4, 32.9, 13.3, 92.5, 59.2, 78.0, 40.7, 45.5, 1~
24 #> $ Height     <dbl> 91.6, 174.0, 136.6, NA, 178.3, 162.0, 162.9, 162.0, 156~
25 #> $ Waist      <dbl> 45.7, 98.0, 64.7, NA, 99.9, 81.6, 90.7, 64.1, 64.6, 108~
26 #> $ WeightMEC  <dbl> 10982.899, 28325.385, 46192.257, 10251.260, 99445.066, ~
27 #> $ Strata     <dbl> 5, 1, 7, 2, 8, 2, 4, 6, 9, 7, 1, 6, 13, 12, 11, 11, 5, ~
28 #> $ PSU        <dbl> 1, 3, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, 2, 1, 1, 1, 2~
29 #> $ Gender     <chr> "Female", "Male", "Female", "Male", "Male", "Female", "~
30 #> $ Race       <chr> "Non-Hispanic Black", "Non-Hispanic White", "Non-Hispan~
31 #> $ Education  <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
32 #> $ Marital    <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
33 #> $ Age        <dbl> 2, 77, 10, 1, 49, 19, 59, 13, 11, 43, 15, 37, 70, 81, 3~
34 #> $ PIR        <dbl> 0.86, 5.00, 1.47, 0.57, 5.00, 1.21, NA, 0.53, NA, NA, 1~
35 #> $ EducationClean <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
36 #> $ MaritalClean <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
37 #> $ IncomeGroup <chr> "Low Income (<1.3)", "High Income (>3.5)", "Middle Inco~
38 #> $ WHtR       <dbl> 0.4989083, 0.5632184, 0.4736457, NA, 0.5602916, 0.50370~

```

## In-Class Activity

- Locate the case-study data, README, and snippet folders in the Explorer.
- Explain why the data example uses relative paths instead of machine-specific paths.
- Check that the weekly page includes the shared snippet rather than duplicating instructions.
- Render the book from the repository root and check that this page updates in the rendered output.

Student output: a rendered Quarto page and one commit showing a small, reproducible edit. Assignment link: [../.. / assignments / assignment01-workflows / README.md](#).

## workflows1.qmd The Analogy: Public Computer and Cloud Drive

To understand how we work in this course, forget about “servers” and “clients.” Imagine you are working in a university computer lab.

### The Repository (Your Cloud Drive)

Think of **GitHub** as your personal **Google Drive** or **Dropbox**.

- It is where your files live permanently.
- If your laptop crashes, the files are still here.
- **Key concept:** This is your “Master Copy.”

### The Codespace (The Public Computer)

Think of a **Codespace** as a **public computer in the library**.

- It is a temporary machine you borrow to do your work.
- It comes pre-installed with all the software you need (R, Python, Quarto), so you don’t have to install anything on your own laptop.
- **Crucial difference:** Just like a public library computer, it can be wiped clean. If you close the browser without sending your work back to your Cloud Drive, your changes may be lost.

### The Workflow

1. **Log in:** Launch your Codespace (open the public computer).
2. **Work:** Edit files using the computer’s software.
3. **Save to the cloud:** Commit and push your work back to your Cloud Drive (GitHub).
4. **Log off:** Stop the Codespace to save your free hours.

This cycle — launch, work, commit, stop — is the rhythm of every week in this course.

---

## workflows2.qmd Getting Started: GitHub Account and Classroom

### Step 1: Create a GitHub Account

If you do not already have one:

1. Go to [github.com](https://github.com) and click **Sign up**.
2. Use your UBC email address (this helps with student benefits).
3. Choose a professional username — this will appear on your public portfolio later in the course.

#### Tip

**Portfolio thinking:** Your GitHub profile will become part of your professional presence. Pick a username you would be comfortable putting on a CV (e.g., `jchen-epi` rather than `xXcoolcoder99Xx`).

### Step 2: Join the Course GitHub Classroom

1. Open the **GitHub Classroom** invitation link posted on Canvas.
2. Accept the invitation. This connects your GitHub account to the course roster.
3. Accept the **starter assignment**. GitHub Classroom will automatically create a personal repository for you, preloaded with the files you need for this week.

### Step 3: Verify Your Repository

Navigate to the repository that was just created. You should see:

- A file list including `practice_report.qmd`, `README.md`, and a `data/` folder.
- A formatted README below the file list explaining the project.

If you see these, you are ready to launch your Codespace.

---

### workflows3.qmd Launching Your Codespace

1. On your repository page, click the green `<>` **Code** button.
2. Select the **Codespaces** tab.
3. Click **Create codespace on main**.

The first launch takes 2–3 minutes while the environment builds. Subsequent launches are faster.

#### Note

**Reopening an existing Codespace:** After you stop or close your Codespace, do **not** create a new one each time. Go to [github.com/codespaces](https://github.com/codespaces), find your existing Codespace, and click its name to restart it. Creating multiple Codespaces wastes your hours and causes confusion.

## workflows4.qmd Tour of VS Code: The “Big Three” Zones

The VS Code interface can look intimidating — it is designed for professional software engineers. You only need to know three areas. Safely ignore 90% of the buttons.

### 1. The Explorer (Left Sidebar)

Your **file cabinet**. It shows folders and files in your project. You will use this to open `.qmd` files, navigate the `data/` folder, and create new files.

### 2. The Editor (Center)

Your **work area**. This is where you read and write code and text. You can have multiple files open in tabs.

### 3. The Terminal (Bottom Panel)

The **engine room** where R and Python actually run. You will not type commands here often — most of your work happens in the Editor — but this is where output and error messages appear.

#### Tip

**Lost a panel?** Use the top menu: **View → Explorer** or **View → Terminal** to bring them back. You can also press `Ctrl+`` (backtick) to toggle the Terminal.

## workflows5.qmd Concepts: IDE, Notebook, Script

The syllabus mentions three ways to write code. Understanding the difference helps you choose the right tool throughout the course.

### IDE (Integrated Development Environment)

VS Code is an **IDE** — a text editor with superpowers: file browsing, syntax highlighting, built-in terminal, extensions, and version control. It is the container that holds everything else.

### Script

A plain text file containing only code (e.g., `.R` or `.py`). Runs top-to-bottom. Best for reusable utilities and automation. You will write your first script in Week 5.

### Notebook

A document that **interleaves code and narrative**. You run code in chunks and see results inline. Examples include Jupyter notebooks (`.ipynb`, Week 5) and Quarto documents (`.qmd`, which you use starting today).

**This course primarily uses Quarto notebooks** because they combine the strengths of notebooks (interactivity, narrative) with the ability to produce publication-quality reports, slides, and websites — all from a single file.

---



## workflows6.qmd Concepts: Literate Programming

### The Big Idea

Traditional workflows separate code from writing: you run an analysis in one program, copy results into a Word document, and hope nothing changes. **Literate programming** weaves code, results, and interpretation into a single document. When the data changes, you re-render and the entire report updates automatically.

This is not just convenient — it is a **scientific obligation**. In health research, reviewers and regulators need to see exactly how you went from raw data to conclusions. A literate document is a transparent, auditable record of your reasoning.

### How Quarto Implements This

A `.qmd` file has three ingredients:

1. **YAML header** (metadata at the top, between `---` lines)
2. **Markdown text** (human-readable prose)
3. **Code chunks** (executable R or Python code in fenced blocks)

When you **Render**, Quarto runs every code chunk, captures the output (tables, figures, numbers), and weaves everything into a finished HTML, PDF, or Word document.

You will use literate programming in every subsequent week of this course: your assignments, your EDA (Week 7), your reports (Week 10), and your final portfolio are all Quarto documents.

## workflows7.qmd Project Structure: Files and Folders

A reproducible project needs a consistent folder structure. This course uses the following convention:

```
my-project/  
  data/           ← Raw data. Never edit files here directly.  
  output/         ← Cleaned data, figures, and tables generated by your code.  
  practice_report.qmd ← Your Quarto notebook.  
  README.md       ← Explains what this project is.  
  .gitignore      ← Tells Git which files to ignore (more in Week 4).
```

### Rules

- **data/ is read-only.** Import from it; never save into it.
- **output/ is generated.** Your code writes cleaned data and figures here. In Week 3, you will export `output/clean_data.csv`.
- **Use relative paths.** Always refer to files relative to the project root (e.g., `"data/patients.csv"`), never absolute paths (e.g., `"/home/user/Documents/patients.csv"`). This ensures your code runs on any machine.

**Activity:** In the Explorer pane, click the **New Folder** icon and create the `output/` folder if it does not already exist.

---

## workflows8.qmd Tutorial: Your First Quarto Document

### Step 1: Open the Starter File

In the Explorer, click `practice_report.qmd`. It opens in the Editor.

## Step 2: Understand the YAML Header

At the very top, between `---` lines, you will see something like:

```
1 ---
2 title: "My First Report"
3 author: "Your Name"
4 format: html
5 ---
```

Change `author:` to your actual name. The YAML header controls metadata and output format.

## Step 3: Read the Markdown

Below the YAML, you will see plain text with formatting:

- `#` Heading creates a section title.
- `**bold**` makes text **bold**.
- `- item` creates a bullet point.

This is **Markdown** — a lightweight way to format text without a word processor.

## Step 4: Read the Code Chunks

Look for grey blocks that start with ````{r}` and end with `````. These are **code chunks**. Everything inside them is R code that will execute when you render.

```
1 ```{r}
2 # This is a code chunk
3 2 + 2
4 ```
```

## Step 5: Edit a Code Chunk

Find a code chunk in the starter file. Make a small change — for example, change a number or add a comment. This is your first edit.

## Step 6: Render

Click the **Render** button (blue arrow icon at the top of the editor). A preview window appears showing your formatted report with code output embedded.



**What just happened:** Quarto ran every code chunk, captured the results, combined them with your Markdown text, and produced an HTML document. This is literate programming in action.

---

## workflows9.qmd Tutorial: Stage, Commit, and Sync

Your work currently exists only on the public computer (Codespace). To save it permanently to your Cloud Drive (GitHub), you need to **commit**.

## Step 1: Open Source Control

Click the **Source Control** icon on the left sidebar (it looks like a branching graph). You will see a list of files that have changed since your last commit.

## Step 2: Stage Your Changes

Hover over a changed file name. Click the + icon that appears. This **stages** the file — it tells Git “I want to include this change in my next snapshot.”

### Note

**Staging vs committing:** Staging selects *which* changes to include. Committing takes the snapshot. Think of staging as putting items on the conveyor belt; committing is pressing the “scan” button at checkout.

## Step 3: Write a Commit Message

In the text box at the top of the Source Control panel, type a short, descriptive message:

Updated author name and edited first code chunk  
in practice report

A good commit message answers: *what did I change and why?*

## Step 4: Commit

Click the **Commit** button (checkmark). Your changes are now saved as a snapshot in your local history.

## Step 5: Sync with GitHub

Click **Sync Changes** (in the Source Control panel or the status bar at the bottom). This pushes your commit to GitHub — your Cloud Drive.

**Verify:** Go to your repository on [github.com](https://github.com) and check that your changes appear. You should see your commit message next to the updated file.

 Warning

If you skip the **Sync step**, your commit exists only on the Codespace. If the Codespace is deleted, the commit goes with it. Always Sync.

---

## workflows10.qmd Stopping Your Codespace (Saving Credits)

GitHub Codespaces gives you a monthly allowance of free hours (approximately 60 core-hours). **If you leave the browser tab open, the meter keeps running**, even if you are not actively working.

### The Right Way to Stop

1. Go to [github.com/codespaces](https://github.com/codespaces).
2. Find your active Codespace.
3. Click the **three dots (...)** → **Stop codespace**.

### What Happens If You Just Close the Tab?

The Codespace runs for another 30 minutes before it times out automatically. That is 30 minutes of wasted credits every time.

 Warning

**Policy:** It is your responsibility to manage your compute hours. If you run out before the month resets, you will need to wait or pay out of pocket. Get in the habit of stopping your Codespace at the end of every session.

## workflows11.qmd Troubleshooting

Sometimes things break. Here is a graduated approach — try the simplest fix first.

### Level 1: Reload the Window

Press F1 (or Ctrl+Shift+P / Cmd+Shift+P) to open the **Command Palette**. Type `Reload Window` and select it. This restarts VS Code without rebuilding the entire environment.

### Level 2: Restart the R or Python Session

If R or Python is frozen, open the Command Palette → type `R: Restart R Session` (or the equivalent for Python). This clears the session without touching your files.

### Level 3: Rebuild the Container (Nuclear Option)

If nothing else works:

1. Open the **Command Palette**.
2. Type `Rebuild Container`.
3. Select **Codespaces: Rebuild Container**.

This shuts down and rebuilds the entire environment from scratch. It solves nearly all technical glitches but takes a few minutes.

#### Warning

**Before rebuilding:** Make sure you have committed and synced your work. Uncommitted changes may survive a rebuild, but there is no guarantee.

## workflows12.qmd Knowledge Check

1. In the analogy, what is the “Cloud Drive” and what is the “Public Computer”?
  2. Why don’t you need to install R or Python on your laptop for this course?
  3. Name the “Big Three” zones of the VS Code interface and what each is for.
  4. What is the difference between an IDE, a notebook, and a script?
  5. What is literate programming, and why does it matter for health research?
  6. What are the three ingredients of a .qmd file?
  7. What is the difference between **staging** and **committing**?
  8. Why must you click **Sync Changes** after committing?
  9. How do you properly stop a Codespace to save your free hours?
- 

## workflows13.qmd Assignment 1: Your First Quarto Report

Due Friday 4 pm. Submit via Canvas.

### Task

Edit and render the starter `practice_report.qmd`, demonstrating that you can navigate your Codespace, edit Quarto, and commit your work.

### Step-by-Step Workflow

1. **Launch** your Codespace from the GitHub Classroom repository.
2. **Edit the YAML:** Change `author:` to your name.



3. **Add a Markdown section:** Below the existing content, add a new heading (**## My Notes**) and write 2–3 sentences about what you learned this week. Use at least one Markdown feature (bold, italics, or a bullet list).
4. **Edit a code chunk:** Modify an existing code chunk or add a new one. For example:

```

1 ```{r}
2 # My first R calculation
3 2 + 2
4 ```

```

5. **Render** the document. Check that the output looks correct.
6. **Commit and Sync:**
  - Open Source Control.
  - Stage your changed files (+).
  - Write a commit message: e.g., "Complete Week 2 assignment: edit YAML, add notes, render report".
  - Click **Commit**, then **Sync Changes**.
7. **Stop** your Codespace.
8. **Verify** on github.com that your commit appears in the repository.
9. **Submit** the repository link on Canvas.

---

## workflows14.qmd Quick Reference

| Action           | How                                                                                         |
|------------------|---------------------------------------------------------------------------------------------|
| Launch Codespace | Repository → green <><br><b>Code</b> → Codespaces →<br><b>Create codespace on main</b>      |
| Reopen Codespace | <a href="https://github.com/codespaces">github.com/codespaces</a> →<br>click Codespace name |

| Action            | How                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------|
| Stop Codespace    | <a href="https://github.com/codespaces">github.com/codespaces</a> → ...<br>→ <b>Stop codespace</b> |
| Open Explorer     | <b>View</b> → <b>Explorer</b> or click<br>file icon (left sidebar)                                 |
| Open Terminal     | <b>View</b> → <b>Terminal</b> or<br><b>Ctrl+`</b>                                                  |
| Render Quarto     | Click <b>Render</b> button (blue<br>arrow) at top of editor                                        |
| Stage a file      | Source Control → hover file<br>→ click +                                                           |
| Commit            | Type message → click                                                                               |
| Sync to GitHub    | Click <b>Sync Changes</b>                                                                          |
| Reload window     | Command Palette (F1) →<br><b>Reload Window</b>                                                     |
| Rebuild container | Command Palette (F1) →<br><b>Codespaces: Rebuild<br/>Container</b>                                 |

## 2 The Analogy: Public Computer vs Cloud Drive

To understand the workflow, imagine working in a university computer lab.

### 2.1 Repository (Cloud Drive)

**Important Distinction:** While GitHub is your “Cloud Drive,” it does **NOT** work like Google Drive.

- **Google Drive:** Auto-saves every keystroke.
- **GitHub:** Works like **Email**. You can write a draft (save in Codespace), but the cloud never sees it until you hit “Send” (Commit & Push).
- Stores your files permanently

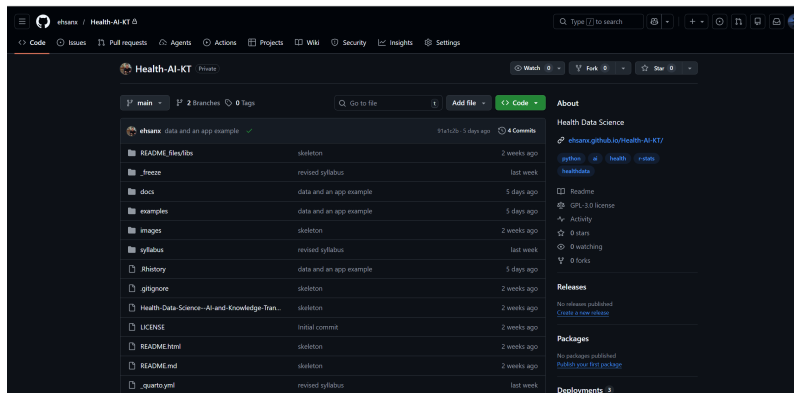


Figure 2.1: The code will be stored and accessible on GitHub as long as the company exists. Given that it’s owned by a large corporation like Microsoft, it will likely last for a long time.

- Safe if your computer crashes
- This is your **master copy**

## 2.2 Codespace (Public Computer)

Codespaces is like a library computer:

- **Temporary machine** It is a disposable environment. If it breaks, we just delete it and get a new one.

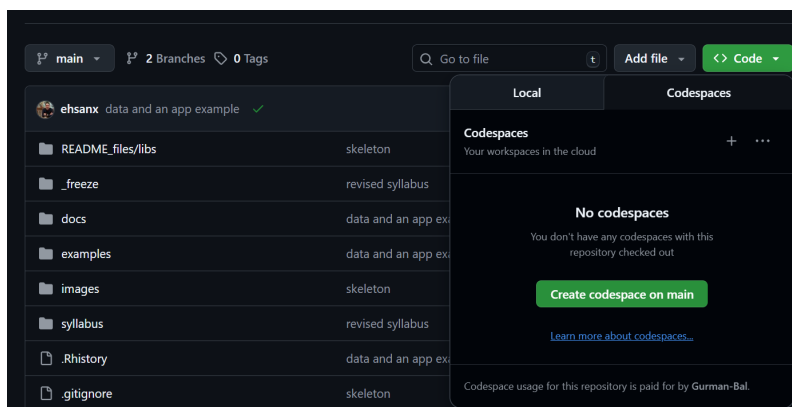


Figure 2.2: Can be accessed by clicking the <> Code button, navigating to the Codespaces tab, and then starting up a Codespace which contains all the file in the Repo.

- **Pre-installed software** It comes with R, Python, and Quarto installed, so you don't have to install anything on your own laptop.
- **The “Two-Save” Rule (Crucial)** Because this is a temporary computer, there are two steps to saving:
  1. **Ctrl+S (Save):** This only saves to the temporary machine. If the machine times out (closes), this is lost.
  2. **Commit & Push:** This sends your work back to the Repository (Cloud Drive). This is the only way to be safe.

**Warning:** If you close the tab without “Committing” (Sending), your work is erased.

### 2.2.1 Check for understanding

Answer:

- Where are your permanent files stored?
- What happens if you press **Ctrl+S** but never commit?

## 3 Tour of the Repository

### 3.1 Step-by-step

1. Open your course repository on GitHub

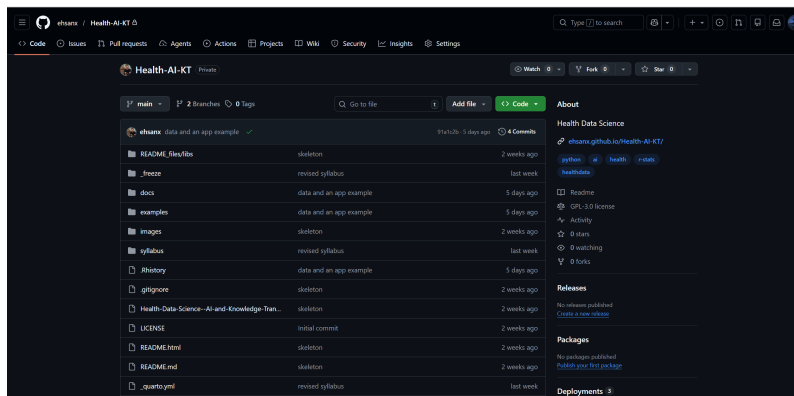
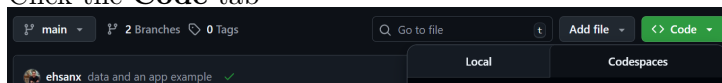


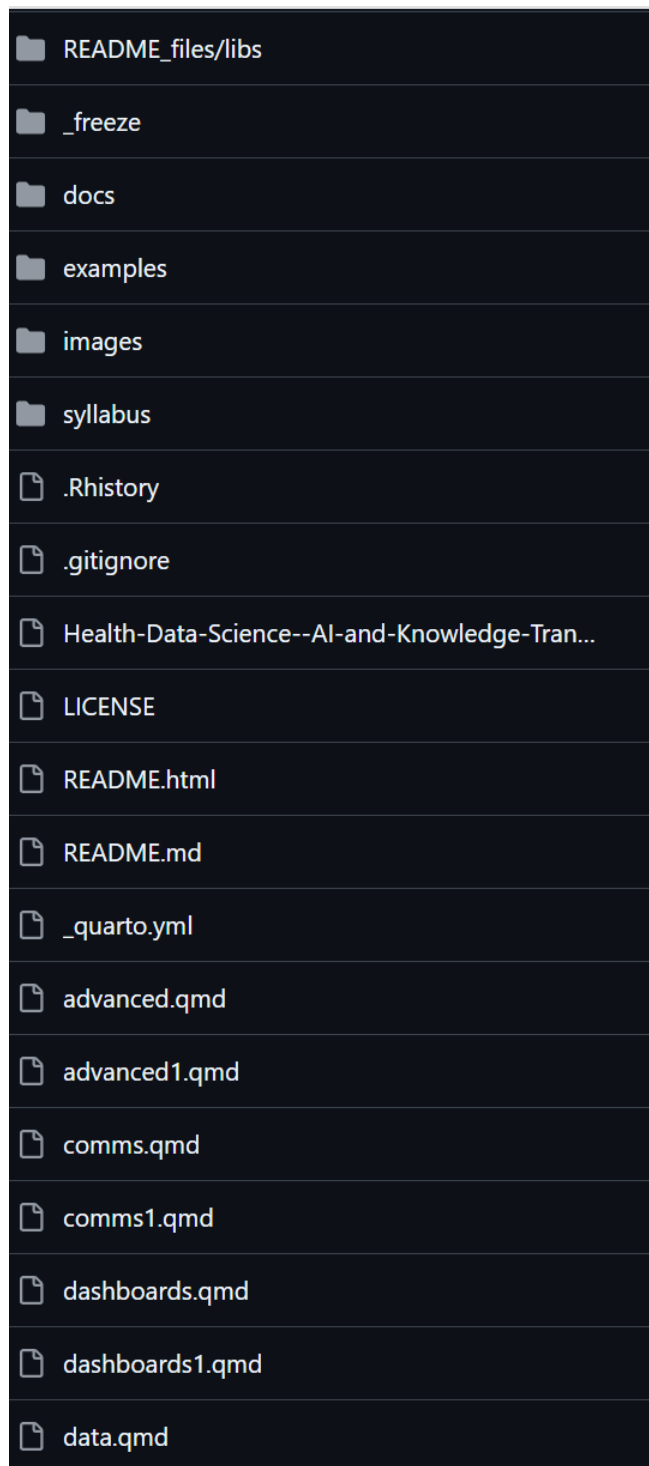
Figure 3.1: Main github page

2. Click the **Code** tab



- **Why?** GitHub has many tabs (Issues, Actions, Settings). The **Code** tab is your “Reset” button. If you ever get lost in a menu, click this to go back to your files.

3. Look at the file list



- This shows the **current state** of the project in the Vault.
- **Crucial:** If a file is listed here, it is safe. If it is *not* here, it only exists on your temporary “Desk” (Codespace) and is at risk of being lost.

#### 4. Scroll to read the README

- The `README.md` file is displayed automatically at the bottom. Think of it as the “Cover Sheet” or “Instruction Manual” for your project.

## 3.2 Task

Find this file:

`practice_report.qmd`

**Verification:** If you can see `practice_report.qmd` in this list, it means the file is safely stored in the cloud (The Vault).

If you can see it, it is safely stored in the cloud.



## 4 Opening Your First Codespace

This process is like walking into the library and turning on a computer. You are creating a fresh, temporary environment in the cloud.

### 4.1 Steps

1. Click the green **Code** button
  - This button is usually for downloading files, but we are using it to launch a computer.

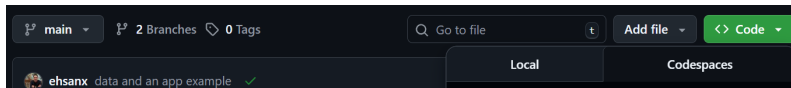


Figure 4.1: Code button

2. Click the **Codespaces** tab
  - **Crucial Choice:** You will see two tabs: “Local” and “Codespaces.”
  - **Local:** Means downloading to your own messy laptop (Hard mode).
  - **Codespaces:** Means using the pre-installed cloud computer (Easy mode). Always choose this.

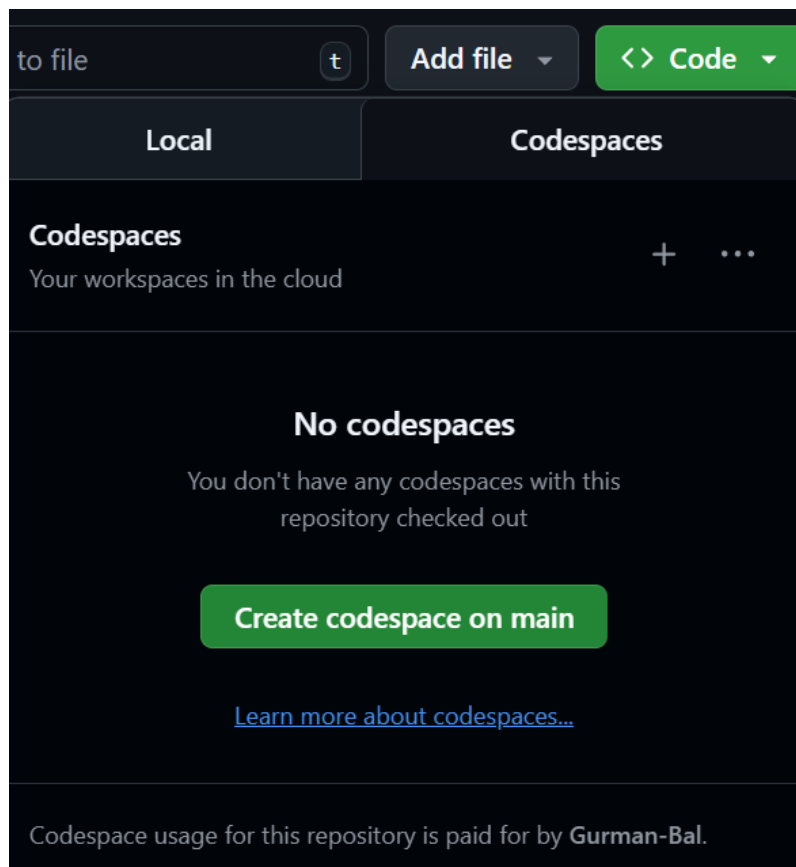


Figure 4.2: Codespaces tab

3. Click **Create Codespace on main**

- This tells GitHub to build a new computer for you from scratch.

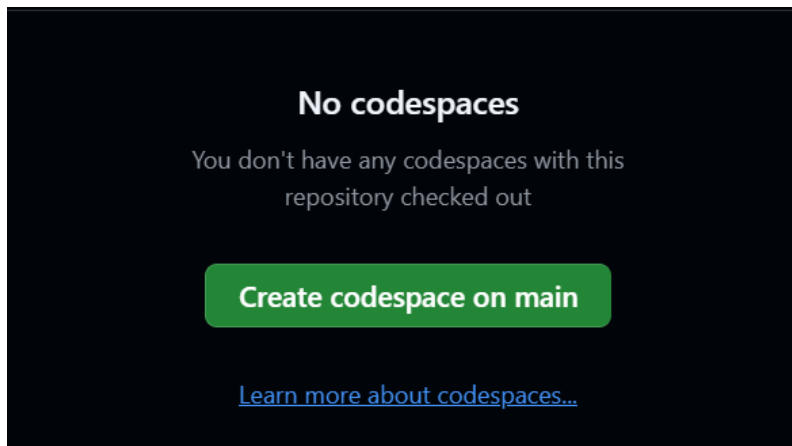


Figure 4.3: Create Codespace

## 4.2 The “Boot Up” Phase

Wait 2–3 minutes.

- **What is happening?** GitHub is silently installing R, Python, Quarto, and all the course libraries for you.
- **The Analogy:** Think of this as the “ventilation system” starting up in a lab. It takes a moment to get the environment ready.

**Success:** You should see **VS Code** (a dark-themed code editor) appear in your browser window.

## 5 Reopening a Codespace

**The Rule:** Do not create a new Codespace each time you log in.

- **Why?** If you click the green “Create” button again, GitHub builds a *second* computer.
- **The Risk:** If you left uncommitted work (Drafts) on “Computer A” and you start “Computer B,” your work will appear lost. It isn’t lost: it’s just on the other computer.

To resume exactly where you left off:

### 5.1 Steps

1. Go to the **Codespace Dashboard**: <https://github.com/codespaces>
  - Think of this as the “Front Desk” where all your active computers are listed.
2. Find your Codespace
  - Look for the box labeled with your course repository name.
  - **Note:** GitHub gives each Codespace a funny random name (e.g., *literate-computing-machine* or *glorious-couscous*). This is the name of your rented desk.

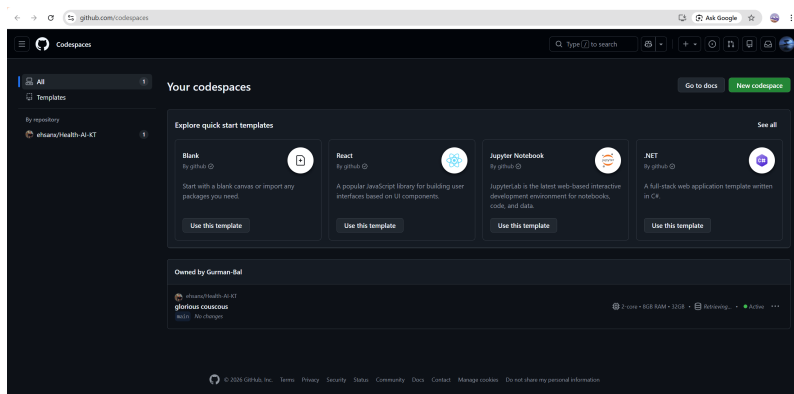


Figure 5.1: GitHub Codespaces

### 3. Click the **Name** of the Codespace

- Do **not** click the “New codespace” button in the corner. Click the text link of the funny name.

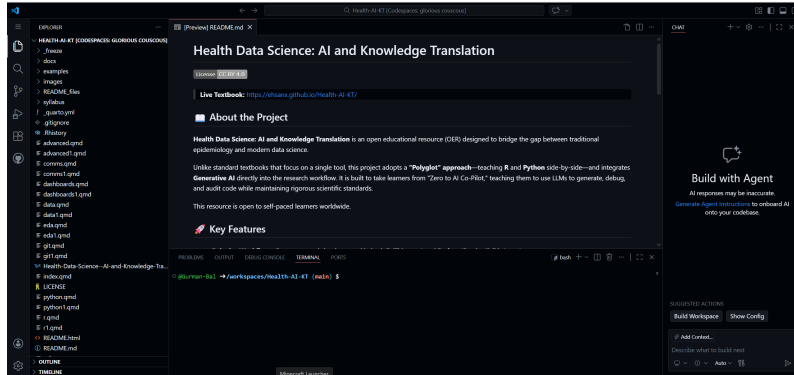


Figure 5.2: Your Codespace

### 4. Wait for it to resume

- **Benefit:** Resuming an existing Codespace is much faster than creating a fresh one because the software is already installed.

## 6 VS Code Orientation



When you open a Codespace, you are using a version of **VS Code** in your browser. It has the same interface as the desktop version used by professional software engineers at Google and Microsoft.

**Don't Panic:** The interface looks complex because it has hundreds of features. **You only need to know three areas.** You can safely ignore 90% of the buttons you see.

### 6.1 Explorer (left)

The “File Cabinet”

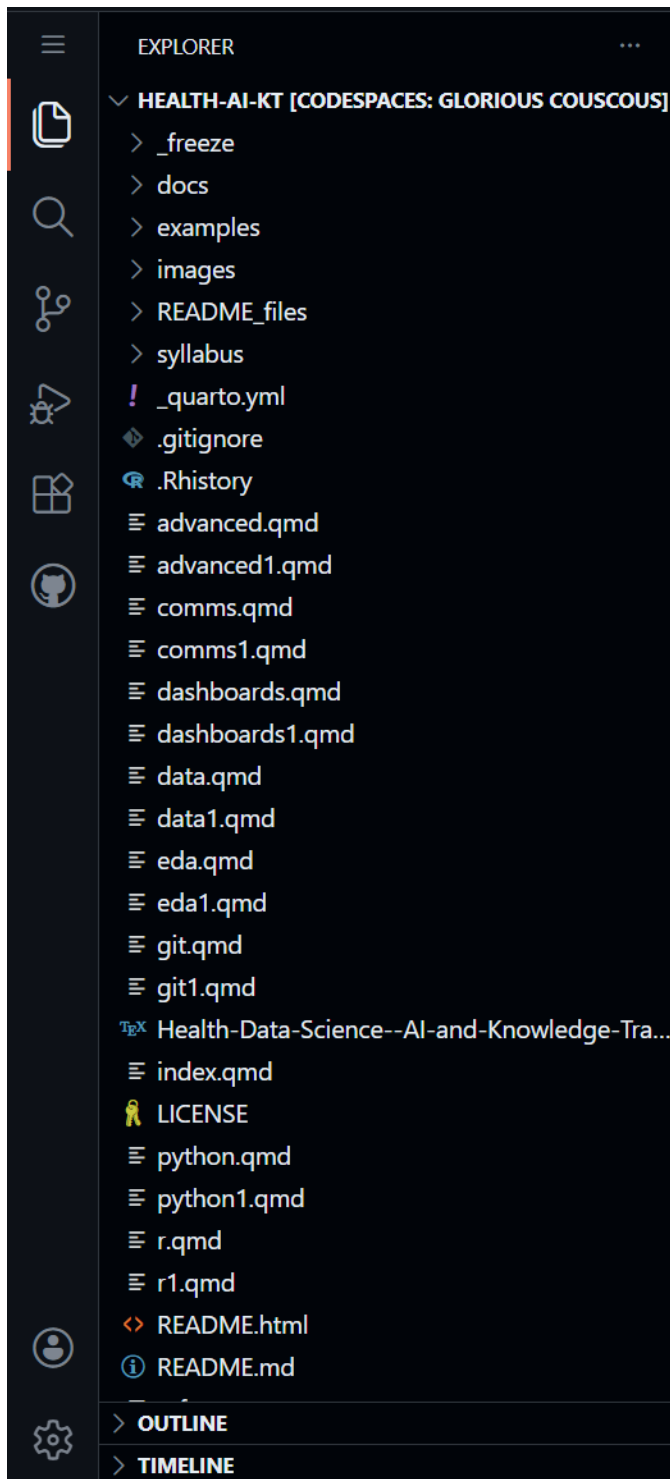


Figure 6.1: Shows files and folders.

- This shows your folders (directories) and files.
- Think of this as your navigation pane. Double-click a file here to open it in the Editor.

## 6.2 Editor (center)

### The “Notebook”

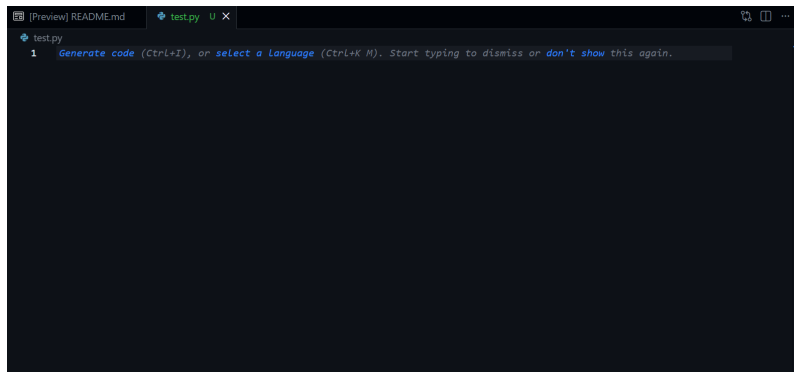


Figure 6.2: Where you type.

- This is where you do your actual work: typing code and writing text.
- In this course, you will mostly be editing `.qmd` (Quarto) files here.

## 6.3 Terminal (bottom)

### The “Engine Room”

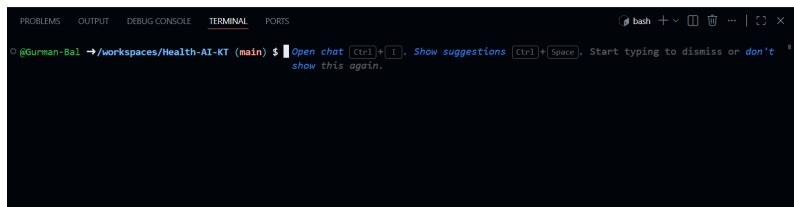


Figure 6.3: Terminal window



- This is where R and Python actually run.
- You will see text scrolling here when you “Render” a document. It looks like “The Matrix,” but it is usually just the computer telling you what it is working on.

If the terminal disappears (or you accidentally close it):

**View → Terminal**

### **6.3.1 Mini Task**

Open and close a few files in the Explorer to get comfortable with the layout.

## 7 Files vs Folders

We use a consistent structure to ensure reproducibility. In Data Science, being “organized” isn’t just about being neat—it is a requirement for your code to work on someone else’s machine.

### 7.1 The Golden Rule of Health Data

We treat data like a **Crime Scene**—never touch the evidence.

- **Start:** `data/` (The Crime Scene). **READ ONLY.**
  - *Rule:* Never open a dataset in Excel and “fix” a typo. That is tampering with evidence.
  - *Correct Way:* Load the “dirty” data into R/Python and fix the typo using code. This leaves a trail of exactly what you changed.
- **End:** `output/` (The Evidence). All graphs and cleaned tables go here.
- **The Detective:** Your code (`.qmd`) moves information from Start to End.

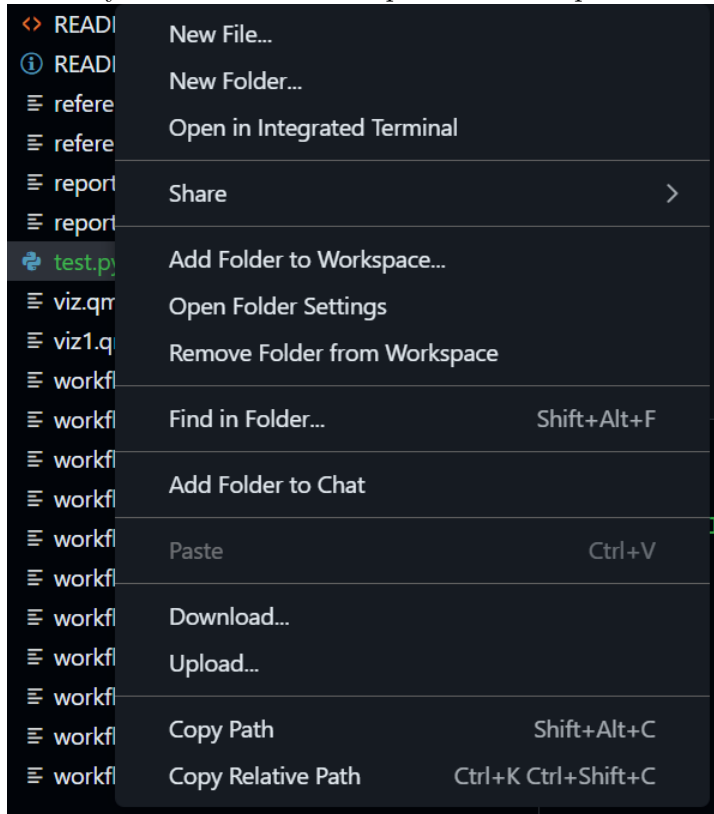
### 7.2 Files vs. Folders (The Mental Model)

- **Files (e.g., `report.qmd`):** These are **sheets of paper**. You write on them.
- **Folders (e.g., `data/`):** These are **labeled boxes**. You store things in them.

## 7.3 Create a folder

In Explorer:

1. Right-click anywhere in the blank space of the Explorer



pane.

2. Select **New Folder**
3. Name it:

images

**Pro Tip:** Always use **lowercase** and **no spaces** for folder names (e.g., `my_data`, not `My Data`). Computers struggle with spaces in file paths.

## 7.4 Typical structure

Your project should always look like this:

|         |                   |
|---------|-------------------|
| data/   | raw data          |
| output/ | generated results |
| images/ | screenshots       |
| .qmd    | reports           |

**Remember:** The `data` folder is for *input*. The `output` folder is for *results*. They should never mix.

## 8 The Magic Moment: Render Quarto

### 8.1 Why do we “Render”?

In the old days, you would run code, copy a graph, save it as a JPG, and paste it into Word. If the data changed, you had to redo every step manually.

**The Better Way:** Think of your code (`.qmd` file) as a **Recipe** and the final report (HTML/PDF) as the **Meal**. \* **Editing:** You don’t change the meal; you change the recipe. \* **Rendering:** This is the “cooking” process. When you click **Render**, the computer follows your instructions from scratch to create a fresh meal.

This is the secret to reproducibility: if you give your recipe to a friend, they can cook the exact same meal.

### 8.2 Step 1 — Open the file

1. Go to the **Explorer** pane (Left side).
2. Locate and click on the file:

```
practice_report.qmd
```

This will open the “Source Code” in the center Editor window. You will see a mix of normal text and code chunks.

## 8.3 Step 2 — Edit name (The YAML Header)

Look at the very top of the file. You will see a block of text sandwiched between three dashes (---). This is called the **YAML Header**. It controls the document’s metadata.

Find the line that says `author:`

```
author: "Your Name"
```

**Task:** Delete “Your Name” and type your actual name inside the quotes. Do not remove the quotes!

## 8.4 Step 3 — Render

1. Look for the **Render** button at the top of the Editor pane (it often looks like a Blue Arrow pointing right, or simply says “Render”).
2. **Click it.**
3. **Observe:**
  - **The Terminal (Bottom):** You will see text scrolling quickly. This is R/Python executing your code chunks.
  - **The Preview (Right):** A new window should appear showing a beautiful, formatted report with your name on it.

**Congratulations!** You just performed “Knowledge Translation”—turning raw code into a human-readable document.

## 8.5 If it fails (Troubleshooting)

Rendering involves many moving parts, so it sometimes gets stuck.

- **The “Wait” Rule:** If nothing happens for 20 seconds, the background process might be initializing. Give it a moment.
- **The “Retry” Rule:** Sometimes the connection hiccups. Click **Render** one more time.
- **The “Red Text” Rule:** Look at the **Terminal** tab at the bottom. If you see red text, it is an error message. Read it—it usually tells you exactly what went wrong (e.g., “File not found” or “Syntax error”).

## 9 Commit Your Work

Your work is currently only on the “Rented Desk” (Codespace). It is a **Draft**. To save it permanently to the “Vault” (GitHub), you must **Commit** and **Push** (Send).

**The Analogy:** Think of this like shipping a package. You have packed the box (saved the file), but now you need to put a label on it and hand it to the courier.

### 9.1 Steps

1. Click the **Source Control** icon
  - It looks like a **blue bubble** or a **connect-the-dots graph** on the far left sidebar.
  - This opens the “Shipping Department.” You will see a list of every file you changed.

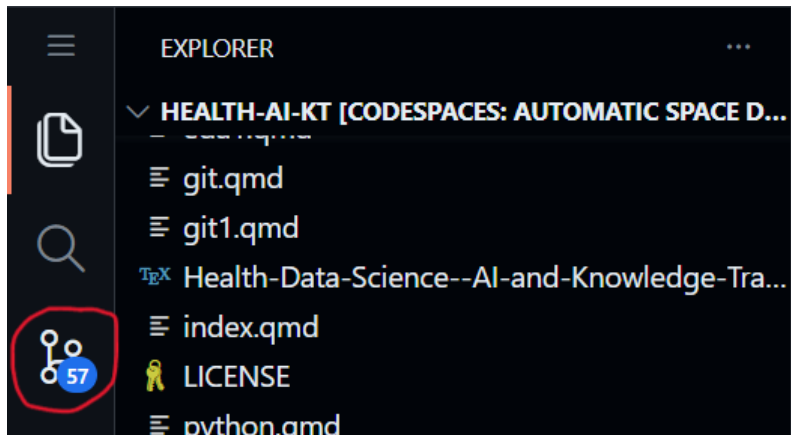


Figure 9.1: Source control icon circled in red



## 2. Type a **Commit Message**

- **The Label:** You cannot ship a package without a label.
- **What to write:** Be descriptive but brief. Imagine you are leaving a note for your future self about *what* you did.
- *Example:* “Updated author name in practice report”.
- *Bad Example:* “stuff” or “fix”.

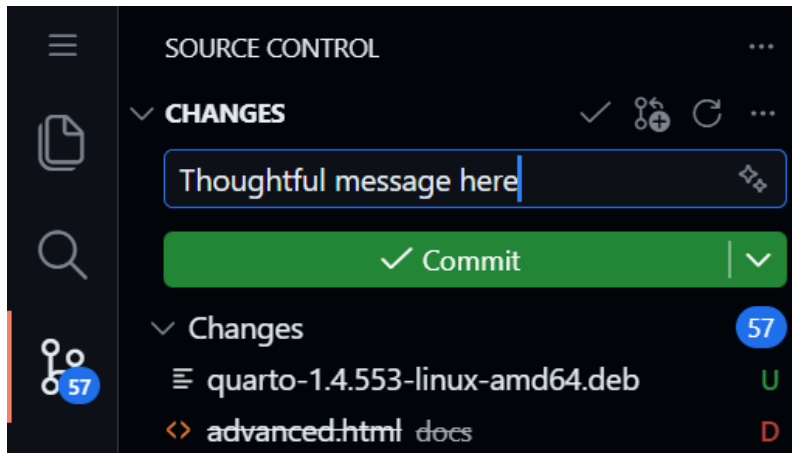


Figure 9.2: This is required

## 3. Click **Commit and Sync** (or Commit and Push)

- This is the “**Send**” button.
- **Action:** It wraps up your changes, stamps them with the time and your message, and teleports them to the Cloud Vault.
- *Note:* You might be asked to click “Always” or “Yes” to approve the sync.

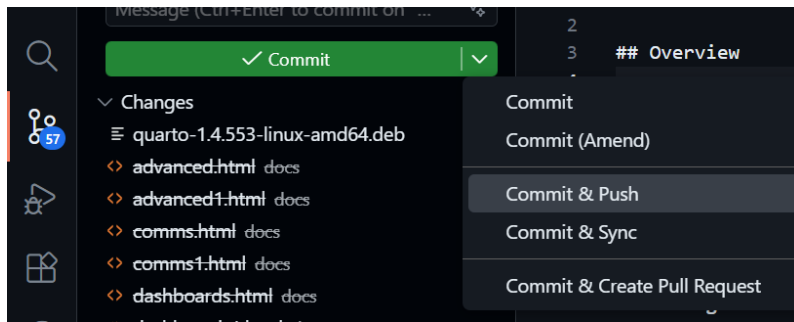


Figure 9.3: This will push your changes to the remote branch

## 9.2 Verify (Trust but Verify)

Did it actually work?

1. Go back to your **GitHub Repository** tab (The Vault).
2. **Refresh** the page.
3. Look at the file list. You should see your “Commit Message” displayed next to the file you just edited.

# 10 Troubleshooting: Rebuild Container

Sometimes the cloud computer gets “stuck.” Maybe the terminal froze, or R crashed.

**The Fix:** We don’t fix broken computers; we get new ones. “Rebuilding the Container” is the cloud equivalent of “**Have you tried turning it off and on again?**”—but better, because it gives you a fresh machine while keeping your files safe.

## 10.1 When to use this?

If things break:

- **Terminal frozen:** You type, but nothing appears.
- **Render fails:** You get weird errors that make no sense.
- **Environment issues:** R or Python can’t find a package that was there yesterday.

## 10.2 Steps

### 1. Open the Command Palette

- Press `F1` (or `Ctrl+Shift+P` on Windows / `Cmd+Shift+P` on Mac).
- A search bar will appear at the very top of the screen. This is the “God Mode” search bar for VS Code.

### 2. Search

- Type: Rebuild Container

### 3. Execute

- Click on the option that says: **Codespaces: Re-build Container**

### 4. Wait (~2-5 minutes)

- The screen will reload.
- **Don't Panic:** You might see a black screen with scrolling text logs. This is normal: it is reinstalling the software.
- When your files reappear in the Explorer, you are ready to go.

### Did I lose my work?

- **Saved files:** No. Anything you pressed **Ctrl+S** on is safe.
- **Unsaved typing:** Yes.
- **R Memory:** Yes. You will need to re-run your code chunks from the top.

# 11 Stop Your Codespace

Codespaces is not free forever. You have a monthly allowance (approx. 60 hours/month for free accounts).

**The Analogy:** Think of a Codespace like a **Taxi**.

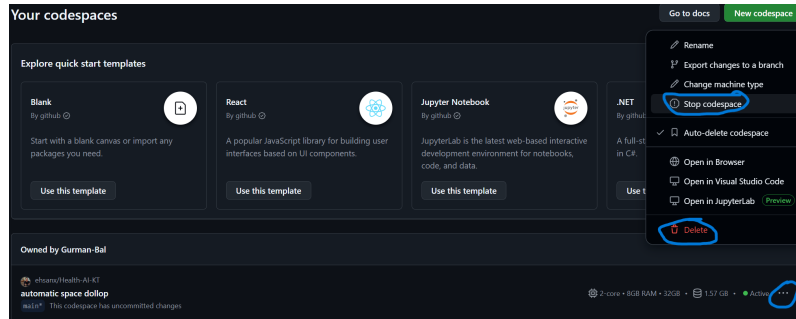
- **Working:** When you are coding, the meter is running.
- **The Trap:** If you just close your browser tab, it is like getting out of the taxi but asking the driver to “wait here.” **The meter keeps running.**
- **The Fix:** You must explicitly “end the ride” (Stop the Codespace).

## 11.1 Correct shutdown

1. Go to the **Codespace Dashboard**: <https://github.com/codespaces>
  - This is your command center.
2. Find your active Codespace
  - Look for the green text that says “Active”.
3. Click ... (The “Meatball” Menu)
  - It is located on the far right side of the Codespace box.
4. Click **Stop Codespace**
  - **Stop:** This “pauses” the computer. It saves your installed packages and RAM state so you can resume quickly next time. **(Recommended)**
  - **Delete:** This destroys the computer. You will have to wait 2-3 minutes for the “Boot Up” phase next time you start.

**Critical Warning:** Closing the browser tab is **NOT** enough.

- GitHub keeps the machine running for **30 minutes** after you close the tab (in case your internet flickered).
- **The Cost:** If you just close the tab every day, you will waste ~15 hours of your monthly quota on empty time.



## 12 Final Checklist

- ☐ I understand repository vs Codespace
- ☐ I rendered a `.qmd` file
- ☐ I committed changes
- ☐ I know how to rebuild container
- ☐ I know how to stop Codespace

### 12.1 Reflection

Write 2–3 sentences:

- What was easiest?
- What was confusing?
- What questions do you still have?

## **Part V**

### **Week 3: R with AI**



# Week 3: R with AI

## Overview

**Summary:** In Week 2 you learned how to use your Codespace, edit a Quarto document, and make your first commit. Now you learn how to speak R—not by memorizing syntax, but by directing an AI co-pilot and *auditing* what it writes. Your job is to be the pilot: you decide what the analysis should do, the AI drafts the code, and you verify every line before it flies.

### Learning Objectives:

- Understand that R is a calculator that needs specific, case-sensitive instructions.
- Load packages with `library()` and explain why this is needed each session.
- Use core constructs: variables (assignment), functions (with arguments), and the pipe (`%>%` / `|>`).
- Perform basic data transformations using `dplyr` verbs: `filter()`, `select()`, `mutate()`, `summarise()`, `group_by()`.
- Inspect a dataset on first contact: structure, dimensions, missingness.
- Identify control flow (`if/else`, loops) in AI-generated code and audit whether it is correct.
- Write and adapt a simple custom function.
- Apply an AI safety checklist to catch hallucinations, wrong column names, and unsafe operations.
- Document your analysis in Quarto by weaving code chunks with Markdown narrative.
- Commit incrementally as you work (reinforcing Week 2 Git basics).

## Connection

Week 2 made the project structure reproducible. This week you use that structure to load, inspect, summarize, and audit R code with AI support. Next week you will put those edits under version control.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
```

```

19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>         <int>         <int>
25 #> 1 105626         9356         9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>         <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## In-Class Activity

- Load the CSV snapshot and inspect column names, row count, and missingness.
- Summarize one numeric variable and one grouping variable with clear NA handling.
- Ask an AI assistant to comment on your code, then audit whether the comments match the actual operations.
- Note one place where an AI assistant could hallucinate a variable name or overstate what the descriptive summary means.

Student output: a short Quarto note with one data inspection, one summary, and one AI-audit comment. Assignment link: [../..../assignments/assignment02-r-with-ai/README.md](#).

---

## r1.qmd Packages: Opening the Toolbox

### Install vs Load

- `install.packages("tidyverse")` — **buying** the toolbox. Done once per environment (already done in your course Codespace).
- `library(tidyverse)` — **opening** the toolbox. Required every time you restart R or reopen your Codespace.

```
1 library(tidyverse)
```

#### Note

If you see an error like `there is no package called 'tidyverse'`, the package is not installed in your Codespace. Ask your TA — this should be pre-configured.

---

## r2.qmd Core Constructs: Boxes, Toasters, and the Assembly Line

### Variables: The Labeled Box

R has no memory unless you store results in a named box using `<-`.

```
1 patient_age <- 55
2 clinic_name <- "Vancouver General"
```

### Warning

**Case sensitivity:** Age, age, and AGE are three different boxes. This is one of the most common bugs in AI-generated code. Always check that column names match your data exactly.

## Functions: The Toaster

A function takes **input** and returns **output**. Arguments (inside the parentheses) control its behaviour.

```
1 mean(c(72, 85, 90), na.rm = TRUE)
```

- `c(72, 85, 90)` is the input (a vector of numbers).
- `na.rm = TRUE` tells R to ignore missing values. **Always look for this** when auditing AI code that works with health data — missingness is the norm, not the exception.

## The Pipe: The Assembly Line

The pipe passes the result of one step into the next. Think of it as “**and then...**”

```
1 data %>%  
2   filter(age > 65) %>%  
3   summarise(mean_bmi = mean(bmi, na.rm = TRUE))
```

Read this as: “*Take the data, and then keep only patients over 65, and then calculate the average BMI.*”

### Tip

You will see two pipe symbols in the wild: `%>%` (from the `magrittr` / `tidyverse` packages) and `|>` (built into R 4.1+). They do the same thing for our purposes. If AI generates the other one, don’t panic — both work in your Codespace.

---

## r3.qmd Data Import and First Contact

### Loading Data

Use relative paths so your code runs on any machine (including your grader's Codespace).

```
1 my_data <- read_csv("data/patients.csv")
```

**How to get the path right:** Right-click the file in the VS Code Explorer → **Copy Relative Path** → paste it into your code.

#### AI Prompt:

Read the CSV file at “data/patients.csv” and save it as `my_data`. Use the `readr` package.

### First Contact: Inspect Before You Analyze

Never start analysis without looking at your data. Run these four checks:

```
1 # How big is it?
2 dim(my_data)           # rows x columns
3
4 # What are the column names and types?
5 glimpse(my_data)
6
7 # Quick summary statistics
8 summary(my_data)
9
10 # How much is missing?
11 colSums(is.na(my_data))
```

Open the **Environment** pane (top-right in RStudio or via the R extension) to see your variables and click a data frame to view it in a grid.

 **Warning**

**Health data reality:** Real datasets like NHANES have substantial missingness. Knowing *where* and *how much* is missing is not optional — it shapes every downstream decision.

---

## r4.qmd Data Transformation: The Five dplyr Verbs

These five verbs are the core of data wrangling in R. You will use them in nearly every assignment and in your term project.

**filter()** — Keep rows that match a condition

```
1 # Patients aged 65 and older
2 seniors <- my_data %>% filter(age >= 65)
```

**select()** — Keep (or drop) columns

```
1 # Keep only ID, age, and BMI
2 slim <- my_data %>% select(id, age, bmi)
```

**mutate()** — Create or modify a column

```

1 # Add a BMI category column
2 my_data <- my_data %>%
3   mutate(bmi_category = case_when(
4     bmi < 18.5 ~ "Underweight",
5     bmi < 25   ~ "Normal",
6     bmi < 30   ~ "Overweight",
7     TRUE      ~ "Obese"
8   ))

```

### `summarise()` — Collapse rows into a summary

```

1 my_data %>% summarise(mean_age = mean(age, na.rm = TRUE))

```

### `group_by()` + `summarise()` — Summaries by group

```

1 my_data %>%
2   group_by(sex) %>%
3   summarise(
4     n = n(),
5     mean_bmi = mean(bmi, na.rm = TRUE)
6   )

```

### AI Prompt:

Using `dplyr`, filter my dataset to patients over 65, group by sex, and calculate the mean BMI for each group. Handle missing values.

#### Tip

**Audit the AI output:** After running the AI-generated code, check that (a) the column names match your actual data, (b) `na.rm = TRUE` is present where needed, and (c) the row counts make sense (e.g., the filtered data should be smaller than the original).



**Git checkpoint:** You have now loaded, inspected, and transformed data. Stage your `.qmd`, write a commit message describing what you did, and commit. Don't wait until the end.

---

## r5.qmd Control Flow: Auditing AI Decisions

You will not write many `if/else` blocks or loops by hand. But AI frequently generates them, so you need to **read** them.

### If/Else: The Fork in the Road

```
1 if (mean_age > 50) {  
2   print("Older cohort")  
3 } else {  
4   print("Younger cohort")  
5 }
```

**Audit question:** Is the threshold (50) appropriate for my research question, or did the AI pick an arbitrary number?

### Loops: The Treadmill

```
1 for (col in c("age", "bmi", "sbp")) {  
2   print(summary(my_data[[col]]))  
3 }
```

**Audit questions:** Does the loop have a clear end condition? Could this be replaced with a simpler vectorized operation (e.g., `summarise(across(...))`)?

### Note

In R, loops are rarely the best tool. If AI gives you a loop, ask: “*Can this be done without a loop using dplyr or purrr?*” The answer is usually yes.

---

## r6.qmd Writing a Simple Function

The syllabus asks you to write at least one custom function. A function packages a repeated task so you don’t copy-paste code.

### Example: A Reusable Summary

```
1 summarise_column <- function(data, column_name) {  
2   data %>%  
3     summarise(  
4       n_total   = n(),  
5       n_missing = sum(is.na(.data[[column_name]])),  
6       mean_val  = mean(.data[[column_name]], na.rm = TRUE),  
7       sd_val    = sd(.data[[column_name]], na.rm = TRUE)  
8     )  
9   }  
10  
11 # Use it  
12 summarise_column(my_data, "bmi")  
13 summarise_column(my_data, "age")
```

### Your Task

1. Ask your AI co-pilot to write a function that takes a data frame and a column name and returns summary statistics.

2. **Audit the code:** Does it handle NA? Does it work on a numeric column? What happens if you pass a categorical column?
3. **Adapt it:** Change one thing (e.g., add a median, or add a `group_by` argument). This is the difference between accepting code blindly and understanding it.

#### AI Prompt:

Write an R function called `summarise_column` that takes a data frame and a column name (as a string) and returns the count, number of missing values, mean, and standard deviation. Then show me how to call it on the “bmi” column of `my_data`.

---

## r7.qmd Debugging: Reading the Red Text

Errors are normal. AI makes mistakes, you make typos, and data surprises you.

### The “Fix It” Loop

1. **Read** the error message. Look for keywords: `not found`, `unexpected`, `non-numeric`.
2. **Copy** the full error text.
3. **Paste** it into your AI chat and ask:

Here is an R error from my Codespace: [paste]. Explain what went wrong in one sentence and give corrected code.

## Common Errors and What They Mean

| Error message                           | Likely cause                                                      |
|-----------------------------------------|-------------------------------------------------------------------|
| object 'X' not found                    | Typo in variable name, or you forgot to run an earlier chunk      |
| could not find function "X"             | Forgot <code>library(tidyverse)</code> or misspelled the function |
| non-numeric argument to binary operator | Trying to do math on a text/factor column                         |
| unexpected symbol                       | Missing comma, parenthesis, or pipe                               |

#### Warning

**AI debugging guardrail:** When AI suggests a fix, re-read the corrected code before running it. Does the fix address the *actual* error, or did AI rewrite the whole chunk and change your logic?

## r8.qmd AI Safety: The Audit Checklist

Every time you accept AI-generated code, run through this checklist:

1. **Column names:** Are they spelled exactly as they appear in `glimpse(my_data)`?
2. **Data types:** Is the code doing math only on numeric columns?
3. **Missingness:** Is `na.rm = TRUE` present where needed?
4. **Privacy:** Does the code avoid printing individual patient rows that could be identifiable?
5. **Logic:** Does the code do what *you* intended, or what the AI *assumed* you meant?

## The Paper Trail Rule

Every time you use AI to write a code chunk, add a `#` comment above it explaining what you asked for and what you verified:

```
1 # AI prompt: "Calculate mean BMI by sex, handling NAs"
2 # Verified: column names match, na.rm = TRUE present, output has 2 rows (M/F)
3 my_data %>%
4   group_by(sex) %>%
5   summarise(mean_bmi = mean(bmi, na.rm = TRUE))
```

This comment trail is your audit log. It demonstrates accountability (required by the course AI policy) and helps you debug later.

---

## r9.qmd Exporting Clean Data

After cleaning and transforming, save your work so future weeks can pick up where you left off.

```
1 # Save a clean version for Week 4 and beyond
2 write_csv(my_data, "output/clean_data.csv")
```

### Tip

**Reproducibility habit:** Your Quarto document should contain *every step* from raw data to clean export. Anyone (including your grader) should be able to render it from scratch and get the same `clean_data.csv`.

**Git checkpoint:** Your analysis is complete. Stage, commit with a clear message (e.g., "Complete Week 3: load, inspect, transform, and export patient data"), and Sync.

---

## r10.qmd Literate Analysis in Quarto

Your `.qmd` file is not just code — it is a **document**. Weave narrative around your code chunks to explain what you are doing and why.

### Pattern

““markdown ## Data Inspection

We begin by examining the structure of the dataset to understand variable types and missingness before any transformations.

```
{r, eval=FALSE} glimpse(my_data) colSums(is.na(my_data))
```

The dataset contains `nrow(my_data)` observations across `ncol(my_data)` variables. Notably, BMI is missing for approximately 12% of records, which we handle with `na.rm = TRUE` in downstream summaries.

This interleaving of code and prose is **literate programming** — the methodology that underpins reproducible research in this course. Practice it now; you will need it for your term project report (Week 10) and final portfolio.

---

## r11.qmd Knowledge Check

1. What is the difference between `install.packages()` and `library()`?
2. Why is `na.rm = TRUE` important when working with health data?
3. What does `my_data %>% filter(age > 65) %>% summarise(n = n())` do? Read it aloud using “and then.”
4. Name the five core `dplyr` verbs and what each does in one sentence.

5. You see the error `object 'BMI' not found` but your column is called `bmi`. What went wrong?
  6. What is the difference between `%>%` and `|>`? Does it matter for this course?
  7. Why should you write `#` comments above AI-generated code?
  8. What four checks should you run immediately after loading a new dataset?
- 

## r12.qmd Assignment: The “Mad Libs” Report

**Task:** Build a short Quarto report that loads, inspects, transforms, and summarizes a health dataset — using your AI copilot for code generation and your own judgment for verification.

### Step-by-Step Workflow

1. Open `week3_assignment.qmd` in your Codespace.
2. **Load packages and data:** Use AI to write the `library()` and `read_csv()` calls. Verify the data loaded correctly with `dim()` and `glimpse()`.
3. **Inspect missingness:** Run `colSums(is.na(...))`. Add a Markdown paragraph interpreting what you see.
4. **Transform:** Use `dplyr` verbs to:
  - Filter to a meaningful subgroup (e.g., `age > 18`).
  - Create at least one new column with `mutate()`.
  - Produce a grouped summary with `group_by()` + `summarise()`.
5. **Write a function:** Create (or adapt from AI) a reusable summary function. Call it on at least two different columns.
6. **Export:** Save your cleaned data to `output/clean_data.csv`.
7. **Document:** Every code chunk should have:

- A `#` comment stating what you asked the AI and what you verified.
  - A Markdown paragraph below interpreting the result.
8. **Render** the document to verify it runs cleanly from top to bottom.
  9. **Git:** Stage, write a descriptive commit message, Commit, and Sync.

## Deliverable

A rendered `.qmd` that a reader can follow as a narrative: what data you started with, what you did to it, what you found, and why your decisions were justified.

## r13.qmd Quick Reference

| Concept           | R code                                 | Notes                                       |
|-------------------|----------------------------------------|---------------------------------------------|
| Load packages     | <code>library(tidyverse)</code>        | Every session                               |
| Read CSV          | <code>read_csv("data/file.csv")</code> | Use relative paths                          |
| Assignment        | <code>x &lt;- 5</code>                 | The <code>&lt;-</code> arrow stores a value |
| Inspect structure | <code>glimpse(df)</code>               | Column names, types, preview                |
| Dimensions        | <code>dim(df)</code>                   | Rows $\times$ columns                       |
| Summary stats     | <code>summary(df)</code>               | Min, median, mean, max, NAs                 |
| Check missingness | <code>colSums(is.na(df))</code>        | Count NAs per column                        |
| Filter rows       | <code>filter(df, age &gt; 65)</code>   | Keep matching rows                          |
| Select columns    | <code>select(df, id, age, bmi)</code>  | Keep named columns                          |
| New column        | <code>mutate(df, bmi_cat = ...)</code> | Create or modify                            |



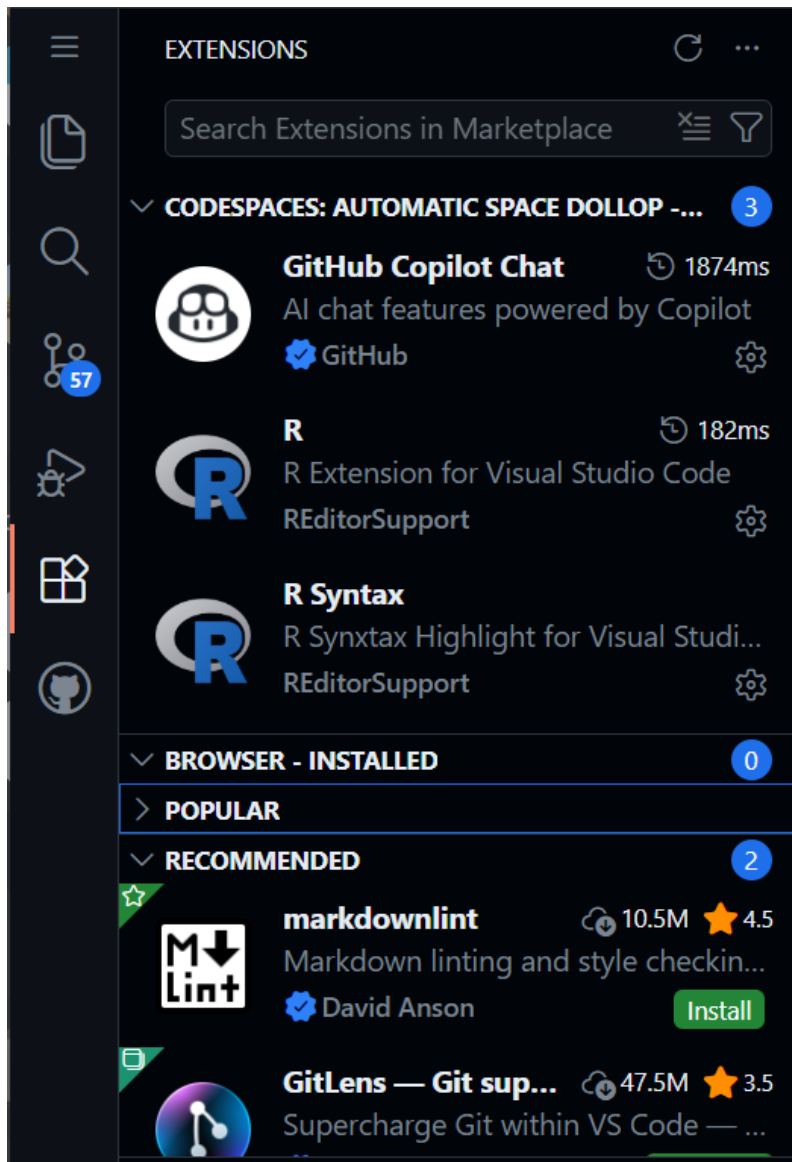
| Concept              | R code                                                        | Notes                  |
|----------------------|---------------------------------------------------------------|------------------------|
| Summarize            | <code>summarise(df, m<br/>= mean(x, na.rm<br/>= TRUE))</code> | Collapse to one<br>row |
| Group +<br>summarize | <code>group_by(df,<br/>sex) %&gt;%<br/>summarise(...)</code>  | Per-group<br>summaries |
| Export               | <code>write_csv(df,<br/>"output/clean.csv")</code>            | Save for later         |
| Pipe                 | <code>%&gt;%</code> or <code> &gt;</code>                     | “And then...”          |

## 13 The Setup (The Cockpit)

Before we start flying, we need to check our instruments in **VS Code**. If your environment isn't set up correctly, your code will not run.

### 13.1 Verify Your AI Extensions

1. Click the **Extensions** icon on the far-left sidebar (it looks like four squares).
2. In the search bar, ensure both **GitHub Copilot Chat** and **R** are installed and active.



## 13.2 Set the AI Model to “0x”

1. Open the Copilot Chat sidebar (Shortcut: **Ctrl+Alt+I** or **Cmd+Option+I** on Mac).
2. Look for the model selector dropdown at the bottom of the chat window.

3. **Crucial Check:** Choose a model with a “0x” **multiplier** (like GPT-4o 0x). If you do not do this, you will quickly run out of your free student quota.
  - *Troubleshooting:* If the Chat asks you to “Sign In” or “Subscribe,” your Student Pack is not active. Use your institutional login at [copilot.microsoft.com](https://copilot.microsoft.com) or authenticate via your university email at <https://m365.cloud.microsoft/chat/>.

### 13.3 Interface Review: Where does R actually live?

To learn R, we are going to start in the “sandbox.” Here is how your screen is laid out:

- **The Console (Bottom):** Your **Engine Room**. *Start here!* This is where you can type R commands directly next to the > prompt and press Enter to see immediate results. If you make a mistake, it will show up here in red text.
- **The Environment (Top Right):** Your **Work Bench**. Whenever you save a piece of data (like a patient’s age), it will appear in a list here so you can verify R actually “remembers” it.
- **The Explorer (Left Sidebar):** Your **File Cabinet**. This is where you find your **data/** folder containing your CSV files.
- **The Editor (Top Center):** Your **Laboratory Notebook**. Typing in the Console is great for testing, but it forgets everything when you close the window. Later in the lesson, we will use the Editor to write and save our permanent code.

## 13.4 Interface Review: Working with .qmd and .Rmd files

When you open a .qmd (Quarto) or .Rmd (RMarkdown) file, you must know where to look. Here is how your coding workflow maps to your screen:

- **The Explorer (Left Sidebar): Your File Cabinet.** This is where you find your reports and your `data/` folder. You cannot run code here; you only click files to open them.
- **The Editor (Top Center): Your Laboratory Notebook.** This is where you write your text narrative and your R code.
  - *Note:* R code must be placed inside grey **Code Chunks** (starting with ````\r{` and ending with `````). To run a chunk, click the small green “Play” triangle in the top right corner of that specific grey box.
- **The Console (Bottom): Your Engine Room.** When you click the green “Play” button in the Editor, the code is sent down here to be executed. If there is an error, it will show up here in red.
  - *Pro Tip:* You can type quick math directly into the console (like `100 / 4`), but it will not be saved in your final report.
- **The Environment (Top Right): Your Work Bench.** Whenever you load data or save a calculation, it will appear in a list here so you can verify R actually “remembers” it.

## 14 Packages (The Toolbox)

In R, tools come in “packages.” Think of `install.packages` as buying a physical toolbox, and `library()` as opening the lid.

- **To open your tools:** Type this in your code and run it:

```
library(tidyverse)
```

**Note:** You must run `library(tidyverse)` every single time you restart your Codespace.

# 15 Basics: Boxes & Toasters

R is a system of storage (Boxes) and transformation (Toasters).

## 15.1 Variables (The Box)

Use `<-` to save data into a name.

Code snippet

```
Save the number 55 into a variable called patient_age patient_age
```

## 15.2 Functions (The Toaster)

Functions take input, perform an action, and return output.  
Look at the snippet below

Code snippet

```
calculate_rectangle_area <- function(length, width) {  
  area <- length * width  
  return(area)  
}
```

```
# Example usage:  
calculate_rectangle_area(10, 5) # Returns 50
```

# 16 Control Flow

## 16.1 The Pipe %>% (The Assembly Line)

The symbol %>% means “**and then...**”.

Both of these lines are identical, which would you prefer to read? (Note |> is another way to write %>%)

```
print(round(sqrt(10), 2))
```

```
10 |> sqrt() |> round(2) |> print()
```

For a real world example of how one may use the pipe

```
library(dplyr)
```

```
mtcars %>%  
  filter(wt > 3) %>%          # 1. Filter for cars heavier than 3,000 lbs  
  group_by(cyl) %>%          # 2. Group by number of cylinders  
  summarize(avg_mpg = mean(mpg)) # 3. Calculate the average MPG
```

## 16.2 If / Else (The Fork in the Road)

An **if/else** statement lets the computer make a decision.

The program checks a condition:

- If it is **TRUE**, it runs one block of code.
- If it is **FALSE**, it runs another.



```
temperature <- 5
if (temperature > 10) {
  print("You can wear a t-shirt")
} else {
  print("Wear a jacket")
}
```

## 16.3 What happens

1. The computer checks: `temperature > 10`
2. `5 > 10` is **FALSE**
3. So the **else** part runs.
4. Output:

"Wear a jacket"

Think of it like a **fork in the road**:

- One path if the condition is true
- Another path if it's false.

## 16.4 Loops (The Treadmill)

A loop repeats code multiple times.

Instead of writing the same instruction again and again, you let the computer do it.

## 16.5 Example

```
for (i in 1:5) {
  print(i)
}
```

## 16.6 What happens

The computer runs the code **5 times**.

Steps:

1. `i = 1` → print 1
2. `i = 2` → print 2
3. `i = 3` → print 3
4. `i = 4` → print 4
5. `i = 5` → print 5

Output:

1 2 3 4 5

## 16.7 Audit Check (Important)

When using loops, ask:

### 1. Does the loop end?

Bad example (infinite loop):

```
while (TRUE) {  print("This never stops") }
```

### 2. Could this be simpler?

Sometimes R has built-in functions that replace loops.

Instead of:

```
total <- 0 for (i in 1:5) {  total <- total + i }
```

You can write:

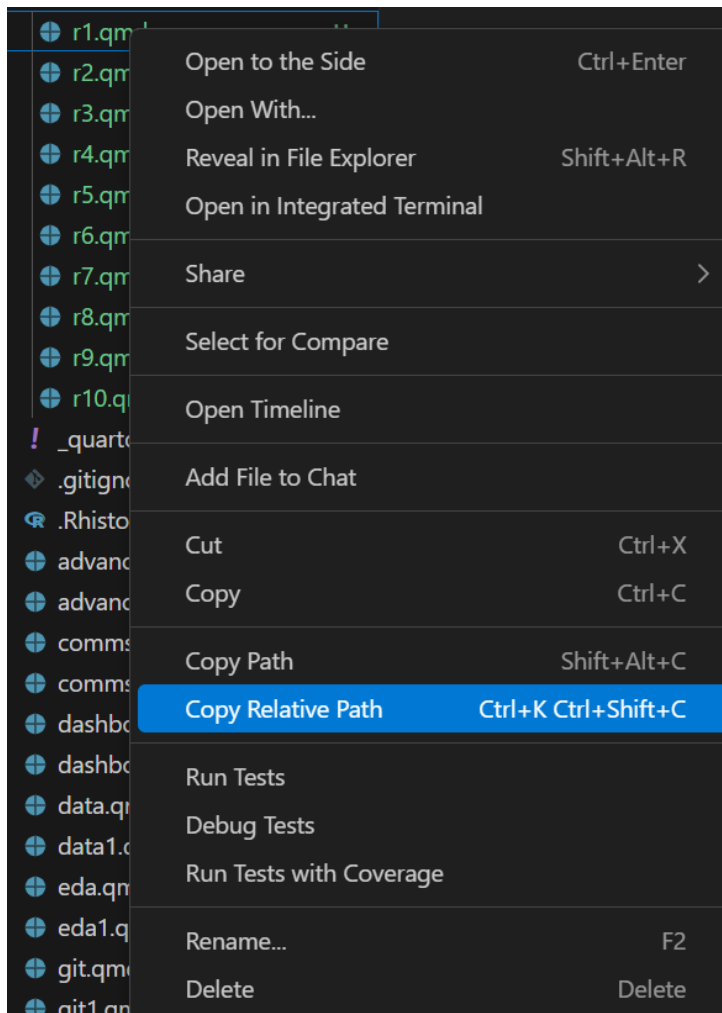
```
sum(1:5)
```

This is **shorter, safer, and faster**.

## 17 The Loading Dock (Data Import)

Don't type file paths from memory; you will likely make a typo.

1. Find a data file in the **Explorer** (left sidebar).
2. **Right-click** the file → Select **Copy Relative Path**.



3. Ask AI: *“Read the CSV at [Paste Path Here] and save it as my\_data.”*

## 18 The Detective (Debugging)

AI makes mistakes. When you see **Red Text** (errors) in the Console:

1. **Copy** the Red Text.
2. **Paste** it into Copilot Chat.
3. **Ask:** *“Explain this error and fix the code below: [Paste Code]”*

Try running this line and ask AI why it doesn't work:

```
"apple" + 5
```

## 19 AI Safety: The Audit Checklist

Use this checklist before accepting any AI code:

- **Spelling:** Are column names spelled exactly as they appear in the data?
- **Case Sensitivity:** Is it `Age` or `age`? (R sees these as different things).
- **Missingness:** Did the AI remember `na.rm = TRUE` for `math`?
- **Privacy:** Did the AI try to print the whole dataset (potentially exposing patient IDs)?

## 20 The Paper Trail (Comments)

**The Rule:** Every time you use AI to write code, write a # comment in plain English above it explaining what you told the AI to do.

Code snippet

```
# I asked AI to filter for patients with BMI over 30 and count them # [AI CODE BELOW]
```

Below is an example of non readable code

```
# What is x? What is 1.05? Why are we subsetting?
f <- function(x, y) {
  z <- x[x$a > y, ]
  r <- sum(z$b) * 1.05
  return(r)
}

res <- f(df, 100)
```

And here is an example of readable code

```
library(dplyr)

# Calculate the total revenue for high-value orders including a 5% tax
calculate_total_revenue <- function(order_data, minimum_price_threshold) {

  sales_tax_rate <- 1.05

  total_revenue <- order_data |>
    filter(price > minimum_price_threshold) |>
```

```
    summarize(total = sum(price)) |>
    pull(total)

    return(total_revenue * sales_tax_rate)
}

# Usage:
final_report_value <- calculate_total_revenue(current_orders, minimum_price_threshold = 100)
```

Just by reading through each line a reader should understand what is going on. One does not need to comment every single line, but rather comment whenever something complex is happening on the code line below it. And as always source work which is not yours.



## 21 Assignment: The “Mad Libs” Report

1. Open `week3_assignment.qmd`.
2. Use Copilot to import the Week 1 dataset.
3. Use the **Pipe** to filter and summarize one variable.
4. **Audit:** Add a `#` comment explaining your prompt.
5. **Render** the file to HTML.
6. **Commit and Push** to GitHub.

## 22 Final Summary Checklist

- ☐ Extensions are active (Copilot + R).
- ☐ Using **0x model** in Chat.
- ☐ Loaded `library(tidyverse)`.
- ☐ Data imported via **Relative Path**.
- ☐ All code chunks have an **Audit Comment (#)**.
- ☐ Handled **NA** values correctly.
- ☐ Document **Rendered** and **Pushed**.

## **Part VI**

### **Week 4: Git & Collaboration**

# Git & Collaboration

## Overview

**Summary:** You already know how to save files and make a basic commit in your Codespace. Now, learn to use Git as a *time machine* and *collaboration engine* for your analysis—tracking every change, safely undoing mistakes, working with teammates through branches and Pull Requests, and keeping a professional audit trail of your research workflow.

### Learning Objectives:

- Distinguish between saving a file and taking a version-control “snapshot” (commit).
- Read code differences (diffs) to see exactly what changed.
- Use the Codespaces Source Control UI to revert to a previous working version.
- Create and switch branches to isolate experimental work.
- Open a Pull Request on GitHub and understand its role in peer review.
- Configure `.gitignore` and use relative paths for repo hygiene.
- Use AI to translate diffs into clear, professional audit logs.
- Set up a group repository and agree on collaboration norms for the term project.

## Connection

Week 3 focused on writing and auditing small R summaries. This week treats each edit as a reviewable change with a clear history. Next week you will compare equivalent summaries across R and Python.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
```

```

28 filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29 group_by(IncomeGroup) |>
30 summarise(
31   n = n(),
32   mean_bmi = round(mean(BMI), 1),
33   .groups = "drop"
34 )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## In-Class Activity

- Create a branch for a tiny dashboard documentation change.
- Edit only a label, caption, or README sentence; do not change analytic logic.
- Open a pull request and use the diff to verify the change is limited.
- In the pull request description, state how you verified the change is documentation-only.

Student output: a documentation-only pull request with a short audit note. Assignment link: [../..../assignments/assignment03-git/README.md](#).

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## git1.qmd Concepts: The “Save Game” of Reproducible Science

### Why Git Matters in Health Research

- **The audit trail:** Why `analysis_final_v3.qmd` is unsafe in health research; Git gives you a permanent, searchable history.

- **The timeline:** Your repository is a sequence of safe checkpoints (commits). Each commit records *who* changed *what* and *why*.
- **Reading the diff:** Red = removed. Green = added. This is the visual language of reproducibility.

## Quick Recap (from Week 2)

You already know how to stage, commit, and sync in your Codespace. This week, we go deeper: branching, collaboration, and building habits that will carry through your term project.

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## git2.qmd Tutorial: Time Travel in Codespaces

*Scenario:* We will intentionally break last week's R script and use Git to rescue it.

### Step 1: Verify Your Baseline

Open your `.qmd` from last week. Make a small, meaningful edit (e.g., improve a comment). Stage (+), write a descriptive commit message, and commit (.). This is your safety net.

### Step 2: The Disaster

Delete a crucial line of R code in a chunk (e.g., the line that loads the data). **Save** the file (Ctrl/Cmd + S).

### Step 3: Assess the Damage

Open **Source Control** → click the modified file → view the side-by-side **diff** to see exactly what changed. Practice reading the red (removed) and green (added) lines.

## Step 4: The Undo Button

Hover over the file name in the Source Control panel and click **Discard Changes** (the curved arrow) to restore the file to the **last committed** version.

### Tip

**Key takeaway:** Saving writes to disk. Committing writes to *history*. Only committed work can be recovered by Git.

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## git3.qmd Repo Hygiene: .gitignore and Relative Paths

### Why This Matters

As your projects grow, you will generate files that should *not* be tracked: rendered HTML, temporary data caches, `.DS_Store`, API keys. Tracking these clutters your history and can leak sensitive information.

### .gitignore Walkthrough

1. Open (or create) the `.gitignore` file in the root of your repository.
2. Add common patterns:

```
# Rendered outputs
*.html
*_files/

# OS junk
.DS_Store
Thumbs.db
```



```
# Sensitive
.env
```

3. Stage, commit with a message like "Add .gitignore to exclude rendered files".

## Relative Paths

Always use **relative paths** in your code (e.g., `read.csv("data/nhanes.csv")`), never absolute paths (e.g., `read.csv("/home/user/Documents/nhanes.csv")`). This ensures your analysis runs on *any* machine—including your grader's Codespace.

### Warning

**Reproducibility rule:** Your term project must run from start to finish in a *fresh* Codespace. Absolute paths guarantee failure.

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## git4.qmd Branching: Safe Experimentation

### Concept

A **branch** is a parallel copy of your project. You can experiment freely on a branch without affecting the stable **main** version. When your experiment works, you **merge** it back.

Think of it like a lab notebook: **main** is your clean, published record. A branch is your scratch pad.

### Tutorial: Create and Merge a Branch in VS Code

#### Step 1: Create a Branch

- Click the branch name in the bottom-left corner of VS Code (it likely says **main**).

- Select **Create new branch...** and name it something descriptive: `add-age-filter`.

### Step 2: Make Changes on the Branch

- Edit your `.qmd` (e.g., add a line of code that filters data by age).
- Stage and commit on this branch.

### Step 3: Switch Back to `main`

- Click the branch name again → select `main`. Notice your recent edits *disappear* from the file. They are safely stored on the other branch.

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## git5.qmd Pull Requests: The Peer Review Gateway

### What Is a Pull Request?

A **Pull Request (PR)** is a GitHub feature that says: *“I have changes on a branch. Please review them before they go into `main`.”* PRs are how professional teams and researchers collaborate. You will use them for peer review later in this course.

### Tutorial: Open Your First PR

1. After committing changes to a branch (see `git4.qmd`), click **Sync Changes** to push the branch to GitHub.
2. Open your repository on **github.com**. You should see a banner: *“add-age-filter had recent pushes.”* Click **Compare & pull request**.
3. Write a title and short description explaining what you changed and why.
4. Click **Create pull request**.

5. For now, review the diff on GitHub, then click **Merge pull request** and complete the merge.

 Tip

**Collaboration norm:** In your term project, agree with your group that changes go through PRs rather than directly to main. This creates a built-in review step.

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## git6.qmd AI Auditing for Version Control

Treat the LLM as a **Git Translator** that explains diffs and drafts audit-quality messages.

### git6a.qmd Scenario A: Diff Translation (Pre-Commit Audit)

**Prompt:**

“I am about to commit these changes. Explain what I changed in plain English. Did I delete anything important?”

Use this to spot accidental deletions or unintended edits before you commit.

### git6b.qmd Scenario B: Auto-Scribe Commit Messages

**Prompt:**

“Based on this diff, write a one-sentence, professional commit message for my Quarto document.”

Check that the message captures the **intent** (e.g., “Filter patient cohort to age 65 per protocol”), not just the action (“changed line 42”).

## git6c.qmd Scenario C: Panic Translator (Error Handling)

### Prompt:

“VS Code in my GitHub Codespace showed this Git error: [paste]. What does it mean in simple terms, and which **buttons in the VS Code UI** should I click to fix it?”

#### Warning

**Strict Guardrail:** Never run random terminal commands suggested by an AI (especially `--force` or `reset --hard`). Prefer the VS Code visual interface, which is safer. If the AI suggests a terminal command, ask it for the GUI equivalent first.

---

## git7.qmd Setting Up Your Group Repository

This week you form your term project groups (**Milestone 0**). Use this session to establish your collaboration workflow.

### Steps

1. **One member** creates the group repository from the GitHub Classroom link (provided on Canvas).
2. All other members accept the invitation and clone the repo by launching a Codespace from it.
3. As a group, agree on norms:
  - All changes go through **branches** → **PRs** (no direct pushes to **main**).
  - Commit messages should be descriptive (use the AI auto-scribe technique).
  - Keep the repo clean: use `.gitignore`, relative paths, and a clear folder structure.

4. Each member makes a small “hello” commit (e.g., add your name to the `README.md`) and pushes it to verify access.

**i** Note

**Milestone 0 due Friday:** Record your group formation on Canvas and ensure everyone can commit to the shared repository.

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## git8.qmd Knowledge Check

1. What is the difference between **saving** a file and **committing** it?
2. What does a red line indicate in a diff?
3. Which button restores the last **committed** version of a file?
4. Why should you use a `.gitignore` file?
5. What is the purpose of a **branch**?
6. How does a **Pull Request** support peer review?
7. Why are relative paths essential for reproducibility?

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## git9.qmd Practice Exercise: The Full Workflow Drill

1. Open your Codespace from last week.
2. Create a `.gitignore` file with sensible defaults for your project. **Commit.**
3. Create a new branch called `practice-edit`.
4. On the branch, make a small improvement to your R script (e.g., add a clarifying comment or improve a variable name).

5. Use an AI model to draft a clear commit message from your diff. **Commit** on the branch.
6. Push the branch and open a **Pull Request** on GitHub.
7. Review the PR diff on GitHub, then merge it.
8. Back on **main**, intentionally introduce a breaking typo and **Save**.
9. Use **Discard Changes** to recover the last working version.

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## git10.qmd Quick Reference: Common Git Actions in VS Code

| Action                           | How                                                                |
|----------------------------------|--------------------------------------------------------------------|
| <b>Stage changes</b>             | Source Control → hover file → click +                              |
| <b>Commit</b>                    | Type message → click                                               |
| <b>View diff</b>                 | Click file name in Source Control                                  |
| <b>Discard uncommitted edits</b> | Hover file → click <b>curved arrow</b> (Discard Changes)           |
| <b>Create a branch</b>           | Click branch name (bottom-left) → <b>Create new branch...</b>      |
| <b>Switch branches</b>           | Click branch name → select target branch                           |
| <b>Merge a branch</b>            | Command Palette → <b>Git: Merge Branch...</b>                      |
| <b>Push / Sync</b>               | Click <b>Sync Changes</b> in status bar                            |
| <b>Open a Pull Request</b>       | Push branch → go to github.com → <b>Compare &amp; pull request</b> |
| <b>Add to .gitignore</b>         | Edit .gitignore in repo root → stage + commit                      |

# Concepts

- **The Audit Trail:** Git records *who* changed *what*.



Selecting any line will show who changed that line and when they did so.

- **The Timeline:** A sequence of safe checkpoints (commits). This shows the reason behind each change and what was change. These checkpoints can always be reverted back to if the need be.

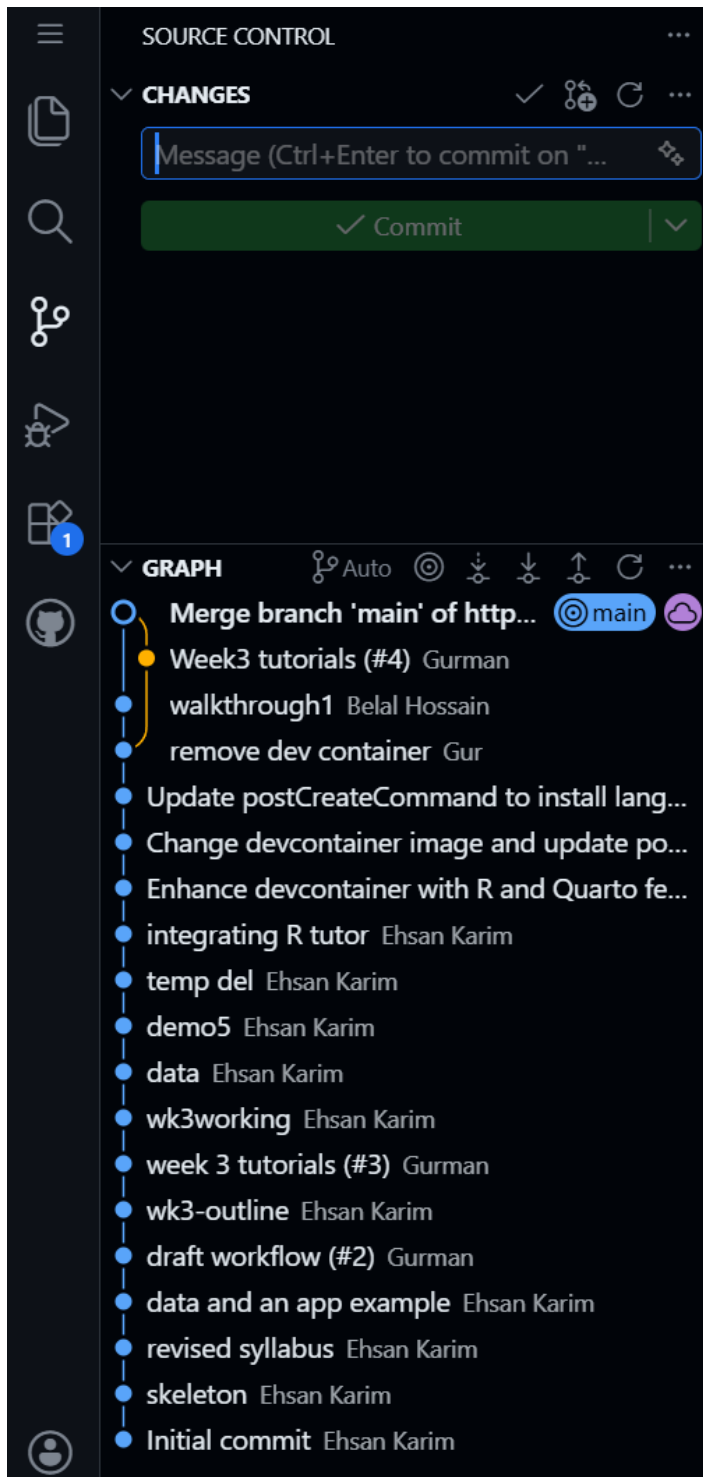
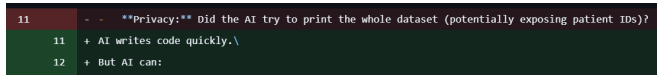


Figure 22.1: This tree shows every commit (checkpoints)



- **The Language of Diffs:**

- Red (-): Removed lines.
- Green (+): Added lines.



```
11 - - **Privacy:** Did the AI try to print the whole dataset (potentially exposing patient IDs)?
12 + AI writes code quickly.
13 + But AI can:
```

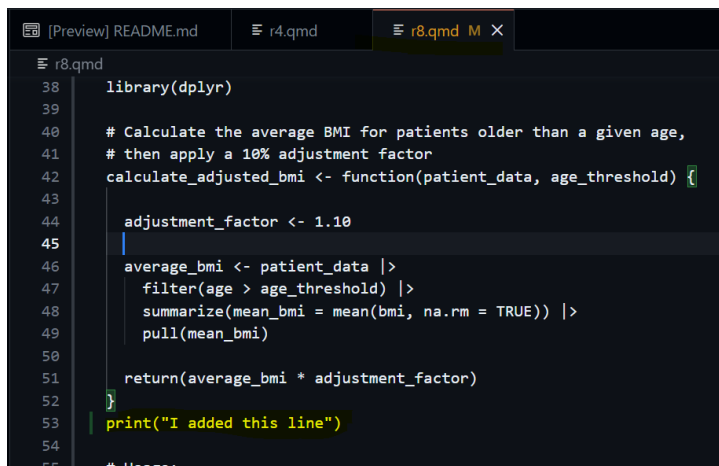
Observing this image, we see that line 11 was replaced, the old data on line 11 is colored red, while the new changes are shown in green.

**Key Rule:** Saving writes to disk on your commuter. Committing and pushing to the repo writes to history.

# Tutorial

Follow these steps

1. **Edit:** Make a change to your code, any file in the repo will do, and save the file.



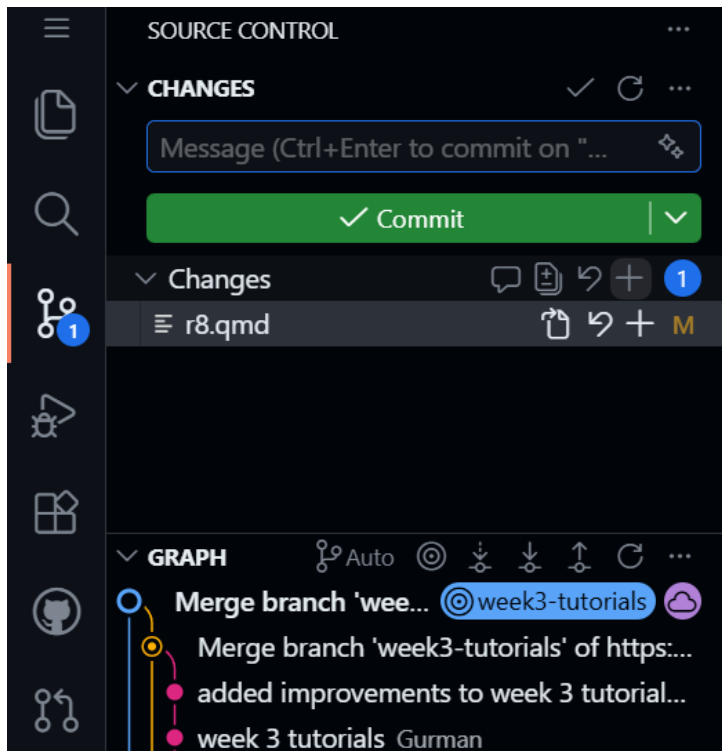
```
38 library(dplyr)
39
40 # Calculate the average BMI for patients older than a given age,
41 # then apply a 10% adjustment factor
42 calculate_adjusted_bmi <- function(patient_data, age_threshold) {
43
44   adjustment_factor <- 1.10
45
46   average_bmi <- patient_data |>
47     filter(age > age_threshold) |>
48     summarize(mean_bmi = mean(bmi, na.rm = TRUE)) |>
49     pull(mean_bmi)
50
51   return(average_bmi * adjustment_factor)
52 }
53 print("I added this line")
54
55 # Usage:
```

Key things to note are that the open file tab has changed colors, this shows that github has acknowledged that the file has been changed.

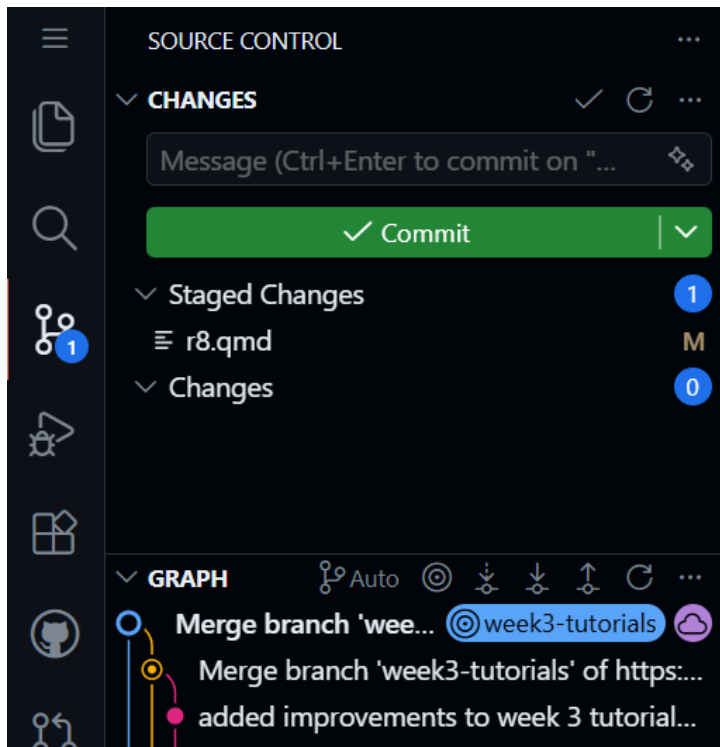
2. **Commit:** Stage and Commit in the Source Control tab.

As we have seen from the previous tutorial, open up the git tab as indicated by the orange-peach highlight. When a file is changed you will be presented with the screen below.

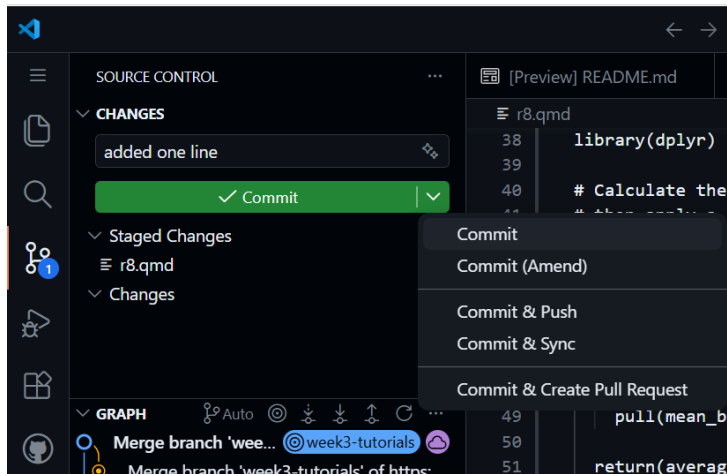
To stage a change, press the + icon.



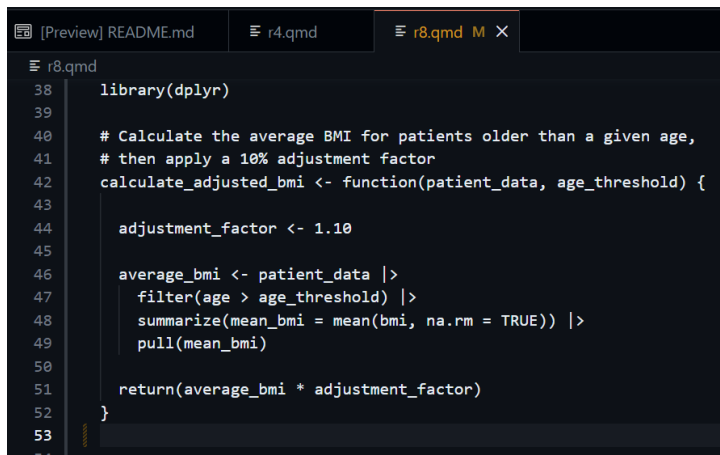
Your menu should look like this now;



Then when you are ready to commit, you can click the big green button! You can also click the down arrow on the big green button to see more options, as well as the commit option. We will discuss those other options later.

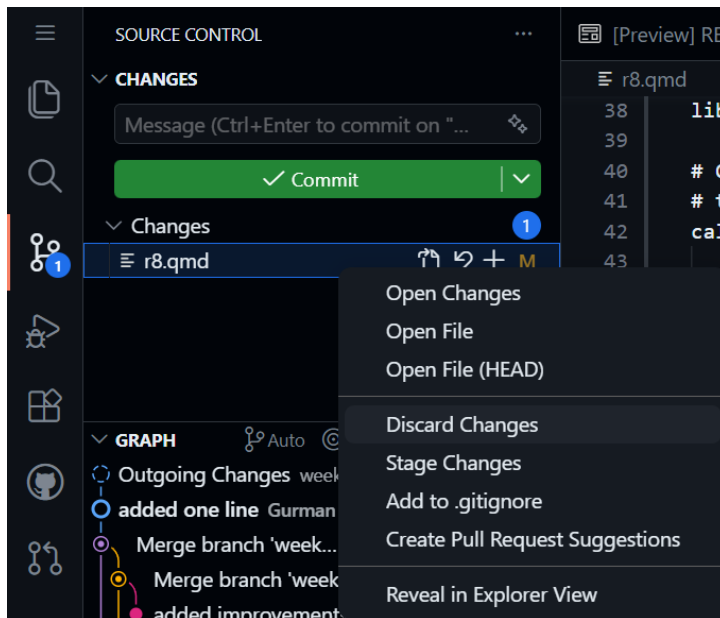


3. **Break:** Delete the code you just added and save.



```
38 library(dplyr)
39
40 # Calculate the average BMI for patients older than a given age,
41 # then apply a 10% adjustment factor
42 calculate_adjusted_bmi <- function(patient_data, age_threshold) {
43
44   adjustment_factor <- 1.10
45
46   average_bmi <- patient_data |>
47     filter(age > age_threshold) |>
48     summarize(mean_bmi = mean(bmi, na.rm = TRUE)) |>
49     pull(mean_bmi)
50
51   return(average_bmi * adjustment_factor)
52 }
53
```

4. **Rescue:** Hover over the file in Source Control again, click the **Discard Changes** option.



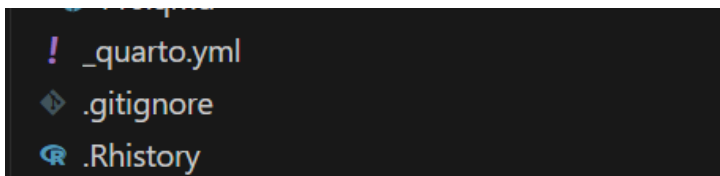
This will revert the file back to what it was before. This can recover data that your computer potentially cannot, its a very powerful in git and showcases one aspect of how git operates. It is a record of all your work and saved chunks.

# Repo Hygiene

## **.gitignore**

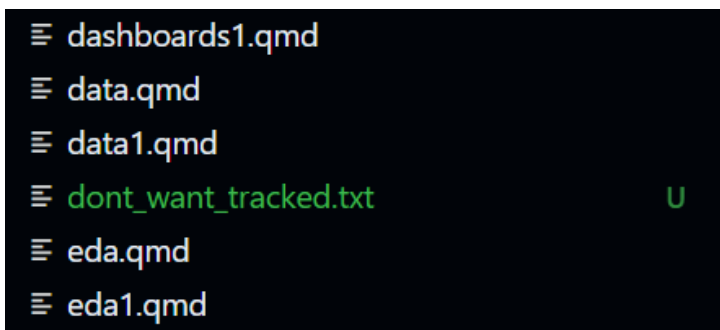
Sometimes there are files you do not want to be tracked by github. This could be anything from files specific to your computer to items like patient data which should not be stored on the cloud unless allowed so by policy.

1. Create a file named **.gitignore** in your root directory:  
It has to be this exact spelling for git to recognize it.

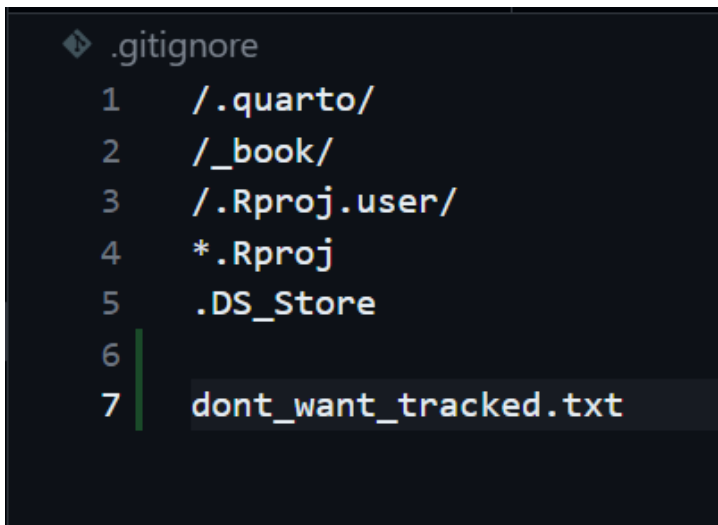


You will know its correct when it has its own icon as seen above

2. Then create a new file, any type will do but for this lesson I will create a text file called dont\_want\_tracked

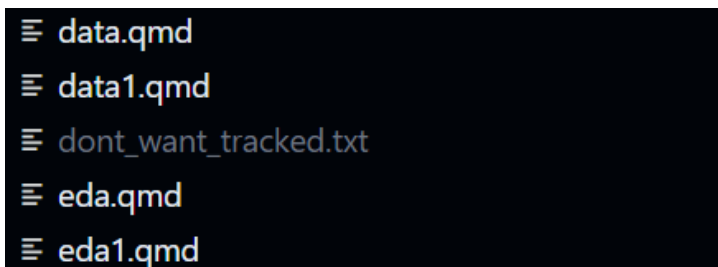


3. Then open the gitignore file and add the name of the file you dont want tracked into it.



```
.gitignore
1  /.quarto/
2  /_book/
3  /.Rproj.user/
4  *.Rproj
5  .DS_Store
6
7  dont_want_tracked.txt
```

4. Now you will see the new file, which was green is now grayed out. This shows that github is no longer tracking it and coloring it green (which it does for new files).



```
≡ data.qmd
≡ data1.qmd
≡ dont_want_tracked.txt
≡ eda.qmd
≡ eda1.qmd
```

This is simple enough right? Whenever in doubt, ask AI how to stop a file from being tracked by git and it will offer some advice. Now stage and commit this like we have done in the previous tutorial.

## Relative Paths

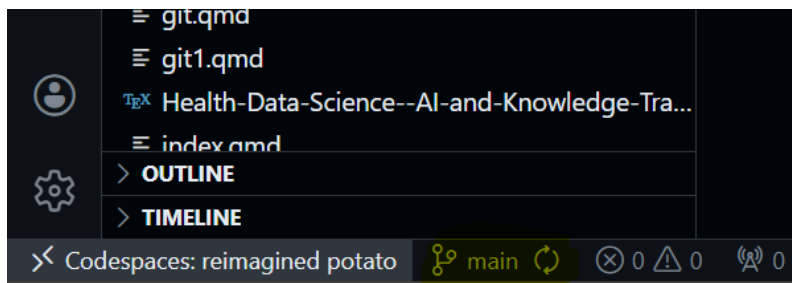
Always use relative paths in your code (e.g., `read.csv("data/nhanes.csv")`), never absolute paths (e.g., `read.csv("/home/user/Documents/nhanes.csv")`). This ensures your analysis runs on any machine—including your grader’s Codespace.

Reproducibility rule: Your term project must run from start to finish in a fresh Codespace. Absolute paths guarantee failure.

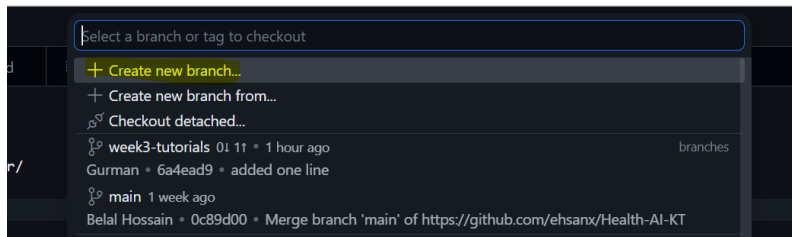
# Branching

## 1. Create:

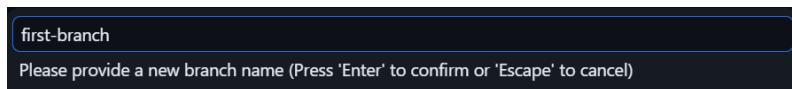
Click branch name (bottom-left).



This will pop up a menu where you have the option to **Create new branch**.



Select this and call it first-branch. Note branch names may not have any spaces and limited special characters, the system will let you know if a name is invalid.

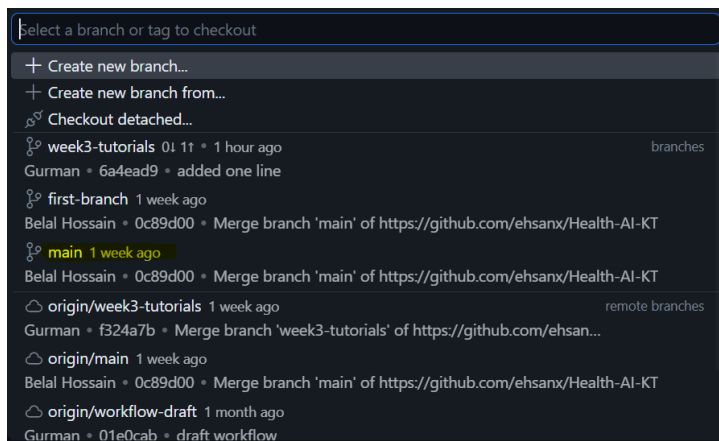


## 2. Work:

Make changes, like those in the previous tutorial and commit your changes on your branch.



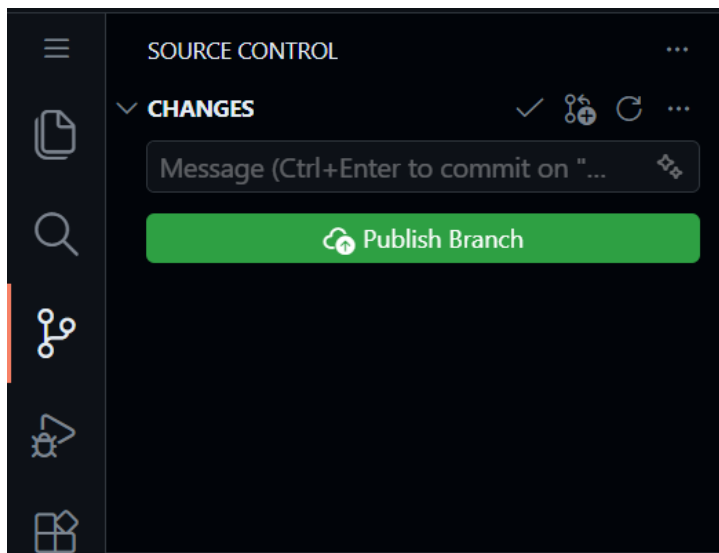
3. **Switch:** Click the branch button on the bottom left again and once this menu pops up, select **main**. Note that your changes vanish (they are safely in the other branch). If you decide to switch back to your branch, your changes will reappear. This is amazing for multitasking on different features and keeping code clean.



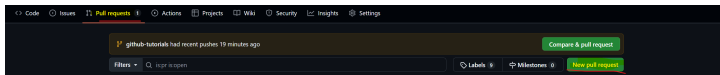
# Pull Requests

A **Pull Request (PR)** is a GitHub feature that says: *“I have changes on a branch. Please review them before they go into **main**.”* PRs are how professional teams and researchers collaborate. You will use them for peer review later in this course. This is also the best way to merge your own changes into main, as it forces you to see what has been change and allows one to check what they are submitting to the main branch, the backbone or “trunk” of the tree

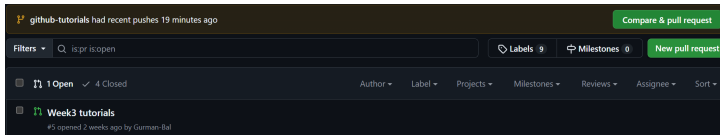
1. **Sync:** Click “Sync Changes” to push your branch to GitHub. This can be blue or green or say publish branch or not. Whichever the case, clicking this button will make everything on your computer update to the online github repo.



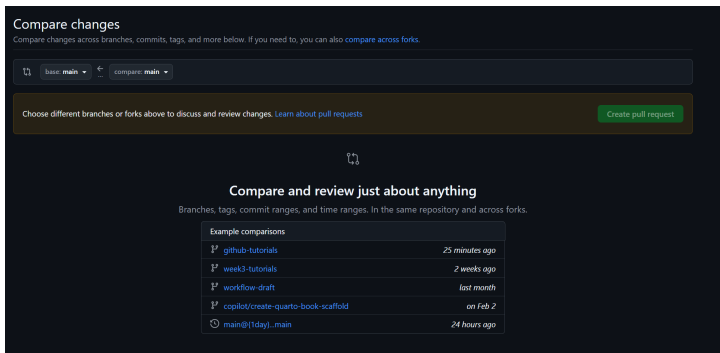
2. **PR:** Go to your repo on github online and click “Compare & pull request.”



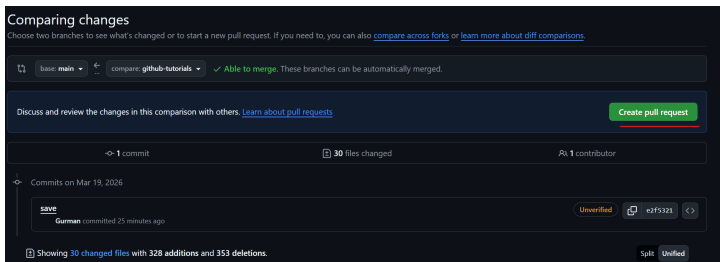
We are then presented with the screen below. Press the big green button “New Pull Request”



Then you are presented this menu. This shows all the different branches that can be merged into main. Select on the one you want to merge into main.



Once you select the one, you can make a pull request! Select this.



3. **Merge:** Review the diff, then click “Merge pull request.” If there are merge conflicts, ask AI the best way to resolve them.

Gurman-Bal and others added 6 commits [last month](#)

week 3 tutorials

Verified8794d84

added improvements to week 3 tutorials

Verified76c47bd

Merge branch 'week3-tutorials' of <https://github.com/ehsanx/Health-AI-KT>

788bc0f

address changes

479773a

Merge branch 'week3-tutorials' of <https://github.com/ehsanx/Health-AI-KT>

f324a7b

Merge branch 'main' into week3-tutorials

Verified17f0242

This branch has not been deployed

No deployments

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress?

[Convert to draft](#)

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# AI Auditing

Use these prompts with your AI assistant:

- **Audit:** “I am about to commit these changes. Explain what I changed in plain English. Did I delete anything important?”

Use this to spot accidental deletions or unintended edits before you commit.

- **Scribe:** “Based on this diff, write a one-sentence, professional commit message for my Quarto document.”

Check that the message captures the **intent** (e.g., “Filter patient cohort to age 65 per protocol”), not just the action (“changed line 42”).

- **Panic:** “VS Code shows this error: [PASTE ERROR]. What does it mean in simple terms and what buttons should I click?”

Never run random terminal commands suggested by an AI (especially `--force` or `reset --hard`). Prefer the VS Code visual interface, which is safer. If the AI suggests a terminal command, ask it for the GUI equivalent first.

# Setting Up Your Group Repository

1. **Invite:** One member creates the repo; others accept the invite.
2. **Clone:** Everyone launches their own Codespace.
3. **Norms:** Agree: “No direct pushes to `main`.”
4. **Test:** Everyone makes a `README.md` update to verify access.

# Knowledge Check

1. **Saving vs. Committing:** Saving = disk; Committing = history.
2. **Red line:** Deletion.
3. **Restore:** Discard Changes button.
4. **.gitignore:** Prevents tracking junk/secrets.
5. **Branch:** Parallel experimental workspace.
6. **PR:** A peer review workflow for merging code.
7. **Relative Paths:** Ensures the project runs on any machine.

## Practice Exercise: Full Workflow

| Step | Task           | Done                     |
|------|----------------|--------------------------|
| 1    | Edit Code      | <input type="checkbox"/> |
| 2    | Create Branch  | <input type="checkbox"/> |
| 3    | AI Draft Msg   | <input type="checkbox"/> |
| 4    | Stage & Commit | <input type="checkbox"/> |
| 5    | Push & PR      | <input type="checkbox"/> |
| 6    | Merge          | <input type="checkbox"/> |



## Quick Reference

| Action  | How-To                                                      |
|---------|-------------------------------------------------------------|
| Stage   | Source Control (+)                                          |
| Commit  | Type msg +                                                  |
| View    | Click file in Source Control                                |
| Diff    |                                                             |
| Discard | Left click file in source control and click discard changes |
| Branch  | Bottom-left status bar                                      |
| Sync    | Sync Changes button                                         |

## **Part VII**

### **Week 5: Polyglot Awareness & R Deepening**

# Polyglot Awareness & R Deepening

## Overview

Week 5 is about becoming a careful bilingual reader of data analysis code. R remains the core language for required course work. Python is introduced so you can read short **pandas** examples, translate small tasks with AI assistance, and verify that a translated result still answers the same question.

The week has two linked goals:

- deepen R fluency with **dplyr**, functions, and clear summaries;
- compare a small R result with a Python/pandas translation using explicit parity checks.

## Learning Objectives

By the end of this week, you should be able to:

- explain when R is the better default and when Python may be useful;
- write a clear R **dplyr** pipeline for a descriptive health-data summary;
- refactor repeated R code into a small function;
- create and run a Python notebook in Codespaces;
- load and inspect a CSV with **pandas**;
- identify common R-to-Python translation traps;
- use AI one step at a time while auditing each translated output;
- document Python dependencies in **requirements.txt**;

- use Git to commit a clean translation workflow.

## Connection

Week 4 made small changes reviewable through Git. This week uses that review mindset inside the analysis itself: first build a trusted R result, then translate one step at a time and compare. Next week you will turn those checked summaries into clearer visual explanations.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
```

```

15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## R Deepening Studio

Start in R. Use the NHANES case-study CSV:

```
1 examples/nhanes-equity/data/nhanes_equity_v6.csv
```

If you are working from the Week 5 folder, the relative path is:

```
1 ../../examples/nhanes-equity/data/nhanes_equity_v6.csv
```

## R Pipeline

```
1 library(readr)
2 library(dplyr)
3
4 nhanes <- read_csv("../..//examples/nhanes-equity/data/nhanes_equity_v6.csv")
5
6 bmi_by_income <- nhanes |>
7   filter(!is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)) |>
8   group_by(IncomeGroup, Gender) |>
9   summarise(
10     n = n(),
11     mean_bmi = round(mean(BMI), 2),
12     sd_bmi = round(sd(BMI), 2),
13     .groups = "drop"
14   ) |>
15   arrange(IncomeGroup, Gender)
16
17 bmi_by_income
```

## R Function

```
1 summarise_mean_sd <- function(data, value, group1, group2) {
2   data |>
3     filter(!is.na({{ value }}), !is.na({{ group1 }}), !is.na({{ group2 }})) |>
4     group_by({{ group1 }}, {{ group2 }}) |>
5     summarise(
6       n = n(),
7       mean_value = round(mean({{ value }}), 2),
8       sd_value = round(sd({{ value }}), 2),
9       .groups = "drop"
10    ) |>
11    arrange({{ group1 }}, {{ group2 }})
12 }
13
14 bmi_function_check <- summarise_mean_sd(nhanes, BMI, IncomeGroup, Gender)
15 bmi_function_check
```

Before translating, compare the pipeline output and the function output. They should have the same grouping labels and the same number of rows.

## In-Class Activity: R First, Python Second

Work in pairs. One partner reads the R output; the other reads the Python output.

1. Load the NHANES CSV in R and Python.
2. Record raw row count, column count, and missing BMI count in both languages.
3. Produce mean and SD of BMI by `IncomeGroup` and `Gender` in R.
4. Translate the same task to Python/pandas one step at a time.
5. Fill in a parity table comparing row counts, grouping labels, and rounded summary values.
6. If something does not match, write the likely reason before changing code.

Student output: a parity-check note with the R pipeline, R function, Python translation, parity table, and a short AI-use note.

## Tutorial Map

Use the child pages in this order:

1. [Python vs. R - When and Why?](#)
2. [Setting Up Your Python Environment](#)
3. [Packages and Reproducibility](#)
4. [Common Translation Traps](#)
5. [Loading and Inspecting Health Data](#)
6. [Reading Python Tracebacks](#)
7. [The One Step, One Cell AI Workflow](#)
8. [Mixing R and Python](#)
9. [Creating Reusable Python Scripts](#)
10. [Tracking Your Progress With Git](#)

11. [Project Proposal Readiness Check](#)
12. [Knowledge Check](#)
13. [The Bilingual Translation Exercise](#)
14. [R To Python Cheat Sheet](#)

For the explicit R mastery exercise, use [R Deepening and Parity Check](#).

## Assignment 4

Complete [Assignment 4: Polyglot Awareness & R Deepening](#).

Submit a repository link on Canvas after committing:

- `assignments/assignment04-polyglot/submission/polyglot-parity.qmd`;
- `assignments/assignment04-polyglot/submission/translation.ipynb`;
- `assignments/assignment04-polyglot/submission/helpers.py`.

## Knowledge Check

Before leaving Week 5, answer these:

1. What is one analysis task you would keep in R?
2. What is one task where Python may be useful?
3. What are three parity checks you ran?
4. What did AI help translate, and how did you verify it?
5. What changed in your Git history from the beginning to the end of the week?



## 23 R Deepening and Parity Check

Week 5 is polyglot-aware, not Python-first. Your strongest analysis path remains R and the tidyverse. Python enters as a comparison language: it helps you read code from other sources, ask better AI translation questions, and verify that a translated result still means the same thing.

### 23.1 Learning Goals

By the end of this activity, you should be able to:

- write a clear R `dplyr` pipeline for a descriptive health-data summary;
- refactor a repeated summary into a small R function;
- translate the same task to Python/pandas one step at a time;
- compare outputs across languages using row counts, grouping levels, and summary values;
- document any mismatch before deciding which code to change.

### 23.2 Dataset

Use the NHANES case-study snapshot:

```
1 examples/nhanes-equity/data/nhanes_equity_v6.csv
```

If you are working from this Week 5 folder, the relative path is:

```
1 ../../examples/nhanes-equity/data/nhanes_equity_v6.csv
```

## 23.3 Part A: R Pipeline

Start with R. The goal is a descriptive, unweighted summary of BMI by income group and gender.

```
1 library(readr)
2 library(dplyr)
3
4 nhanes <- read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
5
6 r_summary <- nhanes |>
7   filter(!is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)) |>
8   group_by(IncomeGroup, Gender) |>
9   summarise(
10     n = n(),
11     mean_bmi = round(mean(BMI), 2),
12     sd_bmi = round(sd(BMI), 2),
13     .groups = "drop"
14   ) |>
15   arrange(IncomeGroup, Gender)
16
17 r_summary
```

## 23.4 Part B: R Function

Refactor the summary so the value and grouping variables are easy to change later.

```
1 summarise_mean_sd <- function(data, value, group1, group2) {
2   data |>
3     filter(!is.na({{ value }}), !is.na({{ group1 }}), !is.na({{ group2 }})) |>
4     group_by({{ group1 }}, {{ group2 }}) |>
5     summarise(
6       n = n(),
7       mean_value = round(mean({{ value }}), 2),
```

```

8         sd_value = round(sd({{ value }}), 2),
9         .groups = "drop"
10    ) |>
11    arrange({{ group1 }}, {{ group2 }})
12 }
13
14 r_function_summary <- summarise_mean_sd(nhanes, BMI, IncomeGroup, Gender)
15 r_function_summary

```

Before moving to Python, check that the function output has the same number of rows and the same grouping labels as the original pipeline.

## 23.5 Part C: Python Translation

Translate one step at a time. Do not ask AI for a full script until you have verified each cell.

```

1 import pandas as pd
2
3 nhanes = pd.read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
4
5 py_summary = (
6     nhanes
7     .dropna(subset=["BMI", "IncomeGroup", "Gender"])
8     .groupby(["IncomeGroup", "Gender"], as_index=False)
9     .agg(
10         n=("BMI", "size"),
11         mean_bmi=("BMI", "mean"),
12         sd_bmi=("BMI", "std")
13     )
14     .sort_values(["IncomeGroup", "Gender"])
15 )
16
17 py_summary["mean_bmi"] = py_summary["mean_bmi"].round(2)
18 py_summary["sd_bmi"] = py_summary["sd_bmi"].round(2)
19
20 py_summary

```

23.6 Part D: Parity Audit

Record the following checks in a short Quarto note:

| Check                                  | R command                                                                           | Python command                                                                | What to record                  |
|----------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------|
| Raw row count                          | <code>nrow(nhanes)</code>                                                           | <code>len(nhanes)</code>                                                      | Same number of rows             |
| Column names                           | <code>names(nhanes)</code>                                                          | <code>nhanes.columns.tolist()</code>                                          | Same list of expected variables |
| Analysis rows after missingness filter | <code>nrow(filter(nhanes, !is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)))</code> | <code>nhanes.dropna(subset=["BMI", "IncomeGroup", "Gender"]).row_count</code> | Same row count                  |
| Group count                            | <code>nrow(r_summary)</code>                                                        | <code>len(py_summary)</code>                                                  | Same number of grouped rows     |
| Summary values                         | <code>inspect mean_bmi and sd_bmi</code>                                            | <code>inspect mean_bmi and sd_bmi</code>                                      | Values match after rounding     |

23.7 AI Audit Prompt

Use this prompt only after you have a working R result:

Translate this R `dplyr` pipeline into Python/pandas one step at a time. For each step, explain the equivalent Python operation, the output I should inspect, and the parity check I should run against my R result. Do not combine steps or add new analysis choices.

23.8 Student Output

Submit a short note with:

- the R pipeline;

- the R function;
- the Python translation;
- a parity table showing the checks above;
- two sentences explaining any mismatch or confirming that the outputs agree.

## 24 Tutorial 1: Python vs. R - When and Why?

### 24.1 The Course Position

R remains the core language for required analysis in this course. Python is introduced so you can read short Python examples, translate small workflows with AI assistance, and recognize when Python may be useful in a health data project.

You are not choosing one language forever. You are learning how to move carefully between them.

### 24.2 Strengths At A Glance

| Task                     | R is often strongest for                            | Python is often strongest for                                  |
|--------------------------|-----------------------------------------------------|----------------------------------------------------------------|
| Descriptive epidemiology | <b>tidyverse</b> , survey workflows, reports        | possible with <b>pandas</b> , but often more verbose           |
| Visualization            | <b>ggplot2</b> , <b>plotly</b> , Quarto integration | <b>seaborn</b> , <b>matplotlib</b> , interactive web libraries |
| Machine learning         | <b>tidymodels</b> , <b>torch</b>                    | <b>scikit-learn</b> , <b>tensorflow</b> , <b>pytorch</b>       |
| Automation               | project scripts and reports                         | general-purpose scripting and APIs                             |
| Communication            | Quarto reports, slides, dashboards                  | web apps, APIs, automation, notebooks                          |

## 24.3 Notebooks vs. Scripts

- A notebook (`.ipynb`) is useful for exploration, teaching, and step-by-step AI translation checks.
- A script (`.py`) is useful for reusable functions and automation.
- A Quarto document (`.qmd`) is best when your analysis needs narrative, code, figures, citations, and a rendered final product.

## 24.4 Check Your Understanding

Write two sentences:

1. One task where you would stay in R.
2. One task where Python might be worth considering.

## 25 Tutorial 2: Setting Up Your Python Environment

### 25.1 Create A Notebook

In your GitHub Codespace:

1. Open the Week 5 folder.
2. Create a new file named `translation.ipynb`.
3. Select the recommended Python kernel when VS Code asks.
4. Run this first cell:

```
1 print("Hello from Python. My Week 5 notebook is running.")
```

### 25.2 Use The Variables Pane

The Variables pane helps you audit Python objects the way the Environment pane helps you audit R objects.

Run this cell:

```
1 age = 25
2 course = "HDSx"
```

Then open the notebook Variables pane and check that both variables appear.

### 25.3 Load A Tiny DataFrame



```

1 import pandas as pd
2
3 demo = pd.DataFrame({
4     "group": ["A", "A", "B"],
5     "value": [10, 15, 20]
6 })
7
8 demo

```

Double-click `demo` in the Variables pane to open the Data Viewer. This is where you will inspect row counts, column names, and translated outputs later in the week.

## 25.4 Common Fixes

| Problem                    | First thing to try                                                 |
|----------------------------|--------------------------------------------------------------------|
| No kernel appears          | Wait for the Codespace to finish starting, then reload the window  |
| Variables pane is missing  | Check that the Jupyter extension is enabled                        |
| Notebook cell will not run | Select a Python kernel from the notebook toolbar                   |
| Package import fails       | Check whether the package belongs in <code>requirements.txt</code> |

## 26 Tutorial 3: Packages and Reproducibility

### 26.1 R `library()` vs. Python `import`

Both languages require you to load packages before using them.

| R                                | Python                                 | Purpose                           |
|----------------------------------|----------------------------------------|-----------------------------------|
| <code>library(readr)</code>      | <code>import pandas<br/>as pd</code>   | read CSV files                    |
| <code>library(dplyr)</code>      | <code>import pandas<br/>as pd</code>   | filter, group, and summarise data |
| <code>library(ggplot2)</code>    | <code>import seaborn<br/>as sns</code> | create plots                      |
| <code>library(reticulate)</code> | not applicable                         | call Python from R                |

### 26.2 Why `requirements.txt` Matters

Python projects record package dependencies in `requirements.txt`. This is the Python-side equivalent of saying, “Here is what another person must install before my code will run.”

For this week, a minimal file is enough:

```
1 pandas
```

If you also create a plot in Python, add:

```
1 seaborn
2 matplotlib
```

## 26.3 Course Rule

If you install or rely on a Python package for submitted work, record it in `requirements.txt`. Do not leave package installation as an undocumented memory from your current Codespace.

## 26.4 R Connection

R reproducibility is handled through project documentation and, later in the course, `renv`. Week 5 is a bridge: you should be able to explain where dependencies are recorded in both languages.

## 27 Tutorial 4: Common Translation Traps

AI translation is useful, but it often hides small language differences. These are the traps to check first.

### 27.1 Trap 1: Indexing

R starts counting at 1. Python starts counting at 0.

```
1 values <- c(10, 20, 30)
2 values[1]
```

```
1 values = [10, 20, 30]
2 values[0]
```

Python slices are end-exclusive:

```
1 values[0:2]
```

This returns the first two values, not the first three.

### 27.2 Trap 2: Indentation

R uses braces to mark code blocks. Python uses indentation.

```
1 if (age >= 20) {
2   "adult"
3 } else {
4   "not in adult cohort"
5 }
```

```

1 if age >= 20:
2     "adult"
3 else:
4     "not in adult cohort"

```

If Python indentation is inconsistent, the code may fail or run with a different meaning.

## 27.3 Trap 3: Assignment vs. Comparison

| Meaning            | R                       | Python                                            |
|--------------------|-------------------------|---------------------------------------------------|
| assign a value     | <code>x &lt;- 10</code> | <code>x = 10</code>                               |
| compare values     | <code>x == 10</code>    | <code>x == 10</code>                              |
| missing/null check | <code>is.na(x)</code>   | <code>pd.isna(x)</code> or <code>x is None</code> |

## 27.4 Trap 4: Missing Values

R and Python both have missing values, but they are handled by different functions. Always check missingness before comparing summaries.

```

1 sum(is.na(nhanes$BMI))

```

```

1 nhanes["BMI"].isna().sum()

```

## 27.5 Audit Habit

When a translated result differs, check indexing, indentation, assignment/comparison, missingness handling, and grouping variables before assuming one language is “wrong.”

## 28 Tutorial 5: Loading and Inspecting Health Data

### 28.1 Load The NHANES Snapshot

In `translation.ipynb`, load the same CSV used in the R example:

```
1 import pandas as pd
2
3 nhanes = pd.read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
```

### 28.2 Inspect Before Summarising

```
1 nhanes.head()
```

```
1 nhanes.shape
```

```
1 nhanes.columns.tolist()
```

```
1 nhanes[["BMI", "IncomeGroup", "Gender"]].isna().sum()
```

### 28.3 Compare To R

| Question       | R                         | Python                   |
|----------------|---------------------------|--------------------------|
| How many rows? | <code>nrow(nhanes)</code> | <code>len(nhanes)</code> |

| Question                     | R                                    | Python                                  |
|------------------------------|--------------------------------------|-----------------------------------------|
| How many columns?            | <code>ncol(nhanes)</code>            | <code>nhanes.shape[1]</code>            |
| What are the column names?   | <code>names(nhanes)</code>           | <code>nhanes.columns.tolist()</code>    |
| How many missing BMI values? | <code>sum(is.na(nhanes\$BMI))</code> | <code>nhanes["BMI"].isna().sum()</code> |

## 28.4 Student Check

Before asking AI for any transformation, write down the raw row count, column count, and missingness count for BMI.

## 29 Tutorial 6: Reading Python Tracebacks

### 29.1 The Bottom Line Is Usually At The Bottom

Python errors often print several lines called a traceback. Read from the bottom upward:

```
1 FileNotFoundError: [Errno 2] No such file or directory: 'Data/patients.csv'
```

This tells you the immediate problem: Python cannot find the file at the path you gave it.

### 29.2 Common Week 5 Errors

| Error                            | Likely cause                                                        | First fix                                                                        |
|----------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>FileNotFoundError</code>   | path is wrong, file is in another folder, or capitalization differs | check the relative path from the notebook location                               |
| <code>ModuleNotFoundError</code> | package is missing                                                  | add the package to <code>requirements.txt</code> and install it in the Codespace |
| <code>KeyError</code>            | column name is misspelled or not present                            | run <code>nhanes.columns.tolist()</code>                                         |



| Error            | Likely cause                                      | First fix                            |
|------------------|---------------------------------------------------|--------------------------------------|
| IndentationError | Python block spacing is inconsistent              | re-indent the block with four spaces |
| NameError        | object has not been created in the current kernel | rerun earlier cells in order         |

## 29.3 AI Debugging Prompt

Read this Python traceback from the bottom up. Identify the error type, the exact object or path causing the issue, and the smallest code change that would fix it. Do not rewrite the full analysis.

## 29.4 Student Check

When you fix an error, add one sentence to your notebook explaining what failed and how you knew the fix worked.

## 30 Tutorial 7: The One Step, One Cell AI Workflow

### 30.1 Why This Rule Exists

Large AI-generated code blocks are hard to audit. A single hidden mistake in loading, filtering, grouping, or missingness handling can cascade into a polished but incorrect result.

Use one cell per analytic step:

1. load data;
2. inspect rows and columns;
3. filter missing values;
4. group and summarise;
5. compare with the R result.

### 30.2 Good AI Prompt

Translate only the data-loading step from R to Python/pandas. Tell me what output I should inspect before moving to the next step.

### 30.3 Risky AI Prompt

Translate my full analysis, make the plots, and write the interpretation.

The risky prompt gives up too much control. It may change the analytic question without telling you.

## 30.4 Parity Audit Checks

| Check          | R                                    | Python                                              |
|----------------|--------------------------------------|-----------------------------------------------------|
| Row count      | <code>nrow(df)</code>                | <code>len(df)</code> or<br><code>df.shape[0]</code> |
| Column names   | <code>names(df)</code>               | <code>df.columns.tolist()</code>                    |
| Unique levels  | <code>unique(df\$IncomeGroup)</code> | <code>df["IncomeGroup"].unique()</code>             |
| Missing values | <code>sum(is.na(df\$BMI))</code>     | <code>df["BMI"].isna().sum()</code>                 |
| Summary table  | <code>nrow(summary_df)</code>        | <code>len(summary_df)</code><br>rows                |

## 30.5 Notebook Habit

After each Python cell, add a short Markdown note:

```
1 Audit check: this cell matches the R output because ...
```

If you cannot fill in that sentence, do not move to the next cell yet.

## 31 Tutorial 8: Mixing R and Python

### 31.1 Why Interoperability Matters

Sometimes a project has one useful Python tool, but the rest of the analysis is already in R. The `reticulate` package lets R call Python code and bring Python objects back into the R session. In this course, `reticulate` is awareness-level: you should know what it does and when it might be useful, but you are not expected to build a full mixed-language pipeline.

### 31.2 Minimal Demo

Run this in an R console or an R Quarto document if `reticulate` is available in your Codespace.

```
1 library(reticulate)
2
3 py_run_string("
4 import pandas as pd
5 numbers = pd.DataFrame({'group': ['A', 'A', 'B'], 'value': [10, 15, 20]})
6 summary = numbers.groupby('group', as_index=False)['value'].mean()
7 ")
8
9 py$summary
```

### 31.3 Health Data Example

The same idea works with the NHANES case-study CSV:

```

1 library(reticulate)
2
3 py_run_string("
4 import pandas as pd
5 nhanes = pd.read_csv('../examples/nhanes-equity/data/nhanes_equity_v6.csv')
6 bmi_rows = nhanes[['BMI', 'IncomeGroup']].dropna()
7 ")
8
9 bmi_rows <- py$bmi_rows
10 head(bmi_rows)

```

## 31.4 When To Use It

Use `reticulate` when:

- a small Python library solves a specific problem better than the R ecosystem;
- the rest of your report, tables, or dashboard are already in R;
- you can document the Python dependency clearly.

Avoid it when:

- a clean R solution already exists;
- mixing languages makes the project harder to reproduce;
- the Python step cannot run in a fresh Codespace.

## 31.5 Check Your Understanding

In one sentence, explain what object moves from Python back into R in the NHANES example above.

## 32 Tutorial 9: Creating Reusable Python Scripts

### 32.1 Notebook or Script?

Use a notebook when you are exploring, explaining, and checking outputs. Use a script when you have a helper function that should be reused without copying and pasting.

This mirrors the R pattern from Week 3: once a chunk of code has a clear job, turn it into a named function.

### 32.2 Create `helpers.py`

In the Week 5 folder, create a file named `helpers.py` with this function:

```
1 from pathlib import Path
2 import pandas as pd
3
4
5 def load_and_preview(path, n_rows=5):
6     """Load a CSV, print its shape, and return the first rows."""
7     csv_path = Path(path)
8     if not csv_path.exists():
9         raise FileNotFoundError(f"Could not find file: {csv_path}")
10
11     data = pd.read_csv(csv_path)
12     print(f"Loaded {data.shape[0]} rows and {data.shape[1]} columns")
13     return data.head(n_rows)
```

## 32.3 Import It In A Notebook

In `translation.ipynb`, import and use the helper:

```
1 from helpers import load_and_preview
2
3 preview = load_and_preview("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
4 preview
```

## 32.4 Reproducibility Check

Before committing:

- verify that `helpers.py` is saved in the same folder as your notebook;
- rerun the notebook from a fresh kernel;
- check that the relative path starts from the notebook location;
- commit both files together with a message such as `Add Week 5 Python helper`.

## 32.5 Student Output

For Assignment 4, your reusable script should contain at least one function that is actually called from the notebook. A function that is written but never used does not demonstrate reproducibility.

## 33 Tutorial 10: Tracking Your Progress With Git

### 33.1 Commit The Audit Trail

Week 4 introduced Git as analysis provenance. Week 5 uses that skill to make translation auditable.

Commit after meaningful checkpoints:

1. R pipeline works.
2. R function reproduces the pipeline.
3. Python notebook loads the same data.
4. Python summary matches the R summary.
5. Final parity note is complete.

### 33.2 Notebook Output Hygiene

Jupyter notebooks can store rendered tables and images inside the `.ipynb` file. That makes Git diffs noisy.

Before committing:

1. Open the Command Palette with `Ctrl+Shift+P`.
2. Run Notebook: Clear All Outputs.
3. Save the notebook.
4. Stage `translation.ipynb`, `helpers.py`, and your Quarto note together.



### 33.3 Example Commit Messages

- 1 Add R BMI summary for Week 5 parity check
- 1 Translate BMI summary to pandas and verify row counts
- 1 Complete Assignment 4 parity audit note

### 33.4 Student Check

Your Git history should show the work developing in steps, not one unexplained final upload.

## 34 Tutorial 11: Project Proposal Readiness Check

### 34.1 Why This Belongs In Week 5

Before the project proposal is due, you should know whether your chosen dataset can be loaded, inspected, and explained in the course workflow. Week 5 adds one extra test: can you verify the basic data shape in both R and Python?

### 34.2 Proposal Readiness Checklist

Your proposal should include:

- a focused health-data question;
- the dataset name and source;
- a brief provenance and stewardship note;
- a feasibility check showing the file can load in a Codespace;
- one or two planned descriptive summaries or visualizations;
- any known privacy, licensing, or access limitations.

### 34.3 Feasibility Check

For your project dataset, record:

| Check       | What to write                    |
|-------------|----------------------------------|
| File path   | relative path from the repo root |
| File format | CSV, RDS, Excel, API, or other   |

| Check                | What to write                                      |
|----------------------|----------------------------------------------------|
| Row and column count | basic shape after loading                          |
| Key variables        | names and meanings of the fields you expect to use |
| Missingness concern  | one variable where missing values may matter       |
| R/Python check       | whether both languages can load the file           |

## 34.4 Student Output

Add this feasibility check to your project proposal notes. If your dataset does not load cleanly yet, describe the blocker honestly instead of hiding it.

## 35 Tutorial 12: Knowledge Check

Answer these in a Markdown cell or short Quarto note.

1. Why is Week 5 called “polyglot awareness” rather than “Python replacement”?
2. Name one task where R is the better default for this course.
3. Name one task where Python might be useful.
4. Why does `my_list[1]` return the second item in Python?
5. What is the difference between `=` and `==`?
6. What does `requirements.txt` document?
7. What is the One Step, One Cell rule?
8. What are three parity checks you should run after translating R code to Python?
9. Why should notebook outputs be cleared before committing?
10. What does `reticulate` allow R to do?

### 35.1 Short Applied Check

Given the R expression:

```
1 mean(nhanes$BMI, na.rm = TRUE)
```

Write the Python/pandas equivalent and name the missingness check you would run before trusting it.

## 36 Tutorial 13: The Bilingual Translation Exercise

### 36.1 Purpose

This exercise turns the week into a concrete artifact: one small R analysis, one Python translation, and one audit note showing that the outputs agree.

### 36.2 Required Task

Translate the Week 5 R-deepening summary into Python/pandas:

1. Start from the R pipeline in [R Deepening and Parity Check](#).
2. Create `translation.ipynb`.
3. Load the NHANES CSV from `../../examples/nhanes-equity/data/nhanes_equity_v6.csv`.
4. Recreate the BMI-by-income-and-gender summary in Python.
5. Compare the Python output to the R output using row counts, grouping labels, and rounded summary values.
6. Extract one Python helper into `helpers.py` and call it from the notebook.
7. Add a Markdown cell explaining what AI helped with and how you verified the result.
8. Clear notebook outputs before committing.

### 36.3 Parity Table Template

| Check                          | R result | Python result | Match? | Notes |
|--------------------------------|----------|---------------|--------|-------|
| Raw row count                  |          |               |        |       |
| Filtered row count             |          |               |        |       |
| Number of grouped rows         |          |               |        |       |
| Mean BMI values after rounding |          |               |        |       |
| Group labels                   |          |               |        |       |

## 36.4 AI Prompt

I have a working R `dplyr` summary. Translate it to Python/pandas one step at a time. After each step, tell me exactly what output to compare with R. Do not add new variables, filters, plots, or interpretation.

## 36.5 Definition of Done

Your exercise is complete when a reader can open the notebook, rerun the cells in order, and see why the R and Python outputs do or do not match.

## 37 Tutorial 14: R To Python Cheat Sheet

Use this as a translation aid, not as a substitute for checking outputs.

| Task           | R                                                       | Python                                                       |
|----------------|---------------------------------------------------------|--------------------------------------------------------------|
| Load package   | <code>library(dplyr)</code>                             | <code>import pandas<br/>as pd</code>                         |
| Read CSV       | <code>read_csv("file.csv")</code>                       | <code>pd.read_csv("file.csv")</code>                         |
| First rows     | <code>head(df)</code>                                   | <code>df.head()</code>                                       |
| Row count      | <code>nrow(df)</code>                                   | <code>len(df)</code> or<br><code>df.shape[0]</code>          |
| Column names   | <code>names(df)</code>                                  | <code>df.columns.tolist()</code>                             |
| Select columns | <code>select(df, BMI,<br/>Gender)</code>                | <code>df[["BMI",<br/>"Gender"]]</code>                       |
| Filter rows    | <code>filter(df, Age<br/>&gt;= 20)</code>               | <code>df[df["Age"] &gt;=<br/>20]</code>                      |
| Missing values | <code>sum(is.na(df\$BMI))</code>                        | <code>df["BMI"].isna().sum()</code>                          |
| Group summary  | <code>group_by(...)<br/> &gt;<br/>summarise(...)</code> | <code>.groupby(...).agg(...)</code>                          |
| New column     | <code>mutate(df, whtr<br/>= Waist /<br/>Height)</code>  | <code>df["whtr"] =<br/>df["Waist"] /<br/>df["Height"]</code> |
| Sort rows      | <code>arrange(df,<br/>IncomeGroup)</code>               | <code>df.sort_values("IncomeGroup")</code>                   |
| Unique values  | <code>unique(df\$Gender)</code>                         | <code>df["Gender"].unique()</code>                           |

## 37.1 Translation Rule

The translation is not finished when the code runs. It is finished when the R and Python outputs have been compared and the comparison is documented.



## **Part VIII**

### **Week 6: Data Visualization**

# Visual Storytelling

## Overview

This chapter challenges students to move beyond default charts and create publication-quality visualizations. You will use AI tools to generate initial code for complex plots in ggplot2 or plotly, and then apply human judgment to refine them for clarity, accuracy, and aesthetics. The lesson emphasizes detecting misleading AI-generated visuals and “steering” the code to tell a truthful and compelling data story.

## Connection

Week 5 checked that summaries agree across languages. This week turns a descriptive summary into a visual claim and asks whether the design helps or misleads the audience. Next week you will document the same dataset with EDA and a Table 1.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
```

```
39 #> 2 Low Income (<1.3) 15293 29.4
40 #> 3 Middle Income 19228 29.3
```

## In-Class Activity

- Work through the [ggplot worked example](#) using the CSV snapshot.
- Audit the flawed plot provided in the visual-audit activity and identify at least three issues.
- Revise the plot so the title, scale, grouping, and caption support descriptive interpretation.
- Explain what changed and why, including any AI assistance you used.

Student output: a corrected plot, caption, audit note, and AI-use note. Assignment link: [../..assignments/assignment05-visualization/README.md](#).

## Definition of Done

- Students can generate, critique, revise, and caption at least one health data visualization.
- The activity includes a concrete visual audit checklist.
- All example plots render in a fresh Codespace.

# Worked Example: Descriptive BMI Visualization

## Goal

This worked example uses the NHANES Health Equity CSV snapshot to build one descriptive visualization. The plot summarizes mean BMI by income group across NHANES cycle. It is **unweighted** and should be interpreted as a classroom visualization exercise, not a population estimate.

```
1 library(readr)
2 library(dplyr)
3 library(ggplot2)
4
5 candidate_paths <- c(
6   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
7   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
8 )
9
10 data_path <- candidate_paths[file.exists(candidate_paths)][1]
11 if (is.na(data_path)) {
12   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
13 }
14
15 nhanes <- read_csv(data_path, show_col_types = FALSE)
```

## Prepare a Descriptive Summary

Keep the summary intentionally simple. We are describing the prepared class dataset, not estimating a national trend.

```

1 bmi_by_income_cycle <- nhanes |>
2   filter(
3     !is.na(BMI),
4     !is.na(IncomeGroup),
5     !is.na(Cycle),
6     Age >= 20,
7     Age <= 80
8   ) |>
9   group_by(Cycle, IncomeGroup) |>
10  summarise(
11    n = n(),
12    mean_bmi = mean(BMI),
13    .groups = "drop"
14  ) |>
15  filter(n >= 30)
16
17 bmi_by_income_cycle
18 #> # A tibble: 33 x 4
19 #>   Cycle      IncomeGroup      n mean_bmi
20 #>   <chr>      <chr>      <int>   <dbl>
21 #> 1 1999-2000 High Income (>3.5) 1123    27.9
22 #> 2 1999-2000 Low Income (<1.3) 1091    28.8
23 #> 3 1999-2000 Middle Income      1327    29.0
24 #> 4 2001-2002 High Income (>3.5) 1491    28.1
25 #> 5 2001-2002 Low Income (<1.3) 1089    28.7
26 #> 6 2001-2002 Middle Income      1561    28.3
27 #> 7 2003-2004 High Income (>3.5) 1284    28.4
28 #> 8 2003-2004 Low Income (<1.3) 1183    28.6
29 #> 9 2003-2004 Middle Income      1573    28.8
30 #> 10 2005-2006 High Income (>3.5) 1480    28.5
31 #> # i 23 more rows

```

## Plot

```

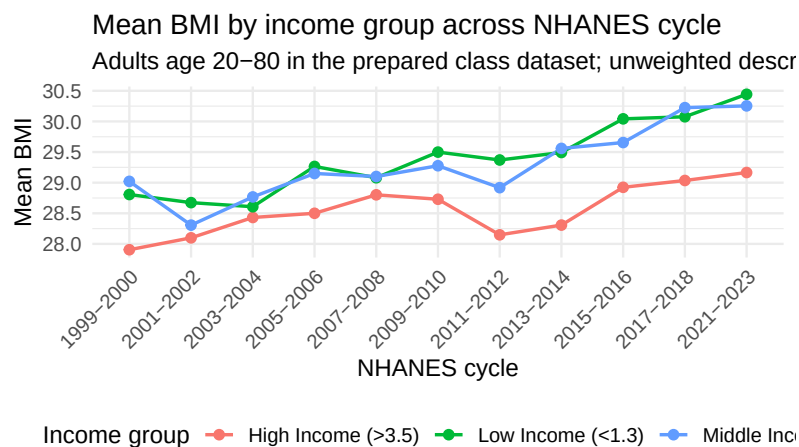
1 ggplot(
2   bmi_by_income_cycle,
3   aes(x = Cycle, y = mean_bmi, color = IncomeGroup, group = IncomeGroup)
4 ) +

```

```

5 geom_line(linewidth = 0.8) +
6 geom_point(size = 2) +
7 labs(
8   title = "Mean BMI by income group across NHANES cycle",
9   subtitle = "Adults age 20-80 in the prepared class dataset; unweighted descriptive summary",
10  x = "NHANES cycle",
11  y = "Mean BMI",
12  color = "Income group",
13  caption = "Caption template: Mean BMI is summarized descriptively from the cached class data",
14 ) +
15 theme_minimal(base_size = 12) +
16 theme(
17   axis.text.x = element_text(angle = 45, hjust = 1),
18   legend.position = "bottom"
19 )

```



ne/cycle values, and should not be interpreted as causal effects or population estimates.

## Caption Template

Use this structure when revising your own plot:

This figure shows [measure] by [group] across [time/category] using the prepared NHANES Health Equity classroom dataset. The summary is descriptive and unweighted. Records with missing

[variables] were excluded. The plot supports comparison of patterns, not causal claims or population-level estimates.

## **Mini Visual Audit Checklist**

- Does the title describe what is plotted without overstating the claim?
- Is it clear whether the statistic is weighted or unweighted?
- Are axes, groups, and units labeled plainly?
- Are missing values or exclusions disclosed?
- Is the scale chosen to support honest comparison rather than exaggeration?
- Does the caption state the limitation that this is descriptive, not causal?



# Activity: AI Visual Audit

## Goal

Students evaluate AI-generated visualizations for correctness, clarity, bias, aggregation choices, and misleading design.

## Inputs

- Student worked example: `weeks/week06-visualization/worked-example-ggplot.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Flawed plot/code: provided by your instructor or TA during the activity

## Activity Steps

1. Review the worked example and note how it labels the data source, statistic, and limitation.
2. Inspect the flawed plot or code provided by the instructor.
3. Identify at least three issues that could mislead a reader.
4. Revise the plot so the title, scale, grouping, and caption match the evidence.
5. Write a short note explaining what changed and why.

## Required Outputs

- `visual-audit-note.md`: an audit note listing at least three issues.
- `corrected-plot.qmd`: a rerunnable Quarto file that creates the corrected plot.

- `corrected-plot.png`: the exported corrected plot.
- `caption.md`: a caption that states the statistic, data source, exclusions, and limitations.
- `ai-use-note.md`: if you used AI, explain what it helped with and how you checked the result.

## Audit Categories

Use these categories in your audit note:

- **Correctness:** Does the plot show what the title and labels claim?
- **Design:** Do scale, color, ordering, and geometry support fair comparison?
- **Interpretation:** Does the caption avoid causal or population-level claims that the plot cannot support?
- **Reproducibility:** Can another person rerun the code from the repository root using relative paths?

## Completion Check

Your submission is complete when a peer can rerun your corrected plot, identify the main comparison without reading your code, and see at least one explicit limitation in the caption.

## **Part IX**

### **Week 7: EDA, Table 1 & AI Auditing**

# Exploring Data

## Overview

Before modeling, you must understand your data. This chapter guides students through the rigorous process of Exploratory Data Analysis (EDA) and the creation of the standard “Table 1” for demographic characteristics. You will learn to document every cleaning decision and exclusion criteria in a reproducible Quarto log, ensuring that your path from raw data to analysis dataset is transparent and repeatable by anyone.

## Connection

Week 6 focused on whether visual design makes a descriptive claim honest and clear. This week documents the analysis cohort, missingness, and Table 1 so that later dashboard and report outputs have an auditable foundation.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```

1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## In-Class Activity

- Work through the [Table 1 worked example](#) using the CSV snapshot.
- Define and document an analysis cohort before summarizing.
- Create a Table 1-style summary with N, BMI mean (SD), and age mean (SD).
- Audit the planted-error starter and categorize at least five issues as correctness, reproducibility, or interpretation problems.

Student output: a rerunnable EDA note, Table 1, planted-error audit, and AI-use note. Assignment link: [../.. / assignments / assignment06-eda-ai-audit / README.md](#).

## Definition of Done

- Students can document cleaning decisions from raw or starter data to an analysis-ready table.
- The planted-error exercise has an instructor key.
- The EDA renders without manual intervention.

# Worked Example: EDA and Table 1

## Goal

This worked example creates a small Table 1-style descriptive summary from the NHANES Health Equity CSV snapshot. The goal is transparent EDA: define the cohort, check missingness, summarize key variables, and write a methods note that does not overclaim.

```
1 library(readr)
2 library(dplyr)
3 library(knitr)
4
5 candidate_paths <- c(
6   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
7   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
8 )
9
10 data_path <- candidate_paths[file.exists(candidate_paths)][1]
11 if (is.na(data_path)) {
12   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
13 }
14
15 nhanes <- read_csv(data_path, show_col_types = FALSE)
```

## Define the Analysis Cohort

Decision: restrict to adults age 20-80. This keeps the example focused on adult BMI summaries and avoids mixing children,

adolescents, and the oldest age-coded records into one classroom table.

```
1 analysis_df <- nhanes |>
2   filter(Age >= 20, Age <= 80)
3
4 cohort_counts <- tibble(
5   step = c("Prepared CSV rows", "Rows after age 20-80 restriction"),
6   n = c(nrow(nhanes), nrow(analysis_df))
7 )
8
9 kable(cohort_counts)
```

| step                             | n      |
|----------------------------------|--------|
| Prepared CSV rows                | 105626 |
| Rows after age 20-80 restriction | 57137  |

## Missingness Check

```
1 key_vars <- c("BMI", "Age", "IncomeGroup", "Gender", "Race", "Cycle")
2
3 missingness <- tibble(variable = key_vars) |>
4   mutate(
5     missing_n = vapply(analysis_df[key_vars], function(x) sum(is.na(x)), numeric(1)),
6     missing_percent = round(100 * missing_n / nrow(analysis_df), 1)
7   )
8
9 kable(missingness)
```

| variable    | missing_n | missing_percent |
|-------------|-----------|-----------------|
| BMI         | 1009      | 1.8             |
| Age         | 0         | 0.0             |
| IncomeGroup | 5395      | 9.4             |
| Gender      | 0         | 0.0             |
| Race        | 0         | 0.0             |



| variable | missing_n | missing_percent |
|----------|-----------|-----------------|
| Cycle    | 0         | 0.0             |

## Table 1-Style Summary

This example stratifies by `IncomeGroup` and reports N plus mean (SD) for BMI and age. It is unweighted and descriptive.

```

1 mean_sd <- function(x) {
2   sprintf("%.1f (%.1f)", mean(x, na.rm = TRUE), sd(x, na.rm = TRUE))
3 }
4
5 table1 <- analysis_df |>
6   filter(!is.na(IncomeGroup)) |>
7   group_by(IncomeGroup) |>
8   summarise(
9     N = n(),
10    `BMI, mean (SD)` = mean_sd(BMI),
11    `Age, mean (SD)` = mean_sd(Age),
12    .groups = "drop"
13  )
14
15 kable(table1)

```

| IncomeGroup        | N     | BMI, mean (SD) | Age, mean (SD) |
|--------------------|-------|----------------|----------------|
| High Income (>3.5) | 16520 | 28.6 (6.3)     | 49.7 (16.0)    |
| Low Income (<1.3)  | 15646 | 29.4 (7.5)     | 47.4 (18.1)    |
| Middle Income      | 19576 | 29.3 (7.0)     | 50.0 (18.4)    |

## Methods Note Template

We used the prepared NHANES Health Equity classroom CSV snapshot. The analysis cohort included records with age 20-80. We summarized BMI and age by income group using unweighted

descriptive statistics. Missing values were checked before summarizing. These summaries describe the prepared class dataset and should not be interpreted as causal effects or population estimates.

## **Reproducibility Notes**

- The data path is relative to the repository.
- The cohort restriction is stated before the table.
- Missingness is reported before complete-case summaries.
- The table labels distinguish N, mean, and SD.
- The methods note states that the analysis is descriptive and non-causal.

# Activity: Table 1 and AI Audit

## Goal

Students create a descriptive summary and audit AI-generated code or interpretation for planted errors.

## Inputs

- Student worked example: `weeks/week07-eda-ai-audit/worked-example-table1.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Planted-error starter: distributed by the instructor or TA during the activity

## Activity Steps

1. Review the worked example and identify the cohort definition, missingness check, and Table 1 labels.
2. Open the planted-error starter provided by the instructor.
3. Run or read the starter code and annotate at least five issues.
4. Categorize each issue as **correctness**, **reproducibility**, or **interpretation**.
5. Produce a corrected EDA note and Table 1.

## Required Outputs

- `eda-note.qmd`: a rerunnable Quarto note with cohort definition, missingness table, and Table 1.
- `eda-note.html`: rendered output from the Quarto note.

- `table1.csv`: exported Table 1 or equivalent descriptive summary.
- `audit-note.md`: at least five detected issues with category, why it matters, and corrected approach.
- `ai-use-note.md`: if you used AI, explain what it helped with and how you verified the result.

## Minimum Table 1 Requirements

- Cohort: age 20-80, unless your instructor approves a different cohort.
- Stratifier: `IncomeGroup`, unless your instructor assigns a different grouping variable.
- Variables: N, BMI mean (SD), and age mean (SD).
- Methods note: state that the table is descriptive, unweighted, and non-causal.

## Completion Check

Your submission is complete when another student can rerun your `eda-note.qmd`, see exactly who is included, understand how missingness was handled, and identify why the planted-error starter was misleading.

## **Part X**

### **Week 8: Dashboard Prototypes for KT**

# Interactive Dashboards

## Overview

Static PDF reports often sit unread, so this chapter introduces dynamic Knowledge Translation through Shiny apps and dashboards. Students will take their prior EDA results and transform them into interactive web tools that allow users to filter and explore the data themselves. You will learn the best practices of user interface design for health data and deploy your first live dashboard to the web via GitHub Pages or Shinylive.

## Connection

Week 7 established an auditable descriptive foundation. This week is the first app-forward week: students run the NHANES dashboard, identify an audience, and improve one audience-facing element. Next week turns one dashboard finding into a short scientific communication product.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`

Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Run the app and describe the intended audience in one sentence.
- Make one audience-facing improvement to a label, help text, or interpretation note.
- Verify the app still loads the cached RDS after your change.
- Write one limitation statement that belongs inside or beside the dashboard.

Student output: a local app run note, one audience-facing text improvement, and a limitation statement. Assignment link: `../../assignments/assignment07-dashboard/README.md`.

## Definition of Done

- Students can produce a small dashboard-style KT artifact from prior EDA outputs.
- The dashboard runs or renders in the supported workflow.
- The page distinguishes required core work from optional deployment extensions.

# Starter Dashboard

## Goal

Students adapt a minimal dashboard-style knowledge translation product by changing text and interpretation before changing analytic logic.

## Required Elements

- One audience-specific question.
- One interactive or dashboard-style control.
- One visualization.
- One plain-language interpretation.
- One limitation or caution.

## Minimal Review Workflow

1. Run the NHANES dashboard from the repository root.
2. Choose a stakeholder audience, such as a public-health analyst, student team, or community-facing briefing group.
3. Identify one label, caption, help text, or interpretation note that could be clearer for that audience.
4. Make a documentation-only or UI-text-only change.
5. Rerun the dashboard and verify the cached dataset loads.
6. Record the change, audience, and limitation in a short note.



## Student Output

Submit a short dashboard adaptation note containing:

- Audience sentence.
- Before/after text change.
- One screenshot or description of the affected dashboard area.
- Limitation statement.
- Reproducibility note describing how the app was run locally.

## Privacy-Safe Publishing Check

- Do not publish row-level data.
- Avoid small-cell claims.
- State that dashboard outputs are descriptive.
- Keep the cached classroom data separate from raw NHANES retrieval work.

## **Part XI**

### **Week 9: Communication of Scientific Findings**

# Dynamic Presentations

## Overview

Data analysis is useless if it cannot be understood by key stakeholders. This chapter focuses on “Audience-Specific KT,” teaching you to convert technical findings into plain language summaries and engaging slide decks using Quarto. You will practice peer review and learn to use LLMs to simplify jargon without losing scientific accuracy, ensuring your work can impact policy and practice.

## Connection

Week 8 made the dashboard audience-facing. This week turns one dashboard plot or table into a short slide story with a clear takeaway, evidence, and limitation. Next week you will expand that communication into a reproducible report.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`

Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Build a 3-5 slide outline explaining one dashboard plot.
- Include one plain-language takeaway and one limitation.
- Add a source note for CDC NHANES and state whether you used the cached data or rebuilt it.
- Ask a peer whether the claim is understandable without seeing the code.

Student output: a 3-5 slide outline plus peer feedback notes. Assignment link: `../.. / assignments / assignment08-communication / README.md`.

## Definition of Done

- Students can render a short Quarto slide deck.
- Each slide has one clear purpose.
- Peer feedback checks clarity, evidence, limitations, and KT value.

# Revealjs and Peer Feedback

## Goal

Students turn one dashboard finding into a short scientific communication product and practice structured peer feedback.

## Minimal Revealjs Starter

```
1 ---
2 title: "NHANES Equity Dashboard Finding"
3 format: revealjs
4 ---
```

## Three-Slide Structure

1. **Question and audience:** Who needs this finding, and why?
2. **Evidence:** Show one dashboard plot or table and state the descriptive pattern.
3. **Limitation and action:** Name what the data can and cannot support, then state the next decision or question.

## Peer Feedback Form

| Prompt                     | Peer note |
|----------------------------|-----------|
| What is the main message?  |           |
| What evidence supports it? |           |

| Prompt                           | Peer note |
|----------------------------------|-----------|
| What is unclear or overclaimed?  |           |
| Is the limitation visible?       |           |
| What would improve audience fit? |           |

## Student Output

Submit a rendered slide deck or slide outline with the completed peer feedback form. Include a source note for CDC NHANES and an AI-use note if AI helped draft the wording.

## **Part XII**

### **Week 10: Writing & Publishing Reports**

# Reproducible Reporting

## Overview

This chapter simulates the final stretch of a research project: writing the manuscript. Students learn to assemble their analysis, code, and writing into a single “dynamic report” that automatically updates numbers and figures if the data changes. You will explore advanced publishing workflows to create professional, citation-ready portfolios that demonstrate your ability to execute end-to-end data science.

## Connection

Week 9 turned one result into a short slide story. This week expands the same evidence into a reproducible report with code-generated outputs, citations, and limitations. Next week you will stress-test whether the report and dashboard work in a fresh environment.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`



Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Draft a short Quarto report that complements the dashboard rather than duplicating it.
- Include a data source paragraph, one figure or table, and a limitations note.
- Add a source note for CDC NHANES and the `nhanesA` retrieval pathway.
- Verify that the report uses relative paths and documents whether results came from the cached data.

Student output: a rendered Quarto report with one figure, one table, citations, limitations, and an AI-use note. Assignment link: `../../assignments/assignment09-reporting/README.md`.

## Definition of Done

- The report renders from source.
- Citations resolve.
- Outputs are regenerated by code rather than pasted manually.
- The README explains how to reproduce the report.

# Report Template: NHANES Equity Summary

## Purpose

This runnable template shows how to pair a dashboard with a short reproducible report. The dashboard supports exploration; the report fixes one question, documents the data, and cites the source.

## Question

How does mean BMI vary by income group among adults age 20-80 in the prepared NHANES Health Equity classroom dataset?

## Data Source

The report uses the cached CSV snapshot at `examples/nhanes-equity/data/nhanes_equity_v6.csv`. The source data are derived from CDC/NCHS NHANES public-use files (Centers for Disease Control and Prevention, National Center for Health Statistics 2026).

```
1 library(readr)
2 library(dplyr)
3 library(ggplot2)
4 library(knitr)
5
6 candidate_paths <- c(
7   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
8   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
```

```

9 )
10
11 data_path <- candidate_paths[file.exists(candidate_paths)][1]
12 if (is.na(data_path)) {
13   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
14 }
15
16 nhanes <- read_csv(data_path, show_col_types = FALSE)

```

## Methods

We restricted the classroom dataset to adults age 20-80 and summarized BMI by income group. The summaries are unweighted and descriptive. They should not be interpreted as causal effects or national population estimates.

```

1 analysis_df <- nhanes |>
2   filter(Age >= 20, Age <= 80)
3
4 missingness <- analysis_df |>
5   summarise(
6     rows = n(),
7     missing_bmi = sum(is.na(BMI)),
8     missing_income = sum(is.na(IncomeGroup)),
9     missing_age = sum(is.na(Age))
10  )
11
12 kable(missingness, caption = "Missingness check for key report variables.")

```

Table 37.6: Missingness check for key report variables.

| rows  | missing_bmi | missing_income | missing_age |
|-------|-------------|----------------|-------------|
| 57137 | 1009        | 5395           | 0           |

## Results

```
1 income_summary <- analysis_df |>
2   filter(!is.na(BMI), !is.na(IncomeGroup)) |>
3   group_by(IncomeGroup) |>
4   summarise(
5     n = n(),
6     mean_bmi = mean(BMI),
7     sd_bmi = sd(BMI),
8     .groups = "drop"
9   ) |>
10  mutate(
11    `BMI, mean (SD)` = sprintf("%.1f (%.1f)", mean_bmi, sd_bmi)
12  ) |>
13  select(IncomeGroup, n, `BMI, mean (SD)`)
14
15 kable(income_summary, caption = "Descriptive BMI summary by income group.")
```

Table 37.7: Descriptive BMI summary by income group.

| IncomeGroup        | n     | BMI, mean (SD) |
|--------------------|-------|----------------|
| High Income (>3.5) | 16330 | 28.6 (6.3)     |
| Low Income (<1.3)  | 15293 | 29.4 (7.5)     |
| Middle Income      | 19228 | 29.3 (7.0)     |

```
1 plot_df <- analysis_df |>
2   filter(!is.na(BMI), !is.na(IncomeGroup)) |>
3   group_by(IncomeGroup) |>
4   summarise(
5     n = n(),
6     mean_bmi = mean(BMI),
7     .groups = "drop"
8   )
9
10 ggplot(plot_df, aes(x = IncomeGroup, y = mean_bmi)) +
11   geom_col(fill = "#2C7FB8") +
12   geom_text(aes(label = paste0("n=", n)), vjust = -0.4, size = 3.5) +
13   labs()
```

```

14 title = "Mean BMI by income group",
15 subtitle = "Adults age 20-80; unweighted descriptive summary",
16 x = "Income group",
17 y = "Mean BMI"
18 ) +
19 expand_limits(y = 0) +
20 theme_minimal(base_size = 12)

```

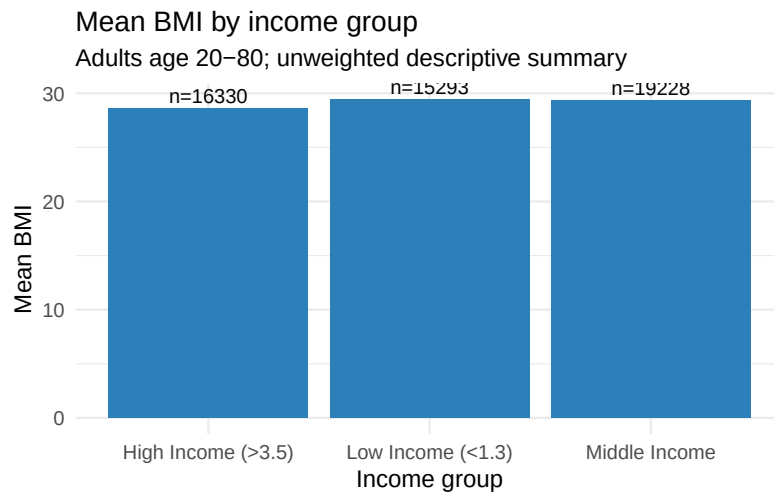


Figure 37.1: Mean BMI by income group among adults age 20-80 in the prepared NHANES Health Equity classroom dataset. Values are unweighted and descriptive.

## Limitations

- The table and figure use a prepared classroom dataset rather than raw NHANES files.
- The summaries are unweighted.
- The comparison is descriptive and does not establish causality.
- Missing BMI or income group values are excluded from the grouped summary.

## **Plain-Language Summary**

In this classroom dataset, mean BMI differs across income groups. The result is useful for practicing reproducible reporting and dashboard interpretation, but it should be described as a descriptive pattern rather than a causal finding.

## **AI-Use and Audit Note**

Document any AI assistance used to draft code, captions, or interpretation. State exactly which output you checked against the code, data dictionary, or rendered report.

## **Part XIII**

# **Week 11: Portfolio Surgery**

# Advanced Topics

## Overview

Before the final submission, students must “bulletproof” their work. This chapter focuses on advanced project hygiene, dependency management (ensuring your code runs on someone else’s machine), and auditability. You will act as a peer reviewer to identify reproducibility failures in others’ code, fixing broken paths and missing seeds, to ensure your final project stands up to scientific scrutiny.

## Connection

Week 10 produced a report with code-generated outputs and citations. This week tests whether the report, dashboard, and instructions survive a fresh environment. The final portfolio should leave a reviewer able to rerun the core work without private setup knowledge.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`



Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Run a fresh-session test: can the app launch from the repository root?
- Verify the cached RDS path works before attempting a CDC rebuild.
- Optionally run the rebuild script only if internet access and time are available.
- Optionally draft a static dashboard extension using a CSV snapshot and no Shiny server.
- Log any dependency, path, or README issues as reproducibility risks.

Student output: a reproducibility test log, repaired README or path issues, and a final risk list. Assignment link: `../../assignments/assignment10-portfolio-surgery/README.md`.

## Static Dashboard Demo

The completed [static dashboard source file](#) demonstrates a no-server alternative that reads the NHANES CSV snapshot in the browser. Use it as an exemplar for optional portfolio extensions after the core Shiny and report workflows are reproducible.

## Definition of Done

- A peer can clone or open the repository in a fresh Codespace and reproduce the core outputs.
- Broken paths, missing dependencies, unclear READMEs, and ungrounded AI claims are fixed.
- The final portfolio checklist is linked from this page.

# Reproducibility Stress Test

## Goal

Students test whether a project can run in a fresh environment and document every failure before final submission.

## Fresh Environment Protocol

Complete these steps in a new Codespace or equivalent clean environment.

1. Open the repository from GitHub.
2. Locate the project README and follow only the documented setup steps.
3. Render the Quarto book or report.
4. Run the NHANES dashboard from the repository root.
5. Open the Week 11 static dashboard source and render it as a standalone Quarto dashboard.
6. Check that all data paths are relative.
7. Record every failure, warning, unclear instruction, and repair.
8. Commit repairs with clear messages.

## Failure Log

| Check                                 | Result | Fix made | Owner |
|---------------------------------------|--------|----------|-------|
| Repository opens in fresh environment |        |          |       |
| Packages install or load              |        |          |       |
| Report renders                        |        |          |       |
| Dashboard runs locally                |        |          |       |

| Check                            | Result | Fix made | Owner |
|----------------------------------|--------|----------|-------|
| Static dashboard renders         |        |          |       |
| README instructions are accurate |        |          |       |
| Relative paths work              |        |          |       |

## Peer Review Roles

- **Driver:** Runs the project exactly as documented.
- **Navigator:** Reads instructions aloud and logs ambiguity.
- **Auditor:** Checks paths, data provenance, citations, and AI-use notes.

Rotate roles after the first major failure or after 20 minutes.

## Final Repair Checklist

- ☐ All required outputs render or run from a fresh environment.
- ☐ README instructions match the actual workflow.
- ☐ Data paths are relative.
- ☐ Cached data files needed for offline work are present.
- ☐ Citations resolve.
- ☐ AI-use notes identify what was drafted and how it was checked.
- ☐ Remaining risks are documented in the failure log.

## **Part XIV**

# **Weeks 12-13: Project Presentations**

# Week 12: Term Project Presentations, Part 1

## Overview

This session is the first presentation block for Milestone 5. Students present their portfolio work with an emphasis on scientific clarity, knowledge translation, reproducibility, and auditability.

## TODO Before Release



### Content TODO

- Add final presentation timing rules.
- Add slide submission instructions.
- Add the peer feedback form or link.
- Add the presentation rubric from [milestones/m5-presentation/](#).

## Definition of Done

- Slides render before class.
- The project repository link is available to reviewers.
- The presenter can explain the dataset, workflow, major findings, limitations, and AI-use audit trail.

# Week 13: Term Project Presentations, Part 2

## Overview

This session completes Milestone 5 presentations and gives students a final opportunity to collect feedback before the Final Portfolio deadline.

## TODO Before Release

### Content TODO

- Confirm which groups present in Part 2.
- Add final reminders for the Dec 11 portfolio submission.
- Add a closing reflection prompt on reproducibility and knowledge translation.

## Definition of Done

- Remaining presentations are complete.
- Peer feedback has been returned to presenters.
- Students know exactly what must change before final portfolio submission.

## **Part XV**

# **Assignments and Milestones**



# Assignments

Weekly assignments are scaffolded practice. Assignments 1-3 are foundational gates; Assignments 4-10 support the project portfolio and use the best-5-of-7 grading policy described in the syllabus.

## Assignment Map

| Assignment   | Week | Focus                                        | Template folder                         |
|--------------|------|----------------------------------------------|-----------------------------------------|
| Assignment 1 | 2    | Codespaces, Quarto, commit/sync              | assignments/assignment01-workflows/     |
| Assignment 2 | 3    | R with AI, tidyverse, functions, <b>renv</b> | assignments/assignment02-r-with-ai/     |
| Assignment 3 | 4    | Git collaboration and repo hygiene           | assignments/assignment03-git/           |
| Assignment 4 | 5    | Polyglot awareness and R deepening           | assignments/assignment04-polyglot/      |
| Assignment 5 | 6    | Visualization and AI visual audit            | assignments/assignment05-visualization/ |
| Assignment 6 | 7    | EDA, Table 1, planted-error audit            | assignments/assignment06-eda-ai-audit/  |

| Assignment    | Week | Focus                                      | Template folder                                          |
|---------------|------|--------------------------------------------|----------------------------------------------------------|
| Assignment 7  | 8    | Dashboard-style KT prototype               | <code>assignments/assignment07-dashboard/</code>         |
| Assignment 8  | 9    | Scientific communication and peer feedback | <code>assignments/assignment08-communication/</code>     |
| Assignment 9  | 10   | Reporting, citations, publishing           | <code>assignments/assignment09-reporting/</code>         |
| Assignment 10 | 11   | Portfolio surgery and reproducibility test | <code>assignments/assignment10-portfolio-surgery/</code> |

## TODO

### Warning

Create a student-facing `README.md` in each assignment folder from `assignments/assignment-template.qmd`.

# Assignment Template

Use this template for all weekly applied assignments.

## Student-Facing Brief

- **Purpose:**
- **Inputs:**
- **Expected output:**
- **Submission location:**
- **Due date:**

## Required Files

- README.md
- One rendered Quarto output or dashboard preview.
- Source .qmd, .R, .py, or supporting files needed to rerun the work.
- Data files, or clear instructions for obtaining them, when redistribution is not allowed.

## Reproducibility Contract

- The work runs in a fresh GitHub Codespace.
- Paths are relative to the project root.
- Required packages are declared.
- Outputs can be regenerated without manual editing.
- AI-generated code or text is audited and documented.

## Grading Checklist

- Clear question or task.
- Correct and reproducible code.
- Evidence of verification.
- Clear writing and captions.
- Honest limitations and AI-use note.

## 38 Assignment 4: Polyglot Awareness & R Deepening

### 38.1 Purpose

This assignment asks you to strengthen an R analysis first, then translate a small piece of it to Python/pandas and verify that both languages give the same descriptive answer. The goal is not to become a Python expert. The goal is to become a careful bilingual reader who can audit AI-assisted translation.

### 38.2 Inputs

Use the canonical NHANES case-study CSV:

```
1 examples/nhanes-equity/data/nhanes_equity_v6.csv
```

If your notebook lives in `weeks/week05-polyglot-r-deepening/`, use:

```
1 ../../examples/nhanes-equity/data/nhanes_equity_v6.csv
```

### 38.3 Step-By-Step Tasks

1. Create `assignments/assignment04-polyglot/submission/`.
2. In that folder, create `polyglot-parity.qmd`.
3. In R, load the NHANES CSV and produce a descriptive summary of BMI by `IncomeGroup` and `Gender`.
4. Refactor the R summary into one small function.

5. Create `translation.ipynb` and translate the same summary to Python/pandas.
6. Create `helpers.py` with at least one reusable Python helper and call it from the notebook.
7. Fill in a parity table comparing row counts, filtered row counts, grouping labels, and rounded summary values.
8. Add a short AI-use note describing what the tool helped with and how you verified the output.
9. Clear notebook outputs before committing.
10. Commit and sync your work.

## 38.4 What To Submit

Submit the following files in `assignments/assignment04-polyglot/submission/`:

- `polyglot-parity.qmd`: your R analysis, R function, parity table, and AI-use note.
- `translation.ipynb`: your Python translation notebook.
- `helpers.py`: your reusable Python helper script.

Submission route: submit your repository link on Canvas after committing and syncing the files above.

## 38.5 Grading Checklist

| Criterion                                                      | Complete | Needs work |
|----------------------------------------------------------------|----------|------------|
| R pipeline loads the canonical NHANES CSV with a relative path |          |            |
| R function correctly reproduces the summary task               |          |            |

| Criterion                                                                           | Complete | Needs work |
|-------------------------------------------------------------------------------------|----------|------------|
| Python notebook translates the same analysis without changing the analytic question |          |            |
| Parity table compares row counts, groups, and rounded summary values                |          |            |
| Differences are explained before any code is changed                                |          |            |
| <b>helpers.py</b> contains a reusable function that is called from the notebook     |          |            |
| AI-use note identifies what was generated, changed, and verified                    |          |            |
| Notebook outputs are cleared before commit                                          |          |            |

| Criterion                                                        | Complete | Needs work |
|------------------------------------------------------------------|----------|------------|
| Files are committed and synced in the required submission folder |          |            |



## 39 Assignment 5: Visualization Audit and Revision

### 39.1 Purpose

Create a clear, honest health-data visualization from the NHANES Health Equity classroom dataset. You will audit a flawed or AI-assisted plot, revise it, and explain why the revision is more accurate and useful for a non-technical audience.

### 39.2 Inputs

- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Worked example: `weeks/week06-visualization/worked-example-ggplot.qmd`
- Activity prompt: `weeks/week06-visualization/activity-ai-visual-audit.qmd`

### 39.3 Step-by-Step Tasks

1. Review the Week 6 worked example.
2. Choose one comparison from the NHANES Health Equity CSV snapshot.
3. Draft an initial plot using R and `ggplot2`.
4. Audit the plot for scale choices, labels, missing context, and overclaiming.
5. Revise the plot so it supports descriptive interpretation.
6. Write a caption and short audit note.
7. If AI helped you write or revise code, document what it did and how you checked it.

## 39.4 What To Submit

Place all submission files in:

```
1 assignments/assignment05-visualization/submission/
```

Submit these files:

- `visualization.qmd`: rerunnable Quarto file that loads the CSV with a relative path and creates the final plot.
- `visualization.html`: rendered output.
- `corrected-plot.png`: exported final plot.
- `audit-note.md`: at least three issues you checked or corrected.
- `caption.md`: final figure caption with data source, statistic, exclusions, and limitation.
- `ai-use-note.md`: required if AI was used; otherwise include a one-sentence note saying AI was not used.

## 39.5 Submission Route

Submit a Canvas assignment entry that contains a link to the repository folder `assignments/assignment05-visualization/submission/`.

## 39.6 Grading Checklist

Complete:

- Uses `examples/nhanes-equity/data/nhanes_equity_v6.csv` with a relative path.
- Plot is rerunnable from the submitted Quarto file.
- Title, axes, legend, and caption match the statistic shown.
- Audit note identifies at least three meaningful issues.
- Caption states that the result is descriptive and non-causal.
- AI-use note is present and specific.

Needs work:

- Uses an absolute or machine-specific path.
- Plot title or caption overclaims causality or population inference.
- Missing caption, audit note, or AI-use note.
- Code does not run from the repository.
- Revised plot is not materially different from the draft version.

## 40 Assignment 6: EDA, Table 1, and AI Audit

### 40.1 Purpose

Produce a transparent descriptive EDA note and Table 1-style summary from the NHANES Health Equity classroom dataset. You will also audit a planted-error starter to practice catching correctness, reproducibility, and interpretation problems.

### 40.2 Inputs

- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Worked example: `weeks/week07-eda-ai-audit/worked-example-table1.qmd`
- Activity prompt: `weeks/week07-eda-ai-audit/activity-table1-audit.qmd`
- Planted-error starter: `weeks/week07-eda-ai-audit/instructor/planted-error-starter.qmd`

### 40.3 Step-by-Step Tasks

1. Load the CSV snapshot with a relative path.
2. Define and document an analysis cohort using age 20-80.
3. Create a missingness table for key variables.
4. Produce a Table 1-style summary by `IncomeGroup`.
5. Include N, BMI mean (SD), and age mean (SD).
6. Write a short methods note stating that the analysis is descriptive, unweighted, and non-causal.
7. Audit the planted-error starter and list at least five issues.
8. Categorize each issue as correctness, reproducibility, or interpretation.

## 40.4 What To Submit

Place all submission files in:

```
1 assignments/assignment06-eda-ai-audit/submission/
```

Submit these files:

- `eda-note.qmd`: rerunnable Quarto file with cohort definition, missingness table, Table 1, and methods note.
- `eda-note.html`: rendered output.
- `table1.csv`: exported Table 1 or equivalent descriptive summary.
- `audit-note.md`: at least five planted issues with category, why it matters, and corrected approach.
- `ai-use-note.md`: required if AI was used; otherwise include a one-sentence note saying AI was not used.

## 40.5 Submission Route

Submit a Canvas assignment entry that contains a link to the repository folder `assignments/assignment06-eda-ai-audit/submission/`.

## 40.6 Grading Checklist

Complete:

- Uses `examples/nhanes-equity/data/nhanes_equity_v6.csv` with a relative path.
- Cohort definition is stated before summarizing.
- Missingness is checked and reported.
- Table 1 includes N, BMI mean (SD), and age mean (SD) by group.
- Methods note avoids causal or population-level overclaims.
- Audit note identifies at least five issues across correctness, reproducibility, and interpretation.
- AI-use note is present and specific.

Needs work:

- Uses an absolute or machine-specific path.
- Cohort is unclear or inconsistent with the code.
- Rows are dropped silently without missingness documentation.
- Statistics are mislabeled or missing variability measures.
- Interpretation overclaims causality.
- Code does not render from the repository.

## Project Milestones

The term project is built through staged milestones so students practice reproducibility, feedback, and knowledge translation before the final submission.

| Milestone       | Due            | Deliverable                                                               |
|-----------------|----------------|---------------------------------------------------------------------------|
| M0              | Week 4 Monday  | Group formation and repository setup                                      |
| M1              | Week 6 Monday  | Proposal, dataset selection, provenance/stewardship statement             |
| M2              | Week 7 Monday  | Preliminary analysis, Table 1 or descriptive summary, runnable repository |
| M3              | Week 10 Monday | Draft report and dashboard-style KT product for review                    |
| M4              | Week 11 Monday | Peer review with reproducibility and auditability checks                  |
| M5              | Weeks 12-13    | In-class presentation                                                     |
| Final Portfolio | Dec 11 Friday  | Complete reproducible portfolio                                           |

## TODO

### Warning

Create detailed milestone briefs in each `milestones/` subfolder from `milestones/milestone-template.qmd`.



# Milestone Template

Use this template for project milestones M0-M5 and the Final Portfolio.

## Deliverable

- **Milestone:**
- **Due:**
- **Submit:**
- **Main artifact:**

## Required Evidence

- Repository link.
- Rendered report, dashboard, slides, or review document.
- Commit history showing meaningful progress.
- AI-use note.
- Reproducibility check result.

## Definition of Done

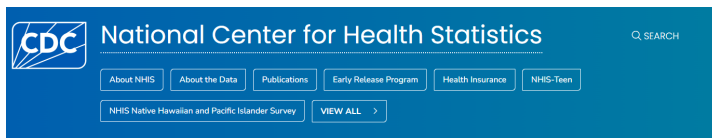
- The artifact opens or renders in the supported workflow.
- Data provenance and stewardship decisions are documented.
- The work avoids unsupported causal or predictive claims.
- A peer or TA can rerun the project from a fresh environment.

## **Part XVI**

# **Reference: Data Sources**

# Data Source: NHIS

The National Health Interview Survey (NHIS) is a cross-sectional household interview survey conducted by the US Centers for Disease Control (CDC). The survey collects information on the health status, health care access, and health behaviors of the US population.



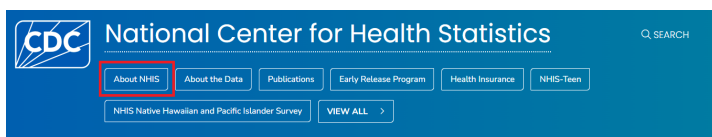
## National Health Interview Survey



[CDC website for NHIS](#)

## About NHIS

You can find more information about the NHIS survey by clicking on [About NHIS](#):



## About NHIS

For Everyone  
NOVEMBER 20, 2024

On the **About NHIS** page, you will find information about what the survey collects, data and documentation, the instructions on how to use the data, and information on related NHIS surveys.

### Overview

The National Health Interview Survey (NHIS) monitors the health of people across the country—the U.S. population—by collecting and analyzing data on a broad range of health topics. NHIS focuses on the health of U.S. children and adults.

CDC's [National Center for Health Statistics](#) (NCHS) conducts NHIS. It is the nation's largest and oldest national health survey. NHIS began collecting data in 1957.

Each year, NHIS collects data from about 27,000 adults through confidential, face-to-face interviews throughout the year. Many of these adults also provide health information about

#### ON THIS PAGE

##### Overview

[What's collected](#)

[Data and documentation](#)

[How the data are used](#)

[Related NHIS surveys](#)

You can click on each of the links or scroll down to find information on each of these topics one by one. For example, if we click on the **How the data are used** link, we will find information as follows:

### How the data are used

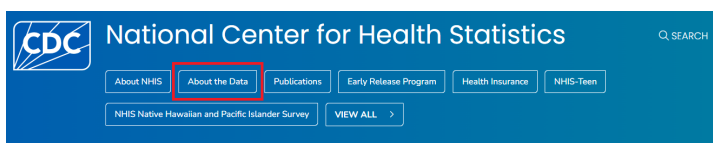
NCHS staff use NHIS data to produce official national health estimates. One of NHIS's key strengths is its ability to produce estimates for many subgroups of the U.S. population. NHIS can do this because it collects data from such a large group of people.

NHIS health statistics provide health information by—

- Demographic characteristics, like age and marital status
- Geographic location, like region and metro area size
- Social and economic circumstances, like employment status and poverty level

## About the Data

If you click on **About the Data** link, you will find information about the NHIS questionnaires, datasets, and documentation:



Under the **Overview** tab, you will find information about different NHIS cycles and the questionnaires, datasets, and documentation available in each cycle. As an example, let's click on the **2023 NHIS Questionnaires, Datasets, and Documentation** link to see the details for the 2023 NHIS:

[NHIS Data, Questionnaires and Related Documentation](#)

## Overview

Resources on this page provide information about the [National Health Interview Survey](#) (NHIS) data collection, its participants, and its protocols. It also provides analytic notes and references. They will help you understand, analyze, and use NHIS data.

Find survey instruments, data, and other resources by year at—

- [2025 NHIS Questionnaires, Datasets, and Documentation](#)
- [2024 NHIS Questionnaires, Datasets, and Documentation](#)
- [2023 NHIS Questionnaires, Datasets, and Documentation](#)
- [2022 NHIS Questionnaires, Datasets, and Documentation](#)
- [2021 NHIS Questionnaires, Datasets, and Documentation](#)
- [2020 NHIS Questionnaires, Datasets, and Documentation](#)
- [2019 NHIS Questionnaires, Datasets, and Documentation](#)

## ON THIS PAGE

### Overview

[Before 2019](#)

[Data linked to NHIS data](#)

[Other NHIS survey data](#)

[Related surveys](#)

[Related pages](#)

## NHIS Questionnaires, Datasets, and Documentation

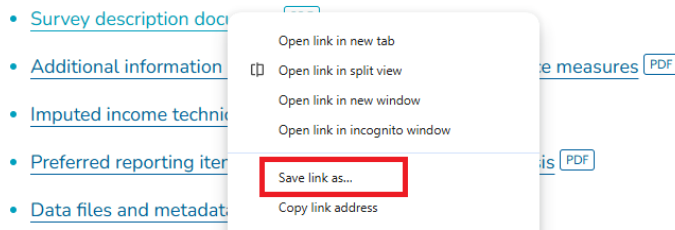
To view the description of the 2023 NHIS, click on the **Survey description document** link:

## Using our data

- [Survey description document](#) [PDF](#)

This will open the PDF of the 2023 NHIS survey description. Alternatively, you can right-click on the link, click on **Save link as...** and save the PDF file on your computer for future reference.

## Using our data



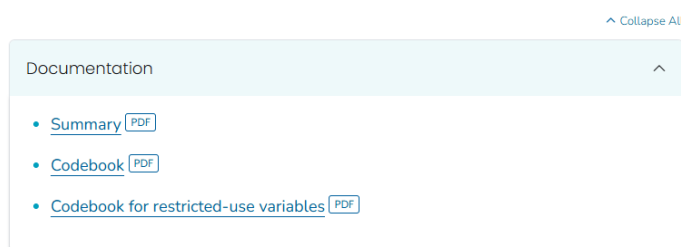
Similarly, you can open and save the questionnaires and related materials under the **Questionnaires and survey materials** tab.

In the recent NHIS, data are available in 5 categories: interview data for adults, interview data for children, data for parent-child pairs, imputed income, and paradata. You can click each link for documentation to see a summary of the data and the codebook.

Data are available in 5 categories:

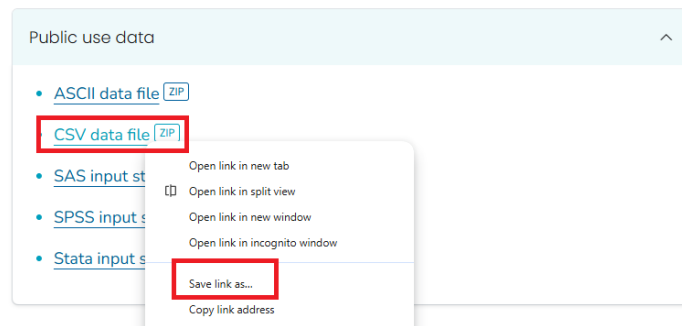
- Sample adult interview
- Sample child interview
- Parent-child pair data
- Imputed income
- Paradata

## Sample adult interview

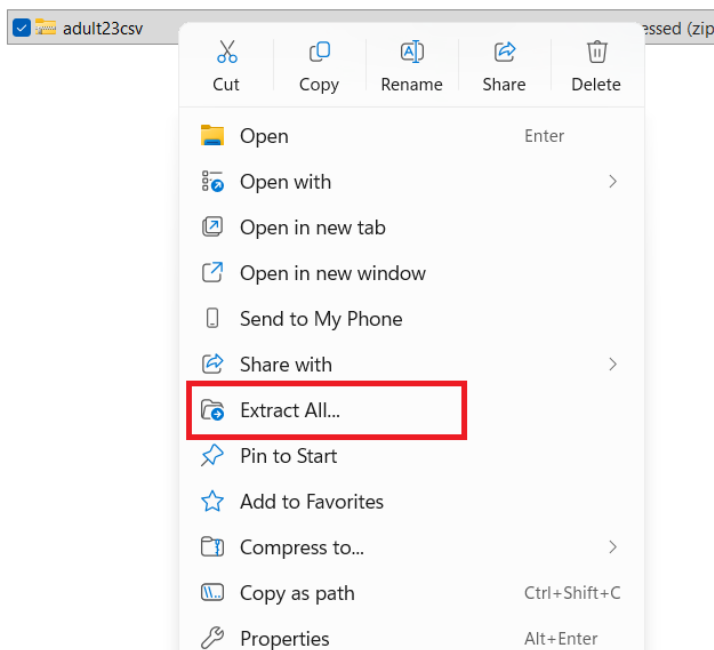


## NHIS Data

The data files are stored on the NHIS website in different formats. You can save the dataset in any statistical package that supports these file formats, e.g., ASCII, CSV, SAS. Let us save the `adult` dataset in the CSV format, which is in a zip file.



The next step is to unzip the file to extract the CSV. Depending on your computer, you can unzip the file with or without software. For example, if you are using Windows, you can right-click the zip file and select **Extract All...** to unzip it.



Once unzipped, you will find two files: the CSV file for the dataset and a readme file with some instructions. Below is the example for the `adult` dataset:

|                                                                                           |                       |           |
|-------------------------------------------------------------------------------------------|-----------------------|-----------|
|  adult23 | Microsoft Excel Co... | 28,738 KB |
|  readme  | Text Document         | 3 KB      |

To open the datafile in Excel, - you can double click on the CSV file, or - right click on the CSV file and click on **Open**.

Later, you will learn how to open the data file in different statistical software. For Excel, it would take a few seconds/minutes because of the large size of the dataset. Once you open the data file, you will see the data in a tabular format with rows and columns.

| URBRRL | RATCAT_A | INCTCFLG_A | IMPINCFLG_A | LANGSPECR_A | LANGSOC_A | LANGDOC_A | LANGMED_A | LANGHM_A | PPSU | PSTRAT |
|--------|----------|------------|-------------|-------------|-----------|-----------|-----------|----------|------|--------|
| 3      | 4        | 0          | 0           |             |           |           |           |          | 2    | 103    |
| 4      | 8        | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 14       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 10       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 5        | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 14       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 3      | 12       | 0          | 0           |             |           |           |           |          | 2    | 103    |

- Each row represents a person.
- Each column represents a variable.
- The first row contains the variable names. The remaining rows contain the data for each person.

## Use of the codebook

To understand the variables in the dataset, you can refer to the codebook. The codebook provides information about the variable names, labels, and values. For example, we want to understand the variable `SEX_A`:



| FQ      | FR    | FS     | FT    | FU       | FV        |
|---------|-------|--------|-------|----------|-----------|
| SRVY_YR | SEX_A | AGEP_A | AGE65 | ASTATNEV | MENTHOL C |
| 2023    | 1     | 67     |       | 1        |           |
| 2023    | 1     | 73     |       | 1        |           |
| 2023    | 1     | 48     |       | 1        |           |
| 2023    | 2     | 42     |       | 1        |           |
| 2023    | 2     | 50     |       | 1        |           |
| 2023    | 2     | 46     |       | 1        |           |
| 2023    | 2     | 36     |       | 1        |           |
| 2023    | 1     | 44     |       | 1        |           |

We can look for `SEX_A` in the codebook, which will provide us with the variable label and the values for that variable (e.g., 1 for Male, 2 for Female, and so on).

17 / 653 | — 117% + |     

---

2023 NATIONAL HEALTH INTERVIEW SURVEY (NHIS)  
Codebook for Sample Adult File (Version: 24 June 2024)  
PUBLIC USE

---

Variable Information:

|                       |                             |
|-----------------------|-----------------------------|
| Variable:             | SEX_A                       |
| Module:               | Roster                      |
| Section:              | HHC : Household Composition |
| File(s):              | Adult                       |
| Data Type:            | Numeric                     |
| Length:               | 1                           |
| Question Text:        |                             |
| Description:          | Sex of Sample Adult         |
| Recode:               | Yes                         |
| Universe:             | HHSTAT_A=1                  |
| Universe Description: | Sample adults 18+           |
| Sources:              |                             |
| Question ID:          |                             |
| Keywords:             |                             |
| Notes:                |                             |
| Evaluation Report:    |                             |

Unweighted frequencies:

| SEX_A | Sex of Sample Adult |           |         |
|-------|---------------------|-----------|---------|
| Code  | Description         | Frequency | Percent |
| 1     | Male                | 13457     | 45.58   |
| 2     | Female              | 16059     | 54.40   |
| 7     | Refused             | 4         | 0.01    |
| 8     | Not Ascertained     | 0         | 0.00    |
| 9     | Don't Know          | 2         | 0.01    |

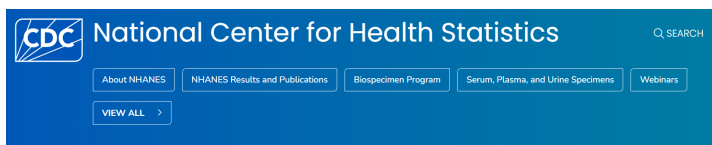
Frequency Missing:

There are so many variables in the dataset, and it is not possible to understand all of them at once. You can refer to the codebook to understand the variables that you are interested in for your analysis. For example, the `WTFA_A` variable is the sample adult weight variable used to calculate weighted estimates

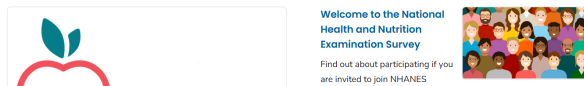
for the US adult population. Similarly, the `HYPEV_A` variable is the hypertension ever variable, indicating whether a doctor or other health professional has ever told the person they have hypertension. Again, you can refer to the codebook to understand the variables that you are interested in.

# Data Source: NHANES

The National Health and Nutrition Examination Survey (NHANES) is a cross-sectional survey conducted by the US Centers for Disease Control (CDC). The survey collects information on the health and nutritional status of the US population. The survey includes interviews, physical examinations, and laboratory tests.



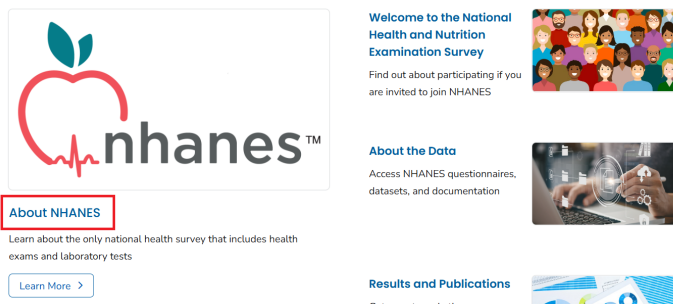
## National Health and Nutrition Examination Survey



[CDC website for NHANES](#)

## About NHANES

You can find more information about the NHANES survey by clicking on [About NHANES](#):



## About NHANES



For Everyone

DECEMBER 18, 2024

AT A GLANCE

- NHANES is the National Health and Nutrition Examination Survey.
- NHANES is a national survey that measures the health and nutrition of adults and children in the United States.
- NHANES is the only national health survey that includes health exams and laboratory tests for participants of all ages.



On the **About NHANES** page, you will find information about what the survey collects, data and documentation, the instructions on how to use the data, and information on related NHANES surveys.

### Overview

The National Health and Nutrition Examination Survey (NHANES) collects data about the health of adults and children in the United States. We also collect data about what participants eat, drink, and take as supplements to determine how many nutrients are in their diet. These dietary interviews and blood tests help us measure the nutritional status of U.S. adults and children.

CDC's [National Center for Health Statistics](#) (NCHS) conducts NHANES. NHANES is the only national health survey that includes health exams, laboratory tests, and dietary interviews for participants of all ages.

#### ON THIS PAGE

- Overview
- What's collected
- Data and documentation
- How the data are used
- Program director
- Contact NHANES

You can click on each of the links or scroll down to find information on each of these topics one by one. For example, if we click on the **Data and documentation** link, we will find information as follows:

### Data and documentation

NCHS [publishes reports](#) featuring analysis of NHANES data. Selected estimates also are available as part of [interactive data visualizations](#) that provide tables and charts.

#### Data files

[NHANES data files](#) and [related documentation](#) are publicly available to download and analyze. The related documentation can help researchers understand and use NHANES data. All potentially identifiable information has been removed to ensure the confidentiality of participants and their households.

[NHANES restricted data files](#) are available through the NCHS Research Data Center (RDC) for a fee. To access restricted NHANES data, researchers must submit requests to the

RELATED PAGES

- What NHANES Covers and How It Works
- Who Participates In NHANES
- How People Use NHANES Data
- What NHANES Data Have Achieved
- Ethics Review Board Approval

VIEW ALL

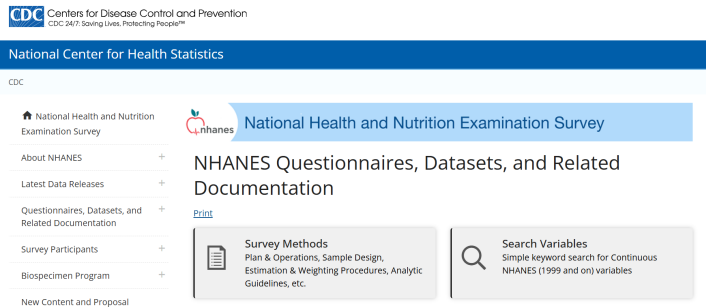
National Health and Nutrition Examination Survey

## Survey Methods and Analytic Guidelines

If you click on **NHANES data files and related documentation** link, you will find information about the NHANES questionnaires, datasets, and documentation:

## Data files

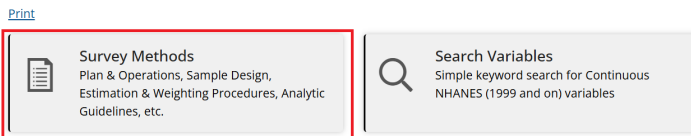
[NHANES data files and related documentation](#) are publicly available to download and analyze. The related documentation can help researchers understand and use NHANES data. All potentially identifiable information has been removed to ensure the confidentiality of participants and their households.



You can click **Survey Methods** to find information on survey methods and analytic guidelines. The **Survey Methods and Analytic Guidelines** page provides information about the plan and operation of the NHANES survey, the sampling design, methods for calculating weights and variance estimation, and guidelines for analyzing NHANES data, along with links to related resources.

[NHANES data files and related documentation](#)

## NHANES Questionnaires, Datasets, and Related Documentation



## Search Variables

If you click on **Search Variables** link, you will find a search tool to search for variables in the NHANES datasets (1999 and on).

## NHANES Questionnaires, Datasets, and Related Documentation

[Print](#)

**Survey Methods**  
Plan & Operations, Sample Design,  
Estimation & Weighting Procedures, Analytic  
Guidelines, etc.

**Search Variables**  
Simple keyword search for Continuous  
NHANES (1999 and on) variables

You can search for variables by entering keywords in the search box. For example, if you enter **diabetes** in the search box, you will find a list of variables related to diabetes in the NHANES datasets.

### Search Variables

[Print](#)

Search for words and "quoted phrases" in selected fields (Data File Name, SAS Label, Variable Description, Variable Name) of Continuous NHANES (1999 and on) variables that have published documentation. *Multiply imputed Dual Energy X-Ray Absorptiometry variables are not included.*

Refer to the [Survey Content Brochure](#) for the full content of Continuous NHANES.

|                         |                                            |
|-------------------------|--------------------------------------------|
| Search Term             | <input type="text" value="diabetes"/>      |
| Fields to Search        | <input type="text" value="All"/>           |
| Sort By                 | <input type="text" value="Variable Name"/> |
| Limited Access          | <input type="text" value="Exclude"/>       |
| Release Cycle           | <input type="text" value="All"/>           |
| Search Result Page Size | <input type="text" value="50"/>            |

Search

Number of variables found: 459

| Variable Name | SAS Label                            | Variable Description                                                                                                                            | Data File Name   | Component Link                          |
|---------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_D) | <a href="#">2005-2006 Questionnaire</a> |
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_E) | <a href="#">2007-2008 Questionnaire</a> |
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_F) | <a href="#">2009-2010 Questionnaire</a> |
| DID040        | Age when first told you had          | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that                                              | Diabetes (DIQ_G) | <a href="#">2011-2012 Questionnaire</a> |

You can restrict your search to a specific NHANES cycle by selecting the cycle from the drop-down menu. For example, if you select the **2017-2018** cycle, you will find a list of variables related to diabetes in the 2017-2018 NHANES dataset.

|                         |           |
|-------------------------|-----------|
| Limited Access          | Exclude   |
| Release Cycle           | 2017-2018 |
| Search Result Page Size | 50        |

Search

Number of variables found: 58

| Variable Name | SAS Label                                | Variable Description                                                                                    | Data File Name              | Component Link                          |
|---------------|------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------|
| DIQ175C       | Age                                      | Why (Do you/Does SP) think (you are/he is/she is) at risk for diabetes or prediabetes? [Anything else?] | Diabetes (DIQ_J)            | <a href="#">2017-2018 Questionnaire</a> |
| RHQ163        | Age told you had diabetes while pregnant | How old (were you/was SP) when (you were/she was) first told (you/she) had diabetes during a pregnancy? | Reproductive Health (RHQ_J) | <a href="#">2017-2018 Questionnaire</a> |

## NHANES Questionnaires, Datasets, and Documentation

Let's explore the questionnaires, datasets, and documentation for the 2017-2018 NHANES cycle. You need to click on **NHANES 2017-2018** under the **Continuous NHANES** tab:

### NHANES Questionnaires, Datasets, and Related Documentation

[Print](#)



**Survey Methods**  
Plan & Operations, Sample Design, Estimation & Weighting Procedures, Analytic Guidelines, etc.



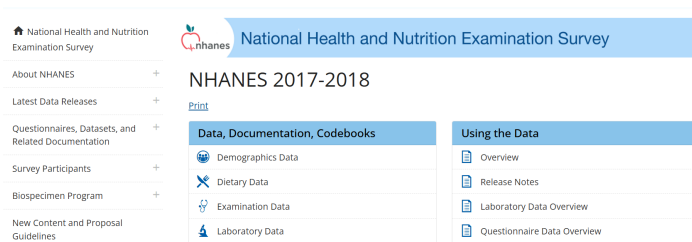
**Search Variables**  
Simple keyword search for Continuous NHANES (1999 and on) variables

[» Guidelines for High Quality Analyses of NHANES Data](#)

#### Continuous NHANES

|                  |                        |                                          |                  |
|------------------|------------------------|------------------------------------------|------------------|
| NHANES 2025-2026 | NHANES 08/2021-08/2023 | NHANES 2017-March 2020 Pre-Pandemic Data |                  |
| NHANES 2019-2020 | NHANES 2017-2018       | NHANES 2015-2016                         | NHANES 2013-2014 |

This will open the page for the 2017-2018 NHANES cycle, where you can find information on the survey description, questionnaires, datasets, and documentation.



There are six categories of data available for the 2017-2018 NHANES cycle: Demographics, Dietary, Examination, Laboratory, Questionnaire, and Limited Access data. You can click each category to find the datasets and documentation available in that category. For example, if you click on **Demographics Data**, you will find a list of datasets and documentation related to demographics data in the 2017-2018 NHANES cycle.

Data are available in 6 categories:

- Demographics
- Dietary
- Examination
- Laboratory
- Questionnaire
- Limited Access

## NHANES 2017-2018

[Print](#)

| Data, Documentation, Codebooks | Using the Data                         |
|--------------------------------|----------------------------------------|
| Demographics Data              | Overview                               |
| Dietary Data                   | Release Notes                          |
| Examination Data               | Laboratory Data Overview               |
| Laboratory Data                | Questionnaire Data Overview            |
| Questionnaire Data             | Examination Data Overview              |
| Limited Access Data            | Survey Methods and Analytic Guidelines |

## 2017-2018 Demographics Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Demographics Variable List](#)
- [Questionnaire Instruments](#)
- [Exam Procedure Manuals](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                           | Doc File                   | Data File                                  | Date Published |
|------------------------------------------|----------------------------|--------------------------------------------|----------------|
| Demographic Variables and Sample Weights | <a href="#">DEMO_1.Doc</a> | <a href="#">DEMO_1 Data [XPT - 3.3 MB]</a> | February 2020  |

There is a **Doc File** available for each dataset that provides information about the dataset, including the variables it contains, their coding, and any special considerations for using it. It is important to read the **Doc** file before using the dataset to understand the data structure and how to use it properly. The **Data File** contains the dataset data in **.XPT** format, which is a SAS Transport file, not a CSV.



## 2017-2018 Demographics Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Demographics Variable List](#)
- [Questionnaire Instruments](#)
- [Exam Procedure Manuals](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                           | Doc File                   | Data File                                  | Date Published |
|------------------------------------------|----------------------------|--------------------------------------------|----------------|
| Demographic Variables and Sample Weights | <a href="#">DEMO_J Doc</a> | <a href="#">DEMO_J Data [XPT - 3.3 MB]</a> | February 2020  |

## Data

You need to click on the [DEMO\\_J Data \[XPT - 3.3 MB\]](#) link to download the dataset. Note that you cannot easily open an .XPT file in Excel or Google Sheets. In the real world, data isn't always in a friendly format. For this week, let's download the `NHANES_Demo_Converted.csv` file from our Canvas page to look at the rows. Later, we will use an AI co-pilot to write R code that reads .XPT files automatically.

Let's open the `NHANES_Demo_Converted.csv` file in Excel and see the first few rows:

|    | A     | B        | C        | D        | E        | F        | G        |
|----|-------|----------|----------|----------|----------|----------|----------|
| 1  | SEQN  | SDDSRVYR | RIDSTATR | RIAGENDR | RIDAGEYR | RIDAGEMN | RIDRETH1 |
| 2  | 93703 | 10       | 2        | 2        | 2        | NA       | 5        |
| 3  | 93704 | 10       | 2        | 1        | 2        | NA       | 3        |
| 4  | 93705 | 10       | 2        | 2        | 66       | NA       | 4        |
| 5  | 93706 | 10       | 2        | 1        | 18       | NA       | 5        |
| 6  | 93707 | 10       | 2        | 1        | 13       | NA       | 5        |
| 7  | 93708 | 10       | 2        | 2        | 66       | NA       | 5        |
| 8  | 93709 | 10       | 2        | 2        | 75       | NA       | 4        |
| 9  | 93710 | 10       | 2        | 2        | 0        | 11       | 3        |
| 10 | 93711 | 10       | 2        | 1        | 56       | NA       | 5        |
| 11 | 93712 | 10       | 2        | 1        | 18       | NA       | 1        |
| 12 | 93713 | 10       | 2        | 1        | 67       | NA       | 3        |
| 13 | 93714 | 10       | 2        | 2        | 54       | NA       | 4        |
| 14 | 93715 | 10       | 2        | 1        | 71       | NA       | 5        |

- Each row represents a person.
- Each column represents a variable.
- The first row contains the variable names. The remaining rows contain the data for each person.

For example, `SEQN` is the unique identifier for each person in the dataset, and `RIAGENDR` is the variable for gender.

## Use of the Codebook

To understand the variables in the dataset, you can refer to the codebook. The codebook provides information about the variable names, labels, and values. The Doc File explained earlier contains the codebook for the dataset. You need to click on the DEMO\_J Doc link to open the codebook on a new page. Let's we want to understand the variables SEQN and RIAGENDR:

### National Health and Nutrition Examination Survey

2017-2018 Data Documentation, Codebook, and Frequencies

Demographic Variables and Sample Weights (DEMO\_J)

Data File: DEMO\_J.xpt

First Published: February 2020

Last Revised: NA

#### Component Description

The demographics file provides individual, family, and household-level information on the following topics:

- Survey participant's household interview and examination status;

#### TABLE OF CONTENTS

- Component Description
- Eligible Sample
- Interview Setting and Mode of Administration
- Quality Assurance & Quality Control
- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender
  - RIDAGEYR - Age in years at screening
  - RIDAGEMN - Age in months at screening - 0 to 24 mos

You need to click on the **SEQN - Respondent sequence number** link or scroll down to find the information about the SEQN variable.

#### SEQN - Respondent sequence number

**Variable Name:** SEQN  
**SAS Label:** Respondent sequence number  
**English Text:** Respondent sequence number.  
**Target:** Both males and females 0 YEARS - 150 YEARS

#### SDDSRVYR - Data release cycle

**Variable Name:** SDDSRVYR  
**SAS Label:** Data release cycle  
**English Text:** Data release cycle

#### TABLE OF CONTENTS

- Component Description
- Eligible Sample
- Interview Setting and Mode of Administration
- Quality Assurance & Quality Control
- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender

Similarly, you can click on the **RIAGENDR - Gender** link or scroll down to find the information about the RIAGENDR variable.

#### RIAGENDR - Gender

**Variable Name:** RIAGENDR  
**SAS Label:** Gender  
**English Text:** Gender of the participant.  
**Target:** Both males and females 0 YEARS - 150 YEARS

| Code or Value | Value Description | Count | Cumulative | Skip to Item |
|---------------|-------------------|-------|------------|--------------|
| 1             | Male              | 4557  | 4557       |              |
| 2             | Female            | 4697  | 9254       |              |
| .             | Missing           | 0     | 9254       |              |

#### RIDAGEYR - Age in years at screening

**Variable Name:** RIDAGEYR

#### TABLE OF CONTENTS

- Component Description
- Eligible Sample
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- Quality Assurance & Quality Control
- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender
  - RIDAGEYR - Age in years at screening
  - RIDAGEMN - Age in months at screening - 0 to 24 mos

As you can see, the **RIAGENDR** variable has two values: 1 for Male and 2 for Female, without any missing values. You can also see the frequency of each value in the dataset. These numbers/frequencies should exactly match the frequencies in your dataset. You can use the codebook to understand other variables in the dataset. Similarly, you can use the other data files (e.g., Dietary) and their codebook to understand the variables in those datasets. Unlike the demographic file, which uses only one dataset and one codebook, the other files use multiple datasets and codebooks. As an example, if you click on the **Questionnaire Data** link, you will find multiple datasets and codebooks for questionnaire data:

## 2017-2018 Questionnaire Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Questionnaire Variable List](#)
- [Questionnaire Instruments](#)
- [2017-2018 Questionnaire Data Overview](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                    | Doc File                     | Data File                                   | Date Published |
|-----------------------------------|------------------------------|---------------------------------------------|----------------|
| Acculturation                     | <a href="#">ACO.J.Doc</a>    | <a href="#">ACO.J.Data [XPT - 396.3 KB]</a> | February 2020  |
| Alcohol Use                       | <a href="#">ALO.J.Doc</a>    | <a href="#">ALO.J.Data [XPT - 434.4 KB]</a> | December 2020  |
| Audiometry                        | <a href="#">AUO.J.Doc</a>    | <a href="#">AUO.J.Data [XPT - 3.9 MB]</a>   | July 2020      |
| Blood Pressure & Cholesterol      | <a href="#">BPO.J.Doc</a>    | <a href="#">BPO.J.Data [XPT - 531.8 KB]</a> | February 2020  |
| Cardiovascular Health             | <a href="#">CDO.J.Doc</a>    | <a href="#">CDO.J.Data [XPT - 518.7 KB]</a> | February 2020  |
| Consumer Behavior                 | <a href="#">CBO.J.Doc</a>    | <a href="#">CBO.J.Data [XPT - 435.4 KB]</a> | April 2021     |
| Consumer Behavior Phone Follow-up | <a href="#">CBOPFA.J.Doc</a> | <a href="#">CBOPFA.J.Data [XPT - 2.7]</a>   | June 2021      |

# Data Source: BRFSS

The Behavioral Risk Factor Surveillance System (BRFSS) is a cross-sectional survey conducted by the US Centers for Disease Control (CDC). The survey collects information on health-related risk behaviors, chronic health conditions, and use of preventive services among US adults. BRFSS completes more than 400,000 adult interviews each year, making it the largest continuously conducted health survey system in the world.



[CDC website for BRFSS](#)

## About BRFSS

You can find more information about the BRFSS survey by clicking on [BRFSS FAQs](#):



On the **BRFSS FAQs** page, you will find information about the BRFSS survey, including what the survey collects, how the survey is conducted, and how to use the data.

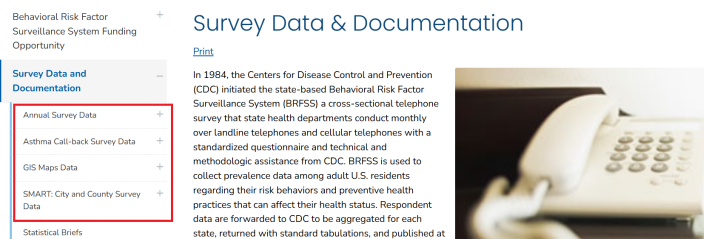
## Survey Data & Documentation

If you click on **Survey Data & Documentation** link, you will find information about the BRFSS questionnaires, datasets, and documentation:



There are four types of data files available for download: 1) annual data, 2) asthma call-back data, 3) GIS map data, and 4) SMART data. The annual data files include information on health-related risk behaviors, chronic health conditions, and use of preventive services among US adults. The asthma call-back data files include information on asthma-related health outcomes and healthcare utilization among adults with asthma. The GIS map data files include information on the geographic distribution of the data. The SMART (Selected Metropolitan/Micropolitan Area Risk Trends) data provide prevalence rates for selected conditions and behaviors for cities and their surrounding counties.

## BRFSS Survey Data & Documentation



For the annual data, you can click on the **Annual Survey Data** link to find information about the annual data files, including how to access the data, how to use the data, and how to analyze the data. There are two types of annual data files available for download: 1) the raw data files and 2) the prevalence and trends database.

## Annual Survey Data

There are separate links for each year of data, and you can click on each link to find information about the data for that year. For example, if you click on the **2023 Annual Survey Data** link, you will find information about the 2023 annual data files, including how to access the data, how to use the data, and how to analyze the data.

[Annual Survey Data](#)

## 2023 BRFSS Survey Data and Documentation

[Print](#)

The 2023 BRFSS data continue to reflect the changes initially made in 2011 for weighting methodology (raking) and adding cell-phone-only respondents. The aggregate BRFSS combined landline and cell phone data set is built from the landline and cell phone data submitted for 2023 and includes data from 48 states, the District of Columbia, Guam, Puerto Rico, and the US Virgin Islands. During 2023, Kentucky and Pennsylvania were unable to collect enough data to meet the minimum requirements to be included in this public data set. The data set has been modified to comply with President Trump's executive orders. There are some missing values which may appear to be inconsistencies in the data based on a respondents' answers to questions that were removed.

### 2023 Survey Data Information

[2023 BRFSS Overview CDC](#) [PDF – 213 KB]

The 2023 BRFSS Survey Data and Documentation page provides information about three resources: 1) data information such as the codebook and sampling weights, 2) data files in ASCII and SAS XPT transport formats, and 3) SAS resources. To download/save the data files, click the specific format you want. For example, if you click on the 2023 BRFSS Data (SAS Transport Format) link, the annual survey data for 2023 will be downloaded in a ZIP file that contains the dataset in a .XPT format.

### Data Files

There are 433,323 records for 2023. More information on participation is available in the [states conducting surveillance, by year table](#). The data files are provided in ASCII and SAS Transport formats. The September update includes updated race/ethnicity calculated variables for Mississippi. The February update includes changes to the data set which has been modified to comply with President Trump's executive orders. There are some missing values which may appear to be inconsistencies in the data based on a respondents' answers to questions that were removed.

[2023 BRFSS Data \(ASCII\)](#) [ZIP – 41.5 MB]

February, 2025

This file for the combined landline and cell phone data set is in ASCII format. It has a fixed record length of 2111 positions.

[2023 BRFSS Data \(SAS Transport Format\)](#) [ZIP – 64.3 MB]

February, 2025

This file for the combined landline and cell phone data set was exported from SAS V9.4 in the XPT transport

You need to unzip the file to extract the data file in .XPT format. However, opening the data file in Excel or Google Sheets is a bit tricky. The SAS XPT (or ASCII) data file needs to be converted/formatted appropriately before it can be opened in Excel or Google Sheets. Many statistical software packages, such as R, SAS, or STATA, can read the data file in .XPT format. Below is the data file, where the column names are the variable names, and the rows are the observations (i.e., survey respondents):

|    | STATE FIPS CODE | FILE MONTH | INTERVIEW DATE | INTERVIEW MONTH | INTERVIEW DAY | INTERVIEW YEAR | FINAL DISPOSITION | ANNUAL SEQUENCE NUMBER | PRIMARY SAMPLING UNIT | CORRECT TELEPHONE NUMBER? | PRIVATE RESIDENCE? | DO YOU LIVE IN COLLEGE HOUSING? | RESIDENT OF STATE | CELLULAR TELEPHONE | ARE YOU 18 YEARS OF AGE OR OLDER? |
|----|-----------------|------------|----------------|-----------------|---------------|----------------|-------------------|------------------------|-----------------------|---------------------------|--------------------|---------------------------------|-------------------|--------------------|-----------------------------------|
| 1  | 1               | 1          | 1 03012023     | 03              | 01            | 2023           | 1100              | 2023000001             | 2023000001            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 2  | 1               | 1          | 1 01062023     | 01              | 06            | 2023           | 1100              | 2023000002             | 2023000002            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 3  | 1               | 1          | 1 03082023     | 03              | 08            | 2023           | 1100              | 2023000003             | 2023000003            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 4  | 1               | 1          | 1 03062023     | 03              | 06            | 2023           | 1100              | 2023000004             | 2023000004            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 5  | 1               | 1          | 1 01062023     | 01              | 06            | 2023           | 1100              | 2023000005             | 2023000005            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 6  | 1               | 1          | 1 01092023     | 01              | 09            | 2023           | 1100              | 2023000006             | 2023000006            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 7  | 1               | 1          | 1 0212023      | 03              | 21            | 2023           | 1100              | 2023000007             | 2023000007            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 8  | 1               | 1          | 1 01062023     | 01              | 06            | 2023           | 1100              | 2023000008             | 2023000008            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 9  | 1               | 1          | 1 01152023     | 03              | 15            | 2023           | 1100              | 2023000009             | 2023000009            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 10 | 1               | 1          | 1 01082023     | 01              | 08            | 2023           | 1100              | 2023000010             | 2023000010            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 11 | 1               | 1          | 1 03082023     | 03              | 08            | 2023           | 1100              | 2023000011             | 2023000011            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 12 | 1               | 1          | 1 01082023     | 01              | 08            | 2023           | 1100              | 2023000012             | 2023000012            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 13 | 1               | 1          | 1 03012023     | 03              | 01            | 2023           | 1100              | 2023000013             | 2023000013            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 14 | 1               | 1          | 1 01062023     | 01              | 06            | 2023           | 1200              | 2023000014             | 2023000014            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 15 | 1               | 1          | 1 0312023      | 03              | 21            | 2023           | 1100              | 2023000015             | 2023000015            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 16 | 1               | 1          | 1 0112023      | 03              | 13            | 2023           | 1100              | 2023000016             | 2023000016            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |

To understand the variables in the dataset, you can refer to the codebook:

### 2023 Survey Data Information

[2023 BRFSS Overview CDC](#) [PDF – 213 KB]

Provides information on the background, design, data collection and processing, and the statistical and analytical issues for the combined landline and cell phone data set.

[2023 BRFSS Codebook CDC](#) [ZIP – 3 MB]

Codebook for the file showing variable name, location, and frequency of values for all reporting areas combined for the combined landline and cell phone data set.

You need to unzip the file to extract the codebook in HTML format. The codebook provides information about the variable names, labels, and values. For example, if you want to understand the variable **Drinking and Driving**, you can find the variable name, label, and values in the codebook:

| Label: Drinking and Driving                                                                              |                                                |           |            |                     |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------|------------|---------------------|
| Section Name: Calculated Variables                                                                       |                                                |           |            |                     |
| Module Number: 15                                                                                        |                                                |           |            |                     |
| Question Number: 3                                                                                       |                                                |           |            |                     |
| Column: 2110                                                                                             |                                                |           |            |                     |
| Type of Variable: Num                                                                                    |                                                |           |            |                     |
| SAS Variable Name: _DRNKDRV                                                                              |                                                |           |            |                     |
| Question Prologue:                                                                                       |                                                |           |            |                     |
| Question: Drinking and Driving (Reported having driven at least once when perhaps had too much to drink) |                                                |           |            |                     |
| Value                                                                                                    | Value Label                                    | Frequency | Percentage | Weighted Percentage |
| 1                                                                                                        | Have driven after having too much to drink     | 6,192     | 1.43       | 1.36                |
| 2                                                                                                        | Have not driven after having too much to drink | 205,541   | 47.43      | 45.81               |
| 9                                                                                                        | Don't know/Not Sure/Refused/Missing            | 221,590   | 51.14      | 52.83               |

You can similarly download the asthma call-back, GIS map, and SMART datasets. To understand the variables in those datasets, you can refer to the codebook for each dataset.

[Asthma Call-back Data](#)

[GIS Data](#)

[SMART Data](#)



## Prevalence and Trends Database

The BRFSS prevalence and trends tool allows users to view and download prevalence estimates through charts, graphs, and maps. You can explore the estimates by selecting the topic or location. For example, if you select by **topic**, e.g., for alcohol consumption for Alabama for the year 2024, you need to select the options from the dropdown menu and click on the **View Results** button to see the results:

**Prevalence and Trends Database**

**CDC** BRFSS Prevalence & Trends Data

Per a court order, HHS is required to restore this website to its version as of 12:00 AM on January 29, 2025. Information on this page may be modified and/or removed in the future subject to the terms of the court's order and implemented consistent with applicable law. Any information on this page promoting gender ideology is extremely inaccurate and disconnected from truth. The Trump Administration rejects gender ideology due to the harms and divisiveness it causes. This page does not reflect reality and therefore the Administration and this Department reject it.

[Print](#)

The enhanced BRFSS Prevalence & Trends Tool allows you to explore prevalence data by location or topic.

- Just like the original tool, this version allows you to view and download prevalence estimates through charts, graphs, and maps.
- In addition, this version also provides access to prevalence estimates from the BRFSS core data at the state level as well as data from Selected Metropolitan/Micropolitan Area Risk Trends (SMART).

These prevalence estimates have been updated to include both crude and age-adjusted prevalence.

The tool will continue to be updated as new functions and data become available.

**Explore by Location** Explore by Topic

Explore bar and trendline graphs for survey questions by selecting a state, territories or Metropolitan/Micropolitan Statistical Areas (MMSAs).

| Dataset    | Location         | Class               | Topic          | Year |
|------------|------------------|---------------------|----------------|------|
| Nationwide | States - Alabama | Alcohol Consumption | Binge Drinking | 2024 |

Question: Binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion) (variable calculated from one or more BRFSS questions)

**View Results** Clear Filters

The results will show the (i) prevalence estimates for alcohol consumption in Alabama for the year 2024, and (ii) the trend of alcohol consumption in Alabama over time.

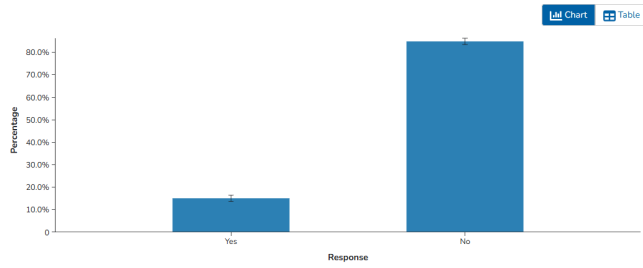
[View Results](#) [Clear Filters](#)

Binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion) (variable calculated from one or more BRFSS questions)

*Alcohol Consumption: Binge Drinking*

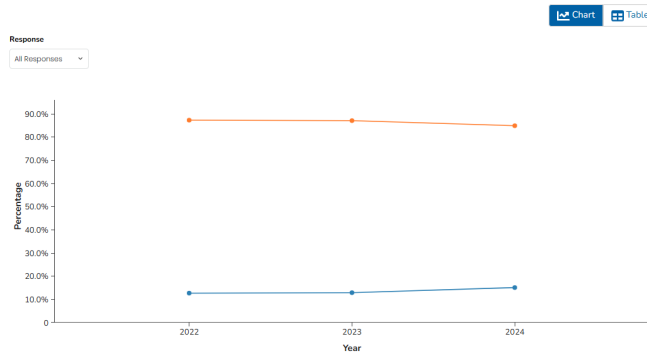
View By: Overall Data Type: Crude Prevalence

2024, Alabama



The ordering of subcategory rows in the table view may vary across responses.

All Available Years, Alabama



# Data Source: CCS

The Canadian Cannabis Survey (CCS) is an annual survey conducted by Health Canada to gather information on cannabis use among people aged 16 years and older living in all Canadian provinces and territories. The survey collects data on various aspects of cannabis consumption, including their knowledge, attitudes and behaviours towards cannabis use.

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## Canadian Cannabis Survey (CCS) 2024 Public Use Microdata File (PUMF)

Have your say

\*\*\*\*\*

[Rate this dataset](#)

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The 2024 Canadian Cannabis Survey (CCS) recruited people aged 16 years and older living in all Canadian provinces and territories. Respondents completed a survey on patterns of cannabis use for medical and non-medical purposes, as well as their knowledge, attitudes and behaviours towards cannabis use. A total of 11,666 respondents completed online surveys between April 4 and July 2, 2024. Summary results are available here: <https://www.canada.ca/en/health-canada/services/drugs-medication/cannabis/research-data/canadian-cannabis-survey-2024-summary.html>

Additional

The Office Cannabis Science and Surveillance releases the PUMF for CCS 2024 to allow

More than 11,000 respondents participated in the 2024 survey. The study provides valuable insights into cannabis use patterns, the cannabis market, and issues of public safety. The survey results are published on Health Canada's website:

[Health Canada website for CCS](#)



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## Canadian Cannabis Survey (CCS) 2024 Public Use Microdata File (PUMF)

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The 2024 Canadian Cannabis Survey (CCS) recruited people aged 16 years and older living in all Canadian provinces and territories. Respondents completed a survey on patterns of cannabis use for medical and non-medical purposes, as well as their knowledge, attitudes and behaviours towards cannabis use. A total of 11,666 respondents completed online surveys between April 4 and July 2, 2024. Summary results are available here:

<https://www.canada.ca/en/health-canada/services/drugs-medication/cannabis/research-data/canadian-cannabis-survey-2024-summary.html>

Additional

The Office Cannabis Science and Surveillance releases the PUMF for CCS 2024 to allow



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## Canadian Cannabis Survey 2024: Summary

On this page

- [Introduction](#)
- [Results by theme](#)
- [Methods](#)
- [Considerations](#)

## Open Licence

The data from the Canadian Cannabis Survey is available under the Open Government Licence - Canada (OGL-Canada). This licence allows users to freely use, modify, and distribute the data, provided that they attribute the source of the data and do not use it for commercial purposes. The OGL-Canada promotes transparency and encourages the use of government data for research, analysis, and public benefit. You can follow the [Open Government Licence - Canada](#) link for more information on the terms and conditions of the licence.

[Open Licence](#)

Additional Information

**Keywords:**  
cannabis | drugs

**Subject:**  
Health and Safety

**Maintenance and Update Frequency:**

[data/cannabis-survey-2024-summary.html](#)

The Office Cannabis Science and Surveillance releases the PUMF for CCS 2024 to allow researchers, provincial governments and others to conduct their own analyses of this important data.

**Publisher - Current Organization Name:** Health Canada

**Publisher - Organization Name at Publication:** Office of Cannabis Science and Surveillance, Strategic Policy Directorate, Controlled Substances and Cannabis Branch, Health Canada

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## Data and Resources

The **Data and Resources** tab on the Health Canada website provides access to the public use data, user guides, and the data dictionary.

Data and Resources

Search resources...

Q

CCS

Canadian Cannabis Survey (CCS) 2024

English dataset CSV

Explore

DOCX

Canadian Cannabis Survey (CCS) 2024 PUMF User Guide

English guide DOCK

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Canadian Cannabis Survey (CCS) 2024 PUMF User Guide

French guide DOCK

Explore

DOCX

Public Use Microdata Files (PUMF) Derived Variable Specifications

English specification DOCK

Explore

DOCX

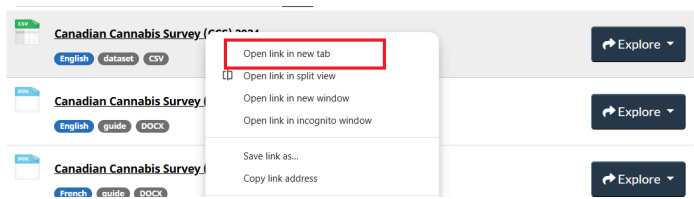
Public Use Microdata Files (PUMF) Derived Variable Specifications

French specification DOCK

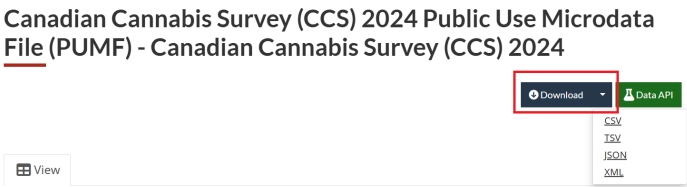
Explore

The CCS 2024 data is available to download in various formats, including CSV, TSV, JSON, and XML. The data can be accessed through the Health Canada website, where you can choose the format that best suits your needs. You need to

open the data link (say, in a new tab in your browser) and select the desired format to download the data.



Under the **Download** tab, you can find the data in different formats. For example, you can download the data in CSV format by clicking on the **CSV** link:



After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual respondent and each column representing different variables:

|    | A      | B    | C             | D     | E   | F               | G              | H    | I          | J               | K             | L          | M            |
|----|--------|------|---------------|-------|-----|-----------------|----------------|------|------------|-----------------|---------------|------------|--------------|
| 1  | seq_id | wave | sample_weight | age65 | sex | sexuality_recod | community_size | prov | race_white | health_physical | health_mental | education3 | income_recod |
| 2  | 1      | 7    | 2622.409      | 2     | 2   | 1               | 1              | 7    | 1          | 2               | 2             | 3          | 3            |
| 3  | 2      | 7    | 1517.9545     | 4     | 1   | 1               | -7             | 5    | 1          | 2               | 3             | 3          | 2            |
| 4  | 3      | 7    | 1547.9545     | 1     | 1   | 1               | 1              | 1    | 0          | 2               | 4             | 1          | 1            |
| 5  | 4      | 7    | 3095          | 1     | 2   | 1               | 2              | 7    | 1          | 4               | 3             | 1          | 2            |
| 6  | 5      | 7    | 2475.0746     | 4     | 1   | 2               | 2              | 2    | 0          | 2               | 4             | 3          | 4            |
| 7  | 6      | 7    | 2306.2095     | 6     | 1   | 1               | 2              | 2    | 1          | 1               | 1             | 2          | 3            |
| 8  | 7      | 7    | 4033.9425     | 4     | 1   | 1               | 1              | 7    | 0          | 4               | 3             | 1          | 2            |
| 9  | 8      | 7    | 1933.1395     | 1     | 2   | 1               | 2              | 9    | 1          | 1               | 1             | 1          | 3            |
| 10 | 9      | 7    | 4273.84       | 3     | 1   | 2               | 9              | 9    | 0          | 4               | 3             | 3          | -8           |
| 11 | 10     | 7    | 1512.9487     | 5     | 1   | -8              | -8             | 6    | -8         | 3               | 2             | 1          | -8           |
| 12 | 11     | 7    | 2438.4091     | 4     | 2   | 1               | 2              | 1    | 0          | 3               | 4             | 2          | 3            |
| 13 | 12     | 7    | 1980.8133     | 1     | 1   | 1               | 2              | 5    | 1          | 3               | 4             | 1          | 1            |

The **Canadian Cannabis Survey (CCS) 2024 PUMF User Guide** gives detailed information on the survey methodology, data collection process, and how to use the public use microdata file (PUMF). You can also find the variable names and their descriptions in this file.

[2024 Public Use Microdata File](#)

## Data and Resources

Search resources...

**Canadian Cannabis Survey (CCS) 2024** [Explore](#)

[English](#) [dataset](#) [CSV](#)

**Canadian Cannabis Survey (CCS) 2024 PUMF User Guide** [Explore](#)

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**Public Use Microdata Files (PUMF) Derived Variable Specifications** [Explore](#)

[English](#) [specification](#) [DOCK](#)

**Public Use Microdata Files (PUMF) Derived Variable Specifications** [Explore](#)

[French](#) [specification](#) [DOCK](#)

Let's find a variable (`access_ease_legal`) in the Excel file and check its description in the data dictionary. You can use the Find function in Excel (Ctrl + F) to search for the variable name:

| cans | risk_cantli | risk_cand | risk_cane | access_ease_legal | access_e | warn_see | warn_crec | warn_kno | warr |
|------|-------------|-----------|-----------|-------------------|----------|----------|-----------|----------|------|
| 3    | 3           | 3         | 3         | 3                 | 5        | 4        | -7        | -7       |      |
| 2    | 2           | 2         | 2         | 4                 | 2        | 4        | -7        | -7       |      |
| 1    | 1           | 1         | 1         | 4                 | 3        | 1        | 1         | 1        |      |
| 4    | 4           | 4         | 4         | 4                 | 3        | 4        | -7        | -7       |      |
| 2    | 3           | 5         | 5         | 4                 | 5        | 1        | 4         | 4        |      |
| 5    | 5           | 5         | 5         | 4                 | 5        | 4        | -7        | -7       |      |
| 2    | 2           | 2         | 2         | 4                 | 4        | 3        | 3         | 2        |      |
| 4    | 4           | 4         | 4         | 1                 | 3        | 4        | -7        | -7       |      |
| 3    | 3           | 3         | 2         | 4                 | 3        | 4        | -7        | -7       |      |
| 2    | 2           | 2         | 2         | 4                 | 4        | 5        | 7         | -7       |      |
| 3    |             |           |           |                   |          |          |           | 1        |      |
| 2    |             |           |           |                   |          |          |           | -7       |      |
| 1    |             |           |           |                   |          |          |           | 2        |      |
| 4    |             |           |           |                   |          |          |           | -7       |      |
| 2    |             |           |           |                   |          |          |           | 3        |      |
| 3    |             |           |           |                   |          |          |           | -7       |      |
| 3    |             |           |           |                   |          |          |           | 1        |      |
| 4    |             |           |           |                   |          |          |           | -7       |      |
| 3    |             |           |           |                   |          |          |           | 3        |      |
| 4    |             |           |           |                   |          |          |           | -7       |      |
| 4    | 4           | 4         | 3         | 4                 | 5        | 4        | -7        | -7       |      |
| 2    | 2           | 2         | 2         | 4                 | 3        | 2        | 1         | 2        |      |

Then you can see the description of the variable in the **Canadian Cannabis Survey (CCS) 2024 PUMF User Guide**, which provides information on what the variable represents and how it was coded:

|                                                        |                                                                                                                                                                                                                                               |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>access_ease_legal</b>                               | How easy or difficult is it for you to get legal cannabis?                                                                                                                                                                                    |
| <i>[variable added in 2024 to replace access_ease]</i> | <input type="radio"/> 1 Very difficult<br><input type="radio"/> 2 Fairly difficult<br><input type="radio"/> 3 Fairly easy<br><input type="radio"/> 4 Very easy<br><input type="radio"/> 5 I don't know<br><input type="checkbox"/> -7 Missing |

In this dataset, -7 generally represents missing. Always check the data dictionary for the specific coding of missing values, “don’t know,” or “refused” responses, as they can vary between datasets.



# Data Source: CADS

The Canadian Alcohol and Drugs Survey (CADS) is a national survey conducted by Statistics Canada that collects data on the use of substances, such as alcohol, cannabis and other drugs. The study was conducted by Statistics Canada in 2019 with the cooperation and support of Health Canada. The target population for the survey was all persons aged 15 years and over living in Canada, excluding residents of the Yukon, Northwest Territories, and Nunavut, full-time residents of institutions, and residents of Native reserves.



[CADS 2019 dataset for CADS](#)

## Open Licence

Similar to the [Canadian Cannabis Survey](#), the CADS is also licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Canadian Alcohol and Drugs Survey: Public Use Microdata File

**Have your say**  
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**Additional Information**

This public use microdata file (PUMF) is from the Canadian Alcohol and Drugs Survey and includes information about the use of substances, including alcohol, cannabis and other drugs. This product is for users who prefer to do their own analysis by focusing on specific subgroups in the population or by cross-classifying variables that were not the focus of our release in The Daily.

**Publisher - Current Organization Name:** Statistics Canada

**Licence:** [Open Government Licence - Canada](#)

 **Government of Canada** / **Gouvernement du Canada**



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## Data and Resources

The **Data** and **Resources** tab provides access to the public use data, user guides, and the data dictionary.

### Data and Resources



 **Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
[English](#) [French](#) [dataset](#) [CSV](#)

 **Explore** ▾

 **Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
[English](#) [French](#) [dataset](#) [SAS](#)

 **Explore** ▾

 **Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
[English](#) [website](#) [HTML](#)

 **Explore** ▾

 **Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
[French](#) [website](#) [HTML](#)

 **Explore** ▾

The public use data file is available in a ZIP format. You need to open the **Canadian Alcohol and Drugs Survey: Public Use Microdata File** link, and click on **Go to resource** to download the ZIP file.

## Data and Resources

Search resources... 



Canadian Alcohol and Drugs Survey: Public Use Microdata File

English French dataset CSV

Explore



Canadian Alcohol and Drugs Survey: Public Use Microdata File

English French dataset SAS

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Canadian Alcohol and Drugs Survey: Public Use Microdata File

English website HTML

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Canadian Alcohol and Drugs Survey: Public Use Microdata File

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### Canadian Alcohol and Drugs Survey: Public Use Microdata File - Canadian Alcohol and Drugs Survey: Public Use Microdata File

 Go to resource

The ZIP file contains the following files: an instruction on how to cite CADS 2019, data documentation, and the dataset in CSV format.

|                                                                                                                                 |                    |                    |        |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------|
|  How to cite CADS2019_Comment citer ECAD2019 | 27-Jul-21 10:42 AM | Adobe Acrobat D... | 388 KB |
|  Documentation                               | 05-Aug-21 6:56 AM  | File folder        |        |
|  Data_donnees                                | 05-Aug-21 6:46 AM  | File folder        |        |

## Data Documentation

The data documentation folder includes three sub-folders: Codebook\_Livre des codes, Guide, and Questionnaires.

|                                                                                                              |                   |             |
|--------------------------------------------------------------------------------------------------------------|-------------------|-------------|
|  Codebook_Livre des codes | 05-Aug-21 6:56 AM | File folder |
|  Guide                    | 05-Aug-21 6:56 AM | File folder |
|  Questionnaires           | 05-Aug-21 6:56 AM | File folder |

The **Guide** folder contains user guides (in English and French) that provide an overview of the survey, including the methodology, data collection process, data processing, data quality, and sampling weights.

|                                                                                                |                |                   |                    |          |
|------------------------------------------------------------------------------------------------|----------------|-------------------|--------------------|----------|
|  ECAD2019_GuF | <b>French</b>  | 14-Jul-21 6:47 AM | Adobe Acrobat D... | 1,653 KB |
|  CADS2019_UgE | <b>English</b> | 14-Jul-21 6:45 AM | Adobe Acrobat D... | 1,306 KB |

The **Codebook\_Livre des codes** folder contains codebooks (in English and French) that provide detailed information about the variables in the dataset, including variable names, labels, and values. There are two types of codebooks: those with and those without variable frequencies. But both codebooks provide the same information about the variables in the dataset.

**Codebook including frequencies for variables**

|                                                                                                               |                |                    |                    |          |
|---------------------------------------------------------------------------------------------------------------|----------------|--------------------|--------------------|----------|
|  ECAD2019_FMGD_lvCd          | <b>French</b>  | 22-Dec-20 11:50 AM | Adobe Acrobat D... | 1,948 KB |
|  CADS2019_PUMF_Cdbk          |                | 22-Dec-20 11:41 AM | Adobe Acrobat D... | 1,925 KB |
|  ECAD2019_FMGD_lvCd_zerofreq |                | 22-Dec-20 11:24 AM | Adobe Acrobat D... | 1,885 KB |
|  CADS2019_PUMF_Cdbk_zerofreq | <b>English</b> | 20 11:16 AM        | Adobe Acrobat D... | 1,863 KB |

**Codebook excluding frequencies for variables**

The **Questionnaires** folder contains the survey questionnaires (in English and French) that were used to collect the data.

## Data

The **Data\_donnees** folder contains the dataset in CSV format. You can open the dataset in Excel or any statistical software to explore the data. The row represents the respondent, and the column represents the variable:

| CADS                                                                              |        |         |         |      |         |          |     |        |       |        |        |        |        |         |       |   |
|-----------------------------------------------------------------------------------|--------|---------|---------|------|---------|----------|-----|--------|-------|--------|--------|--------|--------|---------|-------|---|
| File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat |        |         |         |      |         |          |     |        |       |        |        |        |        |         |       |   |
| A1 PUMFID                                                                         |        |         |         |      |         |          |     |        |       |        |        |        |        |         |       |   |
|                                                                                   | A      | B       | C       | D    | E       | F        | G   | H      | I     | J      | K      | L      | M      | N       | O     | P |
|                                                                                   | PUMFID | REFYEAR | REFMONT | PROV | DVURBAN | MHHTYPEB | SEX | DV_AGE | HHL01 | HDSIZE | HWB_05 | HWB_10 | ALC_05 | ALC_10P | ALC_J |   |
| 1                                                                                 | 1000   | 2019    | 12      | 46   | 2       | 3        | 1   | 8      | 1     | 2      | 2      | 2      | 1      | 16      |       |   |
| 2                                                                                 | 1001   | 2019    | 9       | 46   | 2       | 6        | 1   | 5      | 1     | 4      | 1      | 1      | 1      | 17      |       |   |
| 3                                                                                 | 1002   | 2019    | 6       | 35   | 1       | 3        | 2   | 8      | 1     | 2      | 3      | 2      | 1      | 18      |       |   |
| 4                                                                                 | 1003   | 2019    | 8       | 11   | 1       | 3        | 1   | 8      | 1     | 3      | 2      | 2      | 1      | 18      |       |   |
| 5                                                                                 | 1004   | 2019    | 6       | 12   | 1       | 3        | 1   | 9      | 1     | 3      | 2      | 3      | 1      | 16      |       |   |
| 6                                                                                 | 1005   | 2019    | 10      | 12   | 1       | 3        | 1   | 8      | 1     | 2      | 2      | 1      | 1      | 99      |       |   |
| 7                                                                                 | 1006   | 2019    | 10      | 35   | 1       | 3        | 1   | 6      | 1     | 1      | 1      | 1      | 1      | 18      |       |   |
| 8                                                                                 | 1007   | 2019    | 12      | 48   | 1       | 3        | 1   | 7      | 1     | 2      | 4      | 4      | 1      | 16      |       |   |
| 9                                                                                 | 1008   | 2019    | 10      | 46   | 1       | 6        | 1   | 9      | 1     | 3      | 1      | 1      | 1      | 17      |       |   |
| 10                                                                                | 1009   | 2019    | 6       | 47   | 1       | 3        | 2   | 9      | 1     | 2      | 2      | 2      | 1      | 16      |       |   |
| 11                                                                                | 1010   | 2019    | 11      | 24   | 1       | 8        | 2   | 7      | 1     | 2      | 3      | 2      | 1      | 16      |       |   |

You can refer to the codebook to understand the variables that you are interested in for your analysis. For example, if you are interested in the variable DVURBAN, you can refer to the codebook to find out that the variable (e.g., for the English version of the codebook with frequencies):

|                   |                                                                                                                    |           |                    |           |    |
|-------------------|--------------------------------------------------------------------------------------------------------------------|-----------|--------------------|-----------|----|
| Variable Name:    | DVURBAN                                                                                                            | Length:   | 1.0                | Position: | 14 |
| Question Name:    |                                                                                                                    |           |                    |           |    |
| Concept:          | Characteristics of community where the respondent lives                                                            |           |                    |           |    |
| Question Text:    |                                                                                                                    |           |                    |           |    |
| Universe:         | All respondents                                                                                                    |           |                    |           |    |
| Note:             | The category 'Population centre' is historically comparable to 'Urban' which was used in previous years for CTUMS. |           |                    |           |    |
| Source:           |                                                                                                                    |           |                    |           |    |
| Answer Categories | Code                                                                                                               | Frequency | Weighted Frequency | %         |    |
| Population centre | 1                                                                                                                  | 7,208     | 21,321,106         | 68.8      |    |
| Rural             | 2                                                                                                                  | 2,438     | 5,122,120          | 16.5      |    |
| Not stated        | 9                                                                                                                  | 647       | 4,552,224          | 14.7      |    |
| Total             |                                                                                                                    | 10,293    | 30,995,450         | 100.0     |    |

# Data Source: CTNS

The Canadian Tobacco and Nicotine Survey (CTNS) is a cross-sectional survey that collects information on tobacco and nicotine use among Canadians aged 15 years and older. The survey is conducted by Statistics Canada on behalf of Health Canada and provides valuable insights into the prevalence of cigarette smoking, vaping, cannabis, and alcohol use.



[CTNS 2022 dataset for researchers](#)

## Open Licence

The CTNS is licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Data and Resources

The **Data and Resources** tab provides access to the public use data, user guides, and the data dictionary.

## Data and Resources

Search resources...

|                                                                                                                                                                                 |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English website HTML       | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>French website HTML        | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English French dataset CSV | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English French dataset SAS | Explore |

The public use data file is available in a ZIP format. You can click on the Canadian Tobacco and Nicotine Survey: Public Use Microdata File link, followed by Go to resource link to download the ZIP file.

## Data and Resources

Search resources...

|                                                                                                                                                                                   |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English website HTML         | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>French website HTML         | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English French dataset CSV | Explore |
|  Canadian Tobacco and Nicotine Survey: Public Use Microdata File<br>English French dataset SAS | Explore |

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### Canadian Tobacco and Nicotine Survey: Public Use Microdata File - Canadian Tobacco and Nicotine Survey: Public Use Microdata File

Go to resource

Similar to the [Canadian Alcohol and Drugs Survey \(CADS\)](#) tutorial, the ZIP file for CTNS contains the following files: citation instructions, data documentation, and the dataset in CSV format. The data documentation, questionnaire, and codebook are also the same as those for CADS. The only difference is that the data documentation and questionnaire for CTNS are labelled CTNS rather than CADS.

# Data Source: CSADS

The Canadian Student Alcohol and Drugs Survey (CSADS) is a national survey that collects data on the alcohol and drug use patterns of Canadian students in grades 7 to 12. The survey is conducted every two years and provides valuable insights into the prevalence and trends of substance use among Canadian youth.

## Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File

Have your say

★★★★★

[Rate this dataset](#)

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Every two years, Health Canada conducts the Canadian Student Alcohol and Drugs Survey (CSADS), a cross-sectional survey of students in grades 7 to 12 across the Canadian provinces.

With tens of thousands of participants each cycle, it is one of the largest youth surveys in the world. For over 30 years, the survey—previously known as the Youth Smoking Survey (YSS; pre-2014) and the Canadian Student Tobacco, Alcohol and Drugs Survey (CSTADS;

[CSADS Canada website files CTNS](#)

## Open Licence

The CTNS is licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Data and Resources

The **Data and Resources** tab provides access to the public use data, user guides, the data dictionary, and information on bootstrap weights.



## Data and Resources

Search resources...

|                                                                                   |                                                   |                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|
|  | <b>2021-2022 CSTADS: PUMF — User Guide</b>        | <a href="#">Explore</a> |
|  | <b>2021-2022 ECTADE : FMGD — Mode d'emploi</b>    | <a href="#">Explore</a> |
|  | <b>2021-2022 CSTADS: PUMF — Data</b>              | <a href="#">Explore</a> |
|  | <b>2021-2022 CSTADS: PUMF — Bootstrap Weights</b> | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — User Guide</b>         | <a href="#">Explore</a> |
|  | <b>2023-2024 ECADE : FMGD — Mode d'emploi</b>     | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — Data</b>               | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — Bootstrap Weights</b>  | <a href="#">Explore</a> |

To download the 2021–2022 CSTADS, you can click on the [2021-2022 CSTADS: PUMF – Dataset in CSV format](#) link. The data is available to download in various formats. Under the **Download** tab, you can find the formats. For example, you can download the data in CSV format by clicking on the [CSV](#) link:

**MENU**

[Canada.ca](#) > [Open Government](#) > [Search](#) > [Canadian Student Alcohol and Drugs Survey \(CSADS\): Public Use Microdata File](#)

### Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File - 2021-2022 CSTADS: PUMF — Data

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[View](#)

[CSV](#)  
[TSV](#)  
[JSON](#)  
[XML](#)

After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual respondent and each column representing different variables:

|    | A     | B      | C     | D        | E       | F     | G        | H       | I        | J      | K      | L      | M      | N      | O      |
|----|-------|--------|-------|----------|---------|-------|----------|---------|----------|--------|--------|--------|--------|--------|--------|
| 1  | SEQID | PROVID | GRADE | DVGENDEI | DVURBAN | DVRES | DVORIENT | DVDESCR | WTPUMF   | GH_010 | GH_020 | SS_010 | SS_020 | TS_011 | SS_030 |
| 2  | 18338 | 11     | 9     | 2        | 1       | 3     | 2        | 1       | 1.164716 | 2      | 3      | 2      | 96     | 4      | 96     |
| 3  | 16111 | 24     | 10    | 1        | 1       | 1     | 2        | 4       | 36.14117 | 3      | 3      | 2      | 96     | 4      | 96     |
| 4  | 20587 | 10     | 7     | 2        | 1       | 1     | 2        | 1       | 1.448578 | 2      | 1      | 2      | 96     | 4      | 96     |
| 5  | 54568 | 12     | 8     | 99       | 2       | 1     | 1        | 1       | 3.03666  | 4      | 5      | 2      | 96     | 3      | 96     |
| 6  | 40991 | 10     | 7     | 1        | 2       | 1     | 2        | 1       | 3.745233 | 2      | 3      | 2      | 96     | 4      | 96     |
| 7  | 5254  | 12     | 7     | 1        | 1       | 2     | 2        | 1       | 10.0715  | 3      | 3      | 2      | 96     | 4      | 96     |
| 8  | 10440 | 35     | 9     | 99       | 1       | 1     | 99       | 1       | 75.0031  | 5      | 4      | 2      | 96     | 4      | 96     |
| 9  | 38149 | 48     | 12    | 1        | 1       | 1     | 2        | 1       | 25.83337 | 1      | 3      | 1      | 16     | 4      | 2      |
| 10 | 32243 | 59     | 11    | 2        | 1       | 1     | 2        | 1       | 47.16534 | 1      | 3      | 1      | 16     | 3      | 2      |
| 11 | 5513  | 35     | 11    | 1        | 2       | 1     | 99       | 1       | 99.52681 | 2      | 5      | 2      | 96     | 3      | 96     |
| 12 | 37146 | 48     | 10    | 1        | 1       | 2     | 2        | 4       | 24.62732 | 3      | 2      | 2      | 96     | 4      | 96     |
| 13 | 21979 | 46     | 9     | 1        | 2       | 1     | 2        | 1       | 65.69607 | 1      | 4      | 2      | 96     | 4      | 96     |
| 14 | 33925 | 12     | 7     | 1        | 2       | 3     | 1        | 1       | 3.60877  | 3      | 4      | 2      | 96     | 4      | 96     |
| 15 | 19919 | 94     | 7     | 2        | 1       | 2     | 00       | 00      | 76.14944 | 9      | 9      | 2      | 00     | 4      | 00     |

The user guide gives detailed information on the survey methodology, data collection process, and how to use the public use microdata file (PUMF). You can also find the variable names and their descriptions in this DOCX file.

**2021-2022 CSTADS: PUMF — User Guide**

[English](#)
[guide](#)
[DOCX](#)

Explore

**2021-2022 ECTADE : FMGD — Mode d'emploi**

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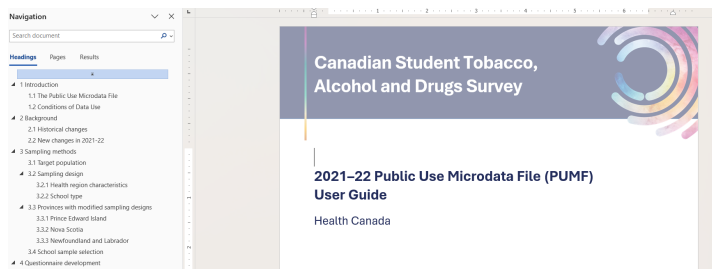
**2021-2022 CSTADS: PUMF — Bootstrap Weights**

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## Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File - 2021-2022 CSTADS: PUMF — User Guide

Download



To find a variable description, you are referred to the **User Guide**. For example, the variable GH\_010 is described as follows:

| Navigation                                      |       |         |   |
|-------------------------------------------------|-------|---------|---|
| 04_010                                          |       | WTPUMP  | 9 |
| 1 result                                        |       |         |   |
| Headings                                        | Pages | Results |   |
| 6.2.1 Skip patterns and inconsistent responses  |       |         |   |
| 6.2.2 Imputation of core smoking responses      |       |         |   |
| 6.2.3 Reducing underestimation risk in the PLME |       |         |   |
| 6.2.4 Standard codes                            |       |         |   |
| 6.3 Survey weights                              |       |         |   |

|        |    |                                                                                                                                                                   |                                                                                                            |
|--------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|        |    | Unknown: if not selected, was assigned to the perturbation to maintain privacy; more information on how these weights were generated can be found in section 6.3. |                                                                                                            |
| 04_010 | 10 | In general, would you say your physical health is excellent, very good, good, fair or poor?                                                                       | 1 = Excellent<br>2 = Very good<br>3 = Good<br>4 = Fair<br>5 = Poor<br>6 = I do not know<br>99 = Not stated |
|        |    |                                                                                                                                                                   | 1 = Excellent<br>2 = Very good                                                                             |

# Data Source: PHAC Health Infobase

The Health Infobase is a comprehensive digital platform maintained by the Public Health Agency of Canada (PHAC). The Infobase provides access to a wide range of health-related data tools, infographics, charts and blogs. There are 182 data sources available on the Infobase. The Health Infobase serves as a central hub for national surveillance data, interactive data tools, and public health reports. The platform is frequently updated with the latest surveillance data.



[PHAC Health Infobase](#)

## Data Categories

The Infobase provides access to health data under six categories:

- **Diseases and disorders:** Comprehensive tracking of chronic conditions such as cancer, cardiovascular diseases, and COVID-19.

- **Drugs and vaccines:** Data on prescription drugs, vaccine coverage, and immunization rates across the country.
- **Mental health, healthy living and environment:** Information on mental health indicators, physical activity levels, nutrition, and environmental factors such as wildfires.
- **Injuries and death:** Data on death, suicide, violence, and drowning across regions.
- **Population groups and demography:** Data on birth, maternity, children, family, and disabilities.
- **Substance use:** Data on substance use such as alcohol, cannabis use, opioids, and smoking.

Under each category, users can select the data for specific health topics. The three top categories are **Respiratory diseases**, **Opioids** and **Mental health**.

### Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

Diseases and disorders

+

Drugs and vaccines

+

Mental health, healthy living and environment

+

Injuries and death

+

Population groups and demography

+

Substance use

+

▼ Filter

Type in keywords to filter results as you type



[Healthy Canadians and Communities Fund](#)

Data exploration tool

Inequalities

Interactive map of funded interventions across Canada that support Canadians who experience health inequalities and are at greater risk of developing chronic disease.

Last updated: 2026-04-07

### Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

Diseases and disorders

—

☐ All diseases and disorders

☒ Blood and metabolic diseases

☐ Chronic diseases

☐ Cancer

☐ COVID-19

▼ Filter

Type in keywords to filter results as you type

Blood and metabolic diseases X

Showing 5 filtered from 182 total products

[Clear all filters](#)



[Canadian Chronic Disease Surveillance System \(CCDSS\)](#)

Data exploration tool

Chronic diseases

Cardiovascular diseases

Musculoskeletal disorders

Neurological diseases

Respiratory diseases

Mental health

Blood and metabolic diseases

Autism

313

You can filter the data by category, by format, or by typing in keywords.

### Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

- Diseases and disorders +
- Drugs and vaccines +
- Mental health, healthy living and environment +
- Injuries and death +
- Population groups and demography +
- Substance use +

Filter covid

Showing 27 filtered from 182 total products  
[Clear all filters](#)

 [COVID-19 Vaccine mistrust among Black Communities in Canada](#)

[Interactive report](#) | [COVID-19](#) | [Vaccines](#)

Interactive report on COVID-19 vaccine mistrust among Black communities in Canada, based on data from the 2022 CanBlack Vax survey.

You can also filter the data by type, such as blog, infographic, or report.

▼ Type

- ☐ Blog
- ☐ Data exploration tool
- ☐ Indicator framework
- ☐ Infographic
- ☐ Interactive report

at greater risk of developing chronic disease.  
Last updated: 2026-04-07

 [Transfusion and transplantation-related adverse events](#)

[Interactive report](#) | [Injuries](#)

Summary of national surveillance systems for transfusion errors, transfusion-transmitted injuries, and cells, tissues and

## Data Availability

The level of information provided in the data sources varies. For some data sources/products, you can find blogs, infographics, and reports along with individual-level data. However, for some data sources/products, you may only find the summary statistics without individual-level data, i.e., only aggregated-level data. For example, the **Human Papillomavirus Indicator Framework** product provides quick statistics, data tools and a list of publications:

Categories

Diseases and disorders

☐ All diseases and disorders
☐ Blood and metabolic diseases
☐ Chronic diseases
☒ Cancer
☐ COVID-19
☐ Cardiovascular diseases
☐ Developmental

Filter

Type in keywords to filter results as you type

Cancer X

Showing 10 filtered from 182 total products

Clear all filters



[Human Papillomavirus Indicator Framework \(HPVIF\)](#)

[Data exploration tool](#)
[Infections](#)
[Vaccines](#)

Cancer

The Human Papillomavirus Indicator Framework (HPVIF) presents HPV vaccination coverage and HPV outcome indicators for Canada.

Last updated: 2025-12-15

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Franglais

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Canada.ca

Health

Health science, research and data

Health Infobase

Human Papillomavirus Indicator Framework (HPVIF)

The Human Papillomavirus Indicator Framework (HPVIF) presents HPV vaccination coverage and HPV outcome indicators for Canada. Sub-national estimates, sociodemographic breakdowns, and trends over time are presented, when possible, as well as international comparisons. Since HPV vaccination can prevent HPV-associated cancers, this framework will serve as a valuable resource for monitoring Canada's progress against HPV.

Last updated: 2025-12-15

Quick stats

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Publications

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About

Quick Stats overview

Suggested citation

The Drug and alcohol use in Canada: Postsecondary students product provides summary statistics as well as links to download data in CSV format:

Substance use

☐ All substance use
☒ Alcohol
☐ Cannabis
☐ Opioids
☐ Smoking and vaping
☐ Stimulants

Key findings from the 2024-2025 Canadian Postsecondary Education Alcohol and Drug Use Survey (CPADS).

Last updated: 2025-12-22



[Drug and alcohol use in Canada: Postsecondary students](#)

[Data exploration tool](#)
[Alcohol](#)
[Cannabis](#)

[Opioids](#)
[Prescription drugs](#)
[Smoking and vaping](#)

Stimulants

Explore Health Canada's latest data on drug and alcohol use by postsecondary students in Canada.

315

## Postsecondary students

### Drug and alcohol use in Canada

Explore Health Canada's latest data on drug and alcohol use by postsecondary students in Canada.

Last updated: 2025-12-31

General population

Youth

Postsecondary students

### About the postsecondary survey

Health Canada monitors the prevalence of substance use and its impact on postsecondary students through the Canadian Postsecondary Education Alcohol and Drug Use Survey (CPADS). Governments and non-governmental organizations use this information to shape policies and programs that address substance use and support students.

In the 2024–25 school year, 29,371 students from 9 provinces and 2 territories completed the survey. The results focus on

#### Additional resources



[Read](#)

[the 2024-2025 summary report](#)



[Read](#)

[previous summary reports](#)



[Download](#)

[dashboard data \(.csv\)](#)

We'd love your feedback! Please take a minute to complete this short and anonymous survey.

[Next](#)



# Data Source: CCDSS

The Canadian Chronic Disease Surveillance System (CCDSS) is a collaborative network of provincial and territorial chronic disease surveillance systems in Canada. The CCDSS is supported by the Public Health Agency of Canada (PHAC) and provincial/territorial governments. The CCDSS was established to provide the estimates of the prevalence and incidence of chronic diseases in Canada, as well as to monitor trends over time for more than 20 chronic diseases to help inform health policies and resource planning. The CCDSS is part of the PHAC Health Infobase that we covered in the previous tutorial on [Health Infobase](#). The CCDSS uses health administrative data, such as hospital discharge records, physician billing claims, and prescription drug data.



[CCDSS website](#)

## Key Indicator Categories

The CCDSS website provides a variety of indicators related to chronic diseases, such as autism, cardiovascular diseases,

chronic respiratory diseases, diabetes, mental health disorders, musculoskeletal disorders, neurological disorders, and multimorbidity.

Diseases, conditions and indicators

Table 1: Diseases, conditions and indicators included in the CCDSS

| Morbidity and mortality                                                                                                                                                                        | Health events and complications                                                                                        | Use of health services                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>incidence/rate of newly identified cases</li><li>prevalence (cumulative and active for select diseases/conditions)</li><li>all-cause mortality</li></ul> | <ul style="list-style-type: none"><li>annual prevalence</li><li>all-cause mortality following a health event</li></ul> | <ul style="list-style-type: none"><li>annual prevalence</li></ul> |

Table 2: CCDSS case definitions and surveillance period reported

| Disease, condition and indicator | Age  | Case definition summary                                                  | First reported year | Last reported year |
|----------------------------------|------|--------------------------------------------------------------------------|---------------------|--------------------|
| Autism                           |      |                                                                          |                     |                    |
| Autism                           | 1-19 | One or more hospital separation records, or two or more physician claims | 2000-2001           | 2023-2024          |
| Cardiovascular diseases          |      |                                                                          |                     |                    |
| Acute myocardial infarction      | 20+  | One or more hospital inpatient admission records                         | 2000-2001           | 2023-2024          |

Data Tool

The CCDSS website provides a data tool that allows users to explore and visualize the data on chronic diseases in Canada. The data tool provides interactive charts and tables that allow users to explore the data by province/territory.

About the Canadian Chronic Disease Surveillance System (CCDSS)

The CCDSS is a collaborative network of provincial and territorial surveillance systems, supported by the Public Health Agency of Canada (PHAC). The system collects data on all residents who are eligible for provincial or territorial health insurance. It can generate national estimates and trends over time for over 20 chronic diseases and conditions, and other selected health outcomes. To identify people with chronic diseases and conditions, provincial and territorial health insurance registry records are linked using a unique personal identifier to the corresponding physician claims, hospital separation records and prescription drug records.

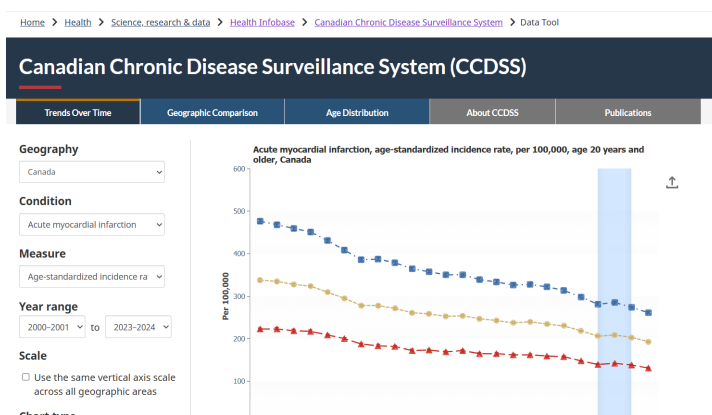
Key information about the CCDSS methods and case definitions is presented below. More detailed documents are available here:

[CCDSS summary of methods](#); [CCDSS case definitions](#); [CCDSS COVID-19 guidance for data users](#).

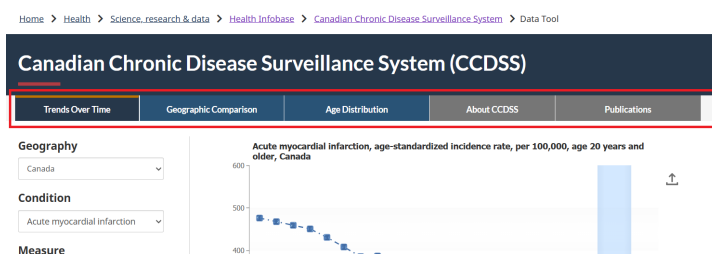
Use the CCDSS Data Tool

Users can use the data tool to explore the prevalence and incidence of cases, mortality rates, and health service utilization, and to track individuals living with two or more chronic conditions.

[CCDSS Data Tool](#)

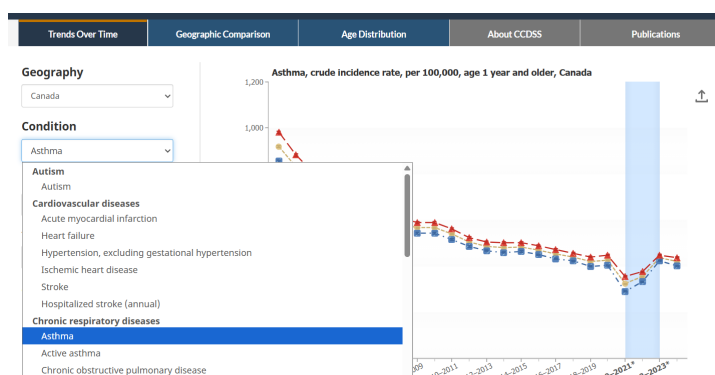


Users can explore trends over time, compare data across provinces/territories, and view the distribution of cases by age group. Under the **Publication** tab, you can find information on the repository for data-driven insights, fact sheets, peer-reviewed publications, and reports.



## Interactive chart example

Consider **Asthma** under **Chronic respiratory diseases** as an example.



You can explore different summary measures by filtering by geography, measure, and year. Again, you can further explore trends over time, compare data across provinces/territories, and view the distribution of cases by age group.



You can also download the summary data in CSV format for further analysis. To do that, you need to scroll down to the bottom of the **Summary Table**, and click on the **Download** button.

|       |            |     |      |         |      |        |
|-------|------------|-----|------|---------|------|--------|
| Males | 2019-2020  | 404 | 1.54 | 401-407 | 0.38 | 68,605 |
| Males | 2020-2021* | 292 | 1.31 | 290-295 | 0.45 | 49,690 |
| Males | 2021-2022* | 333 | 1.39 | 331-336 | 0.42 | 57,745 |
| Males | 2022-2023* | 425 | 1.55 | 422-428 | 0.36 | 75,625 |
| Males | 2023-2024  | 402 | 1.48 | 399-405 | 0.37 | 73,400 |

Download Detailed Table:  
Asthma, crude incidence rate

Save Summary Table

Once you open the CSV file in Excel or other software, you will find that the data is not at the individual or person

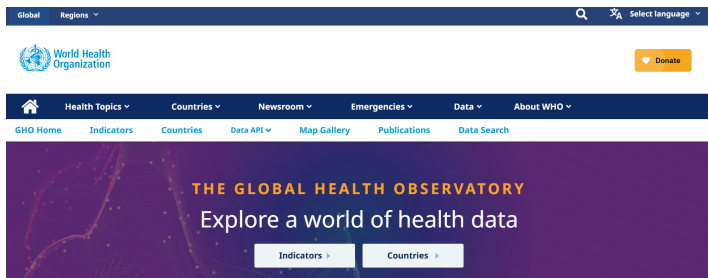
level. Instead, the data are aggregated at the country and province/territory levels, and summary measures are provided for both sexes and for years. For example, see row 4. The **Geography** column says *Canada* and the **Rate (per 100,000)** column says *919*. This is an example of aggregate data.

|    | A                                                     | B          | C              | D         | E           | F                  | G              | H            | I            | J           | K           | L          |
|----|-------------------------------------------------------|------------|----------------|-----------|-------------|--------------------|----------------|--------------|--------------|-------------|-------------|------------|
| 1  | Data Table: Asthma, Crude incidence rate, per 100,000 |            |                |           |             |                    |                |              |              |             |             |            |
| 2  |                                                       |            |                |           |             |                    |                |              |              |             |             |            |
| 3  | Geography                                             | Sex        | Age group type | Age group | Fiscal year | Rate (per 100,000) | Standard Error | Lower 95% CI | Upper 95% CI | Rate CV (%) | Case counts | Population |
| 4  | Canada                                                | Both sexes | Total          | 1+        | 2000-2001   | 919                | 1.8            | 915          | 923          | 0.2         | 260730      | 28367890   |
| 5  | Canada                                                | Females    | Total          | 1+        | 2000-2001   | 961                | 2.6206         | 976          | 986          | 0.27        | 140225      | 14289240   |
| 6  | Canada                                                | Males      | Total          | 1+        | 2000-2001   | 856                | 2.4657         | 851          | 861          | 0.29        | 120505      | 14078655   |
| 7  | Canada                                                | Both sexes | Total          | 1+        | 2001-2002   | 833                | 1.7103         | 830          | 837          | 0.21        | 237405      | 28488200   |
| 8  | Canada                                                | Females    | Total          | 1+        | 2001-2002   | 883                | 2.4814         | 878          | 886          | 0.26        | 126365      | 14337965   |
| 9  | Canada                                                | Males      | Total          | 1+        | 2001-2002   | 763                | 2.3526         | 779          | 788          | 0.3         | 110820      | 14150240   |
| 10 | Canada                                                | Both sexes | Total          | 1+        | 2002-2003   | 762                | 1.6318         | 759          | 765          | 0.21        | 218155      | 28623720   |
| 11 | Canada                                                | Females    | Total          | 1+        | 2002-2003   | 804                | 2.3637         | 800          | 809          | 0.29        | 115840      | 14399350   |
| 12 | Canada                                                | Males      | Total          | 1+        | 2002-2003   | 719                | 2.2487         | 715          | 724          | 0.31        | 102310      | 14224360   |
| 13 | Canada                                                | Both sexes | Total          | 1+        | 2003-2004   | 727                | 1.5893         | 724          | 730          | 0.22        | 209140      | 28774260   |
| 14 | Canada                                                | Females    | Total          | 1+        | 2003-2004   | 773                | 2.3117         | 769          | 778          | 0.3         | 111860      | 14468175   |
| 15 | Canada                                                | Males      | Total          | 1+        | 2003-2004   | 680                | 2.1801         | 676          | 684          | 0.32        | 97275       | 14306080   |
| 16 | Canada                                                | Both sexes | Total          | 1+        | 2004-2005   | 696                | 1.5513         | 693          | 699          | 0.22        | 201445      | 28932495   |
| 17 | Canada                                                | Females    | Total          | 1+        | 2004-2005   | 738                | 2.2531         | 734          | 743          | 0.31        | 107350      | 14541550   |
| 18 | Canada                                                | Males      | Total          | 1+        | 2004-2005   | 654                | 2.1316         | 650          | 658          | 0.33        | 94095       | 14398940   |

Because this data is not at the person-level, and the summary measures are aggregated, you do not need a codebook to translate columns. However, you cannot use this data to calculate the prevalence or incidence of asthma in Canada because it is already aggregated. You can only use this data to explore the trends over time, compare data across provinces/territories, and view the distribution of cases by age group and sex. The individual patients have already been erased to protect their privacy.

## Data Source: GHO

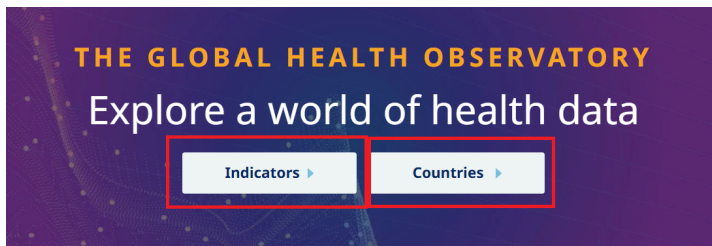
The Global Health Observatory (GHO) is the World Health Organization's (WHO) main health statistics repository. The GHO provides data on a wide range of health topics, such as mortality, morbidity, risk factors, health systems, and health equity. The GHO collects data from member states, international organizations, and other sources to provide a comprehensive picture of global health. The GHO is a valuable resource for researchers, policymakers, and public health professionals who are interested in understanding global health trends and disparities. The GHO provides various ways to interact with its data, including interactive maps and downloadable CSV files.



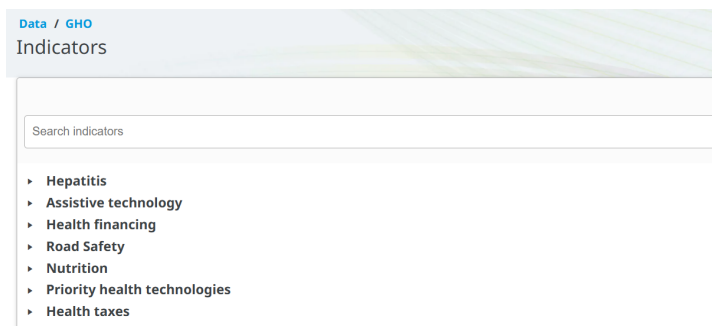
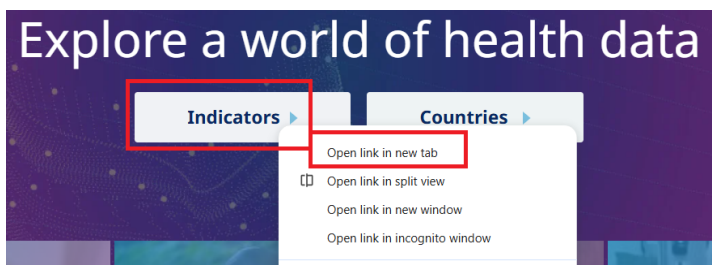
[WHO Global Health Observatory](#)

### World Health Data

You can browse information by broader health indicators or by specific countries.



To explore the data by indicators, click on the **Indicators** tab. You can find a list of indicators across categories such as Hepatitis, Nutrition, Mental health, and so on.



You can also use the search bar to search for specific indicators. For example, if you search for **Life expectancy at birth**, you will find a list of indicators related to life expectancy, such as **Life expectancy at birth (years)**.

Data / GHO

## Indicators

Life expectancy at birth

- World Health Statistics
  - Life expectancy and life tables
    - Life expectancy at birth (years)
- World Health Statistics
  - Life expectancy at birth (years)
- Global Health Estimates: Life expectancy and leading causes of death and disability
  - Mortality and global health estimates
    - Life expectancy at birth (years)

To explore the data by country, click on the **Countries** tab. You can find a list of countries and regions. You can click the country name to see the data for that country.

World Health Organization

**Data**

WHO Glob

Indicators Countries **Dasht**

|     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| All | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| U   | V | W | X | Y | Z |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

**A**

Afghanistan  
Albania  
Algeria  
Andorra  
Angola  
Antigua and Barbuda  
Argentina  
Armenia  
Australia  
Austria  
Azerbaijan

**B**

Bahamas  
Bahrain  
Bangladesh  
Barbados

**Australia**  
Health data overview for Australia

Population 26 451 124 (2023)

Current health expenditure (% of GDP) 10.54 (2021)

WHO region Western Pacific

World Bank income level High income (HIC)

**Population, Australia**

In Australia, the current population is 26,451,124 as of 2023 with a projected increase of 23% to 32,506,969 by 2050.

**Population growth rate**  
Australia, 2023

**0.95%** +0.012 percentage points  
rate change since 2022

**Australia**

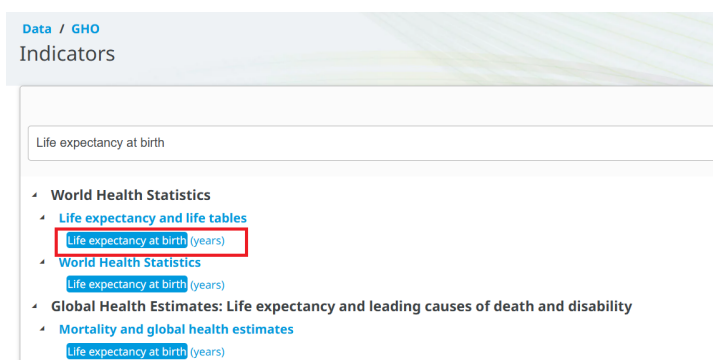
On this page

- Population
- Life expectancy
- Leading causes of death
- Health statistics



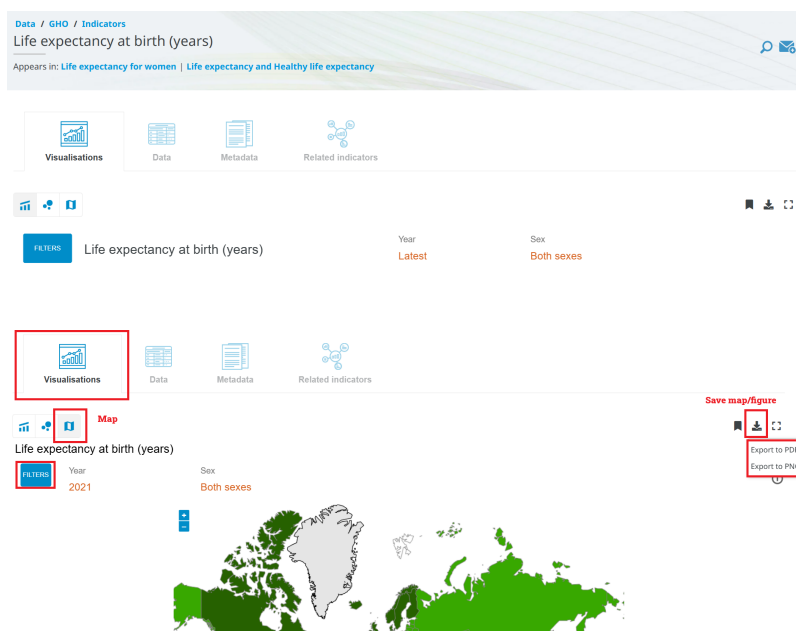
## Data by indicator example

Let's click on the Life expectancy at birth (years) indicator to see the details:



You can find different visualization options, such as charts and maps. You can also download the chart and map to your computer.

[GHO Indicators](#)



To see the data table, you can check the **Data** section. You can also download the data in CSV format for further analysis. To do that, you need to click on the **Download** button at the top right corner of the data table.

Visualisations

Data

Metadata

Related indicators

Filters

Last updated: 2024-08-02

| Indicator   | Life expectancy at birth (years) |                  |                  | Life expectancy at age 60 (years) |                  |                  |
|-------------|----------------------------------|------------------|------------------|-----------------------------------|------------------|------------------|
|             | Both sexes                       | Male             | Female           | Both sexes                        | Male             | Female           |
| Afghanistan | 59.1 [58.3-59.9]                 | 57.4 [56.6-58.2] | 61.0 [60.2-61.7] | 13.7 [13.2-14.2]                  | 13.6 [13.1-14.1] | 13.8 [13.3-14.3] |
| 2021        | 60.5 [59.8-61.5]                 | 59.3 [58.6-60.3] | 61.7 [60.9-62.4] | 13.8 [13.1-14.4]                  | 13.6 [13.0-14.1] | 14.0 [13.3-14.4] |
| 2020        | 61.2 [60.3-62.2]                 | 60.0 [59.3-61.1] | 62.5 [61.7-63.5] | 15.2 [14.5-15.7]                  | 15.4 [14.8-16.0] | 15.0 [14.4-15.7] |
| 2019        | 60.5 [59.7-61.5]                 | 59.0 [58.1-59.9] | 62.1 [61.3-63.0] | 15.1 [14.5-15.6]                  | 15.3 [14.8-15.9] | 14.9 [14.3-15.5] |
| 2018        | 60.9 [60.0-61.7]                 | 59.8 [59.0-60.9] | 62.1 [61.3-63.0] | 15.1 [14.4-15.6]                  | 15.4 [14.8-16.0] | 14.9 [14.3-15.5] |

EXPORT DATA in CSV format

Right-click here & Save link

Similar to the [Canadian Chronic Disease Surveillance System \(CCDSS\) database](#), the life expectancy data is also at an aggregated level, which means that the data is summarized at the country level, and does not contain individual-level data:

| IndicatorCode | Indicator     | ValueType                        | ParentLocationCode | ParentLocation | Location type         | SpatialDimValueCode | Location | Period type     | Period | Dim1       | FactValue  |       |
|---------------|---------------|----------------------------------|--------------------|----------------|-----------------------|---------------------|----------|-----------------|--------|------------|------------|-------|
| 1             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Male       | 48.73 |
| 2             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Male       | 49.57      |       |
| 3             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Both sexes | 51.48 |
| 4             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SWZ      | Eswatini        | Year   | 2021       | Male       | 51.64 |
| 5             | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Male       | 51.75 |
| 6             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Both sexes | 52.31      |       |
| 7             | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Both sexes | 53.95 |
| 8             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Female     | 54.6  |
| 9             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SWZ      | Eswatini        | Year   | 2021       | Both sexes | 54.59 |
| 10            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | MOZ      | Mozambique      | Year   | 2021       | Male       | 54.8  |
| 11            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Female     | 55.38      |       |
| 12            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | GNB      | Guinea-Bi Year  | 2021   | Male       | 56.08      |       |
| 13            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | ZWE      | Zimbabwe        | Year   | 2021       | Male       | 56.19 |
| 14            | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Female     | 56.53 |
| 15            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SSD      | South Sud Year  | 2021   | Male       | 56.48      |       |
| 16            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | NAM      | Namibia         | Year   | 2021       | Male       | 57.27 |
| 17            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | NAM      | Namibia         | Year   | 2021       | Male       | 57.27 |

The data on the website, as well as the downloaded data in CSV format, both provide the point estimate with 95% confidence interval:

Visualisations

Data

Metadata

Related indicators

Filters

Last updated: 2024-08-02

| Indicator   | Life expectancy at birth (years) |                  |                  | Life expectancy at age 60 (years) |                  |                  |
|-------------|----------------------------------|------------------|------------------|-----------------------------------|------------------|------------------|
|             | Both sexes                       | Male             | Female           | Both sexes                        | Male             | Female           |
| Afghanistan | 59.1 [58.3-59.9]                 | 57.4 [56.6-58.2] | 61.0 [60.2-61.7] | 13.7 [13.2-14.2]                  | 13.6 [13.1-14.1] | 13.8 [13.3-14.3] |
| 2021        | 60.5 [59.8-61.5]                 | 59.3 [58.6-60.3] | 61.7 [60.9-62.4] | 13.8 [13.1-14.4]                  | 13.6 [13.0-14.1] | 14.0 [13.3-14.4] |
| 2020        | 61.2 [60.3-62.2]                 | 60.0 [59.3-61.1] | 62.5 [61.7-63.5] | 15.2 [14.5-15.7]                  | 15.4 [14.8-16.0] | 15.0 [14.4-15.7] |
| 2019        | 60.5 [59.7-61.5]                 | 59.0 [58.1-59.9] | 62.1 [61.3-63.0] | 15.1 [14.5-15.6]                  | 15.3 [14.8-15.9] | 14.9 [14.3-15.5] |
| 2018        | 60.9 [60.0-61.7]                 | 59.8 [59.0-60.9] | 62.1 [61.3-63.0] | 15.1 [14.4-15.6]                  | 15.4 [14.8-16.0] | 14.9 [14.3-15.5] |

EXP

Right

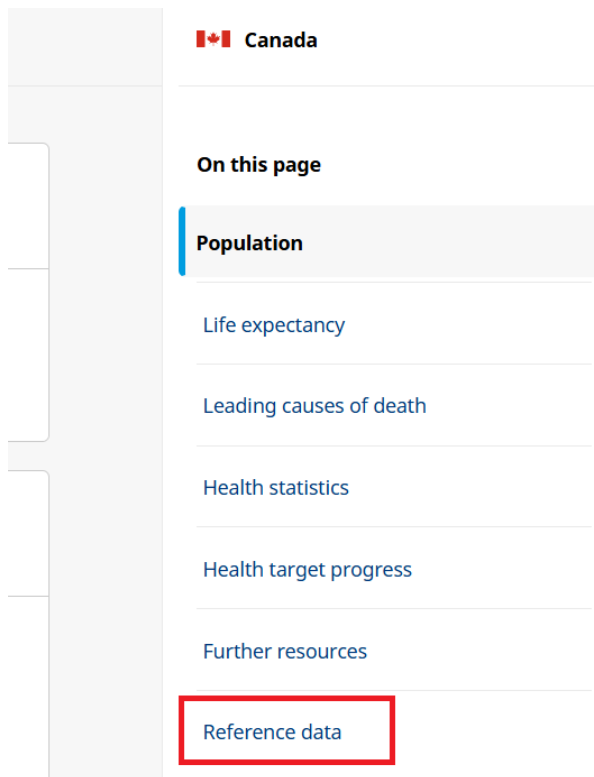
## Data by country example

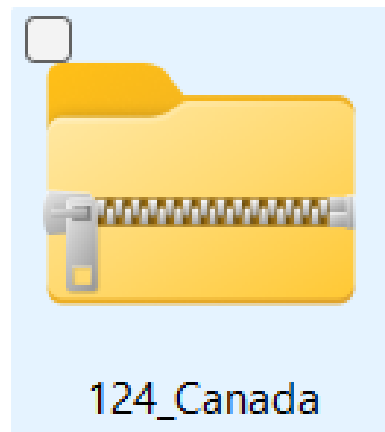
Let's click on **Canada** to explore the data for Canada.



You can explore different indicators for Canada. You can also download all indicators as a ZIP file by clicking **Reference data**.

[GHO Data by Country](#)





Once you unzip the folder, you will find several folders. Each folder contains a codelist, data, a data dictionary, along with a license and terms of use for different indicators.

|                                                                                         |                    |             |
|-----------------------------------------------------------------------------------------|--------------------|-------------|
| 1F96863_3.4.1 - cardiovascular disease, cancer, diabetes or chronic respiratory disease | 03-Mar-26 10:33 AM | File folder |
| 2D6FBE4_3.3.5 - Neglected tropical diseases                                             | 03-Mar-26 10:33 AM | File folder |
| 5C8435F_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 6A64C9A_7.1.2 - Clean fuels                                                             | 03-Mar-26 10:33 AM | File folder |
| 12EE54A_6.2.1 - Safely managed sanitation & hand-washing                                | 03-Mar-26 10:33 AM | File folder |
| 168BF41_3.4.2 - Suicide                                                                 | 03-Mar-26 10:33 AM | File folder |
| 45CA7C8_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 75DDA77_3.A.1 - Age-standardized prevalence of tobacco use                              | 03-Mar-26 10:33 AM | File folder |
| 77D059C_3.3.1 - HIV infections                                                          | 03-Mar-26 10:33 AM | File folder |
| 84FD3DE_3.9.3 - Unintentional poisoning                                                 | 03-Mar-26 10:33 AM | File folder |
| 608DE39                                                                                 | 03-Mar-26 10:33 AM | File folder |
| 1548EA3_6.1.1 - Safely managed drinking water                                           | 03-Mar-26 10:33 AM | File folder |
| 217795A_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 361734E_16.1.1 - Intentional homicide                                                   | 03-Mar-26 10:33 AM | File folder |
| 1772666_3.1.2 - Births attended by skilled health personnel                             | 03-Mar-26 10:33 AM | File folder |

For example, the 77D059C\_3.3.1 - HIV infections folder contains the following files:

|                                                                                                                                 |                    |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------|
|  124_77D059C_Code list_2025-03-26              | 13-Apr-26 11:43 PM |
|  124_77D059C_Dataset_2025-03-26                | 13-Apr-26 11:43 PM |
|  124_77D059C_License & Terms of Use_2025-03-26 | 13-Apr-26 11:43 PM |
|  124_77D059C_Metadata_2025-03-26               | 13-Apr-26 11:43 PM |
|  DDS - Data Dictionary 2024-12-12              | 13-Apr-26 11:43 PM |

Metadata

The GHO provides metadata for each indicator, providing important context for understanding the data and its limitations. For example, the metadata for the **Hepatitis - new infections** indicator includes information on the definition of the indicator, the methods used to calculate the indicator, data sources, and other relevant information.

[Indicator Metadata Registry List](#)

Data / GHO

Indicators

Search indicators

Hepatitis

Hepatitis

Hepatitis - new infections

Hepatitis - number of persons living with chronic hepatitis

Hepatitis - deaths caused by chronic hepatitis, number

Hepatitis - prevalence of chronic hepatitis among the general population

Hepatitis - deaths caused by chronic hepatitis B and C (per 100 000)

Data / GHO / Indicators

Hepatitis - new infections

Appears in: [Hepatitis cases and deaths](#)

Data

Metadata

Related indicators

Indicator name:

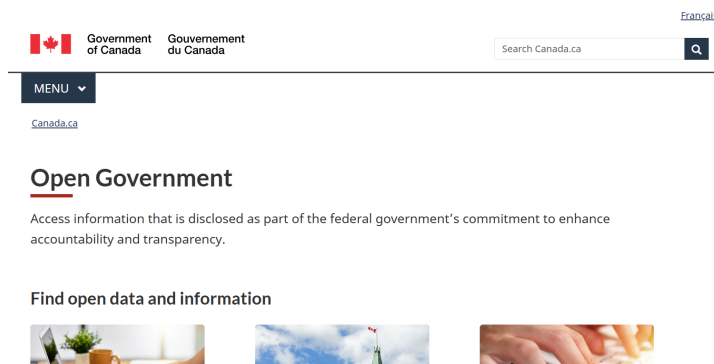
Hepatitis - (chronic HBV and HCV) incidence

Short name:

Chronic hepatitis incidence number

# Data Source: Open Portal

Open Government Portal Canada is a platform that provides access to searching for a wide range of public data. Users can explore and download data in different formats under the [Open Government Licence - Canada](#).

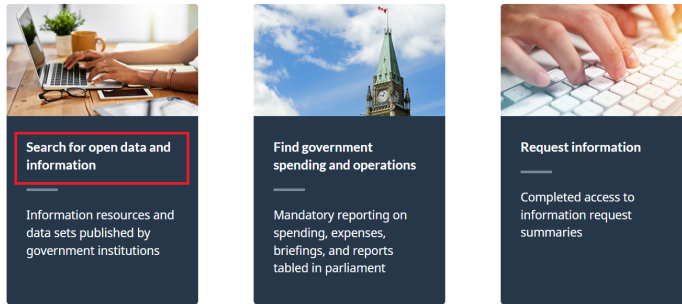


[Open Government Portal Canada](#)

## Search for Open Data

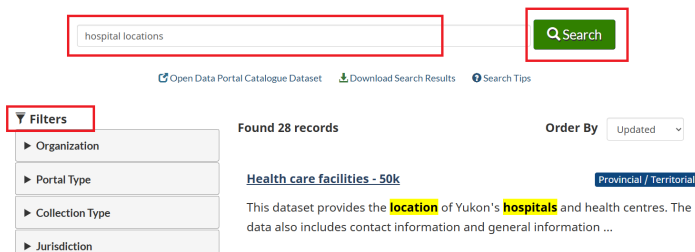
To search for open data, you can use the search bar on the Open Government Portal Canada homepage. For example, if you are interested in finding data on “hospital locations”, you can enter that term in the search bar and click the search icon. There are many filter options to narrow your search results, such as format and organization.

## Find open data and information



[Canada.ca](#) > [Open Government](#)

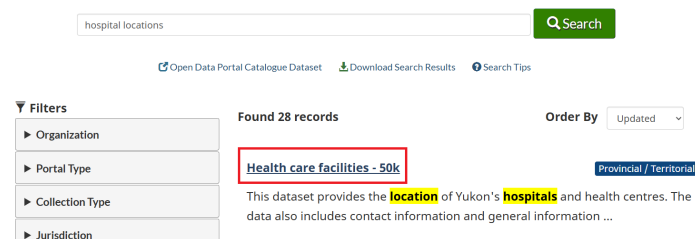
## Open Government Portal



To explore a specific dataset, click its title. This will take you to a page with more information about the dataset, including its description, format options, and download links. For example, if you click on the “Health care facilities - 50k” dataset, you will find detailed information about the location of Yukon’s hospitals and health centres.

[Canada.ca](#) > [Open Government](#)

## Open Government Portal



## Health care facilities - 50k

### Have your say

★★★★★

[Rate this dataset](#)  
[Provide feedback](#)

This dataset provides the location of Yukon's hospitals and health centres. The data also includes contact information and general information about each facility.

Distributed from GeoYukon by the Government of Yukon. Discover more digital map data and interactive maps from Yukon's digital map data collection. For more information: [geomatics.help@yukon.ca](mailto:geomatics.help@yukon.ca)

Under the **Data** and **Resources** tab, you can find the available formats for downloading the location of Yukon's hospitals and health centres.

### Data and Resources

View on Map

Search resources...

**ArcGIS Online layers**

English web service ESRI REST

Explore

**ArcGIS Online layers**

English web service ESRI REST

Explore

**Shape files**

English dataset HTML

Explore

**XML metadata**

English dataset HTML

Explore

**Original metadata (<https://open.yukon.ca>)**

English dataset HTML

Explore

Similarly, say we are interested in “opioid use” and want to find open data related to that topic. We can enter “opioid use” into the search bar and filter by **Open data**:

opioid use

Search

[Open Data Portal Catalogue Dataset](#)
[Download Search Results](#)
[Search Tips](#)

Filters

Portal Type - Open Data

Organization

Portal Type

☒ Open Data 10
 ☐ Open Information 34

Collection Type

Found 10 records

Order By Updated

Opioid agonist treatment archives

Federal

Correctional Service Canada provides Opiate Agonist Treatment to patients with an **opioid use** disorder. Archives of these quarterly web pages ...

Record Modified: Apr 9, 2026 Record Released: Nov 28, 2025

Publisher: Correctional Service of Canada

Formats: [CSV](#) [PDF](#)

Keywords: CSC OAT OUD opiate treatment Suboxone Methadone Sublocade health drugs institutions



To explore the data on opioid agonist treatment, we can click the dataset title. This will take us to a page with more information about the dataset, including its description, format options, and download links.

opiod use Search

[Open Data Portal Catalogue Dataset](#) [Download Search Results](#) [Search Tips](#)

**Filters**

Portal Type: Open Data

Organization

Portal Type

☒ Open Data 10

☐ Open Information 34

Collection Type

**Found 10 records**

Order By Updated

**Opioid agonist treatment archives** Federal

Correctional Service Canada provides Opiate Agonist Treatment to patients with an **opiod use** disorder. Archives of these quarterly web pages ...

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Publisher: Correctional Service of Canada

Formats: [CSV](#) [PDF](#)

Keywords: CSC OAT OUD opiate treatment Suboxone Methadone Sublocade health drugs institutions

## Data and Resources

Search resources... Search

**Opioid Agonist Treatment (OAT) in CSC: July 2019** Explore

English dataset CSV

**Opioid Agonist Treatment (OAT) in CSC: July 2019** Explore

French dataset CSV

**Opioid Agonist Treatment: December 2019** Explore

English dataset CSV

**Opioid Agonist Treatment: December 2019** Explore

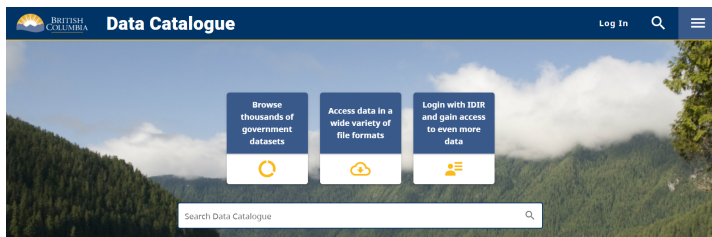
French dataset CSV

**Opioid Agonist Treatment: March 2020** Explore

English dataset CSV

# Data Source: BC Data Catalogue

The British Columbia Data Catalogue is a central hub provided by the Government of British Columbia (BC). Its primary purpose is to provide public access to thousands of datasets managed by the provincial government and its partners. Essentially, it is a digital library designed to make government information transparent and usable for citizens, researchers, and developers.

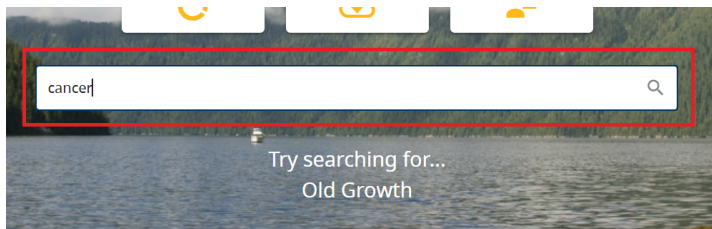


[British Columbia Data Catalogue](#)

## Search for Data

To search for data, you can use the search bar on the BC Data Catalogue homepage. For example, if you are interested in finding data on “cancer”, you can enter that term in the search bar and click the search icon.

[Search Data Catalogue](#)



**BRITISH COLUMBIA Data Catalogue** Log In

Search: cancer

1 dataset found matching "cancer" Order By: Relevance

**Metadata for BC Cancer Registry**  
 Jan 1985 to Dec 2022 the BC Cancer Registry is a population-based dataset of all new reportable cancers diagnosed among BC residents. The Registry contains diagnosis information, as well as personal and demographic information...  
 Data Innovation Program (DIP)  
 First Published: 2020-01-14  
 Data Types: csv

**Filter Search Results**  
 Expand All | Collapse All | Reset  
 Licence  
 Open Government Licence - British Columbia  
 Statistics Canada Open Licence Access Only  
 Open Government Licence - Canada  
 King's Printer Licence - British Columbia

Similarly, say we are interested in “COVID-19” and want to find open data related to that topic. We can enter “covid” into the search bar:

Search: covid

8 datasets found matching "covid" Order By: Relevance

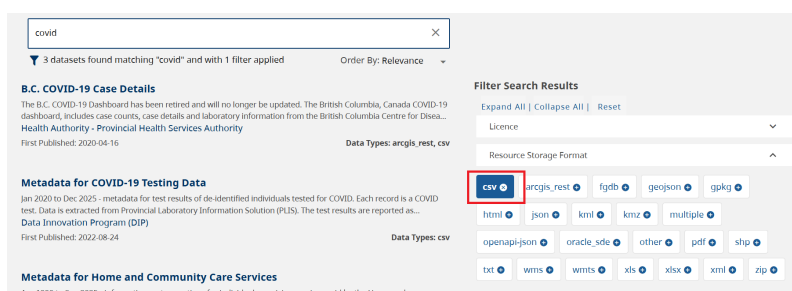
**British Columbia COVID-19 Dashboard**  
 The B.C. COVID-19 Dashboard has been retired and will no longer be updated. The British Columbia, Canada COVID-19 dashboard, includes case counts, case details and laboratory information from the British Columbia Centre for Disease...  
 Health Authority - Provincial Health Services Authority  
 First Published: 2020-04-15  
 Data Types: other

**B.C. COVID-19 Case Details**  
 The B.C. COVID-19 Dashboard has been retired and will no longer be updated. The British Columbia, Canada COVID-19 dashboard, includes case counts, case details and laboratory information from the British Columbia Centre for Disease...  
 Health Authority - Provincial Health Services Authority  
 First Published: 2020-04-16  
 Data Types: arcgis\_rest, csv

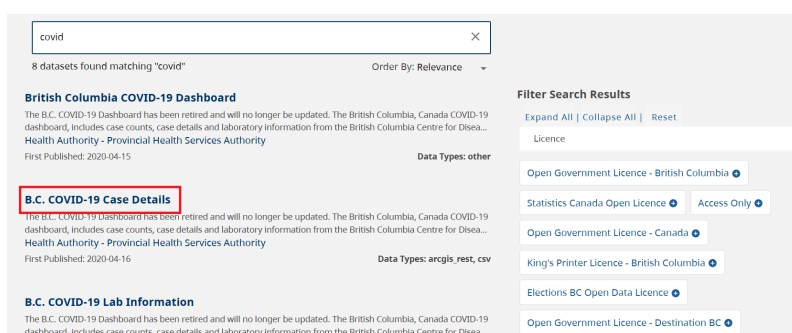
**B.C. COVID-19 Lab Information**  
 The B.C. COVID-19 Dashboard has been retired and will no longer be updated. The British Columbia, Canada COVID-19 dashboard, includes case counts, case details and laboratory information from the British Columbia Centre for Disease...

**Filter Search Results**  
 Expand All | Collapse All | Reset  
 Licence  
 Open Government Licence - British Columbia  
 Statistics Canada Open Licence Access Only  
 Open Government Licence - Canada  
 King's Printer Licence - British Columbia  
 Elections BC Open Data Licence  
 Open Government Licence - Destination BC

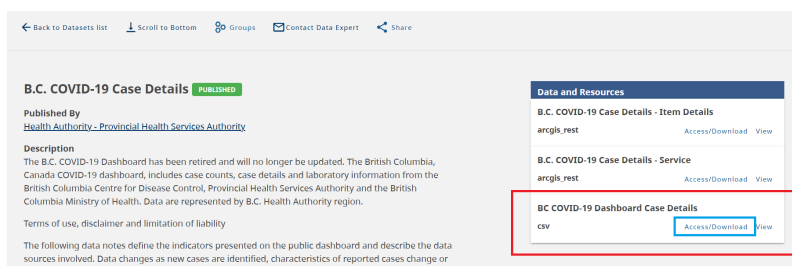
You can filter your search by license, format, group, and so on:



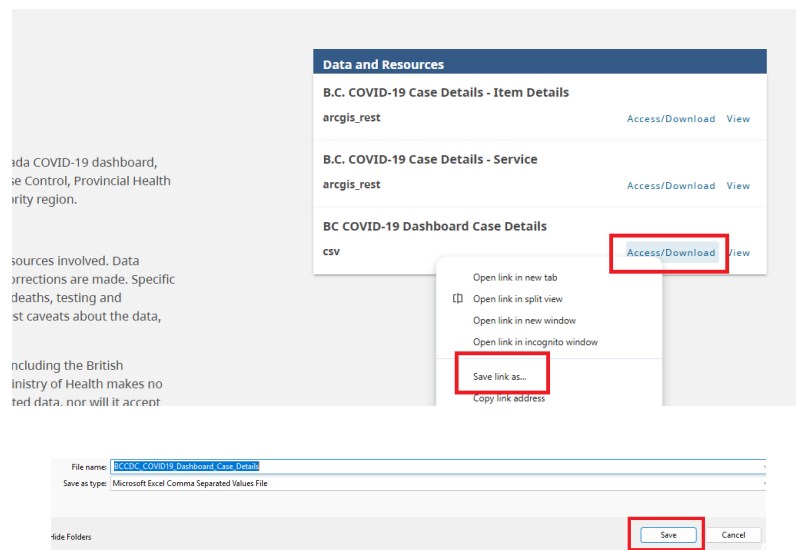
Let's explore the "COVID-19 Case Details" dataset. To do that, we can click the dataset title. This will take us to a page with more information about the dataset, including its description and related links.



You will find information on COVID-19 case details as well as a link to download the data:



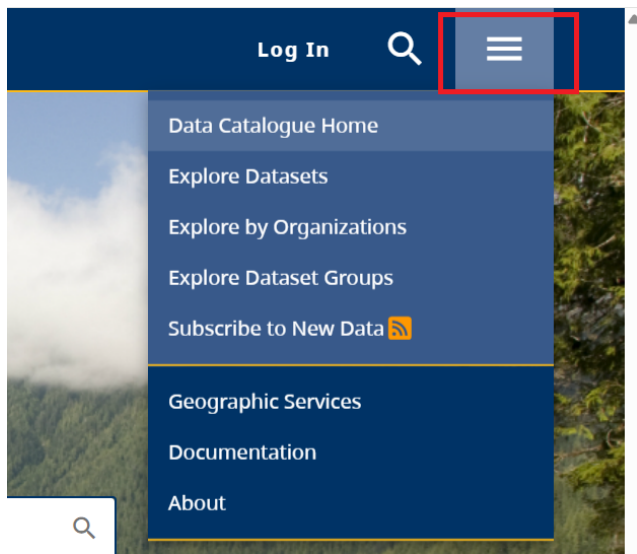
To download the dataset, right-click on the **Access/Download** link and select **Save link as...** This will allow you to save the dataset to your computer.



After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual case and each column representing different variables.

| 1  | Reported_Date | HA                | Sex | Age_Group | Classification_Reported |
|----|---------------|-------------------|-----|-----------|-------------------------|
| 2  | 29-01-20      | Out of Canada     | M   | 40-49     | Lab-diagnosed           |
| 3  | 06-02-20      | Vancouver Coastal | F   | 50-59     | Lab-diagnosed           |
| 4  | 10-02-20      | Out of Canada     | M   | 30-39     | Lab-diagnosed           |
| 5  | 10-02-20      | Out of Canada     | F   | 20-29     | Lab-diagnosed           |
| 6  | 18-02-20      | Interior          | F   | 30-39     | Lab-diagnosed           |
| 7  | 24-02-20      | Fraser            | F   | 30-39     | Lab-diagnosed           |
| 8  | 24-02-20      | Fraser            | M   | 40-49     | Lab-diagnosed           |
| 9  | 03-03-20      | Fraser            | M   | 50-59     | Lab-diagnosed           |
| 10 | 03-03-20      | Vancouver Coastal | F   | 60-69     | Lab-diagnosed           |
| 11 | 03-03-20      | Vancouver Coastal | F   | 30-39     | Lab-diagnosed           |
| 12 | 03-03-20      | Vancouver Coastal | F   | 80-89     | Lab-diagnosed           |
| 13 | 03-03-20      | Vancouver Coastal | M   | 60-69     | Lab-diagnosed           |
| 14 | 05-03-20      | Out of Canada     | F   | 50-59     | Lab-diagnosed           |
| 15 | 05-03-20      | Vancouver Coastal | M   | 60-69     | Lab-diagnosed           |
| 16 | 05-03-20      | Fraser            | F   | 50-59     | Lab-diagnosed           |
| 17 | 05-03-20      | Vancouver Coastal | M   | 30-39     | Lab-diagnosed           |

To learn more about BC Data Catalogue, click on the three bars on the right side of the page:



## **Part XVII**

# **Assistant Playbook**

# Assistant Playbook

This playbook is for instructors, TAs, and AI assistants helping maintain the HDSx course materials.

## Operating Principles

- Treat the syllabus as the source of truth for weeks, deliverables, and deadlines.
- Keep student-facing content beginner-accessible and cloud-first.
- Prefer reproducible Quarto, R, Python, and GitHub Codespaces workflows over local-machine assumptions.
- Keep generated output separate from source content.
- Add content where the normalized structure says it belongs: weeks, assignments, milestones, references, examples, or assets.

## Editing Rules

- Update `_quarto.yml` whenever a page should appear in the book.
- Keep paths relative and test them after moving files.
- Use `assets/` for shared images and data that support teaching pages.
- Use `examples/` for runnable code and sample datasets.
- Do not bury assignment requirements only inside the syllabus; add a student-facing assignment or milestone brief.



## Reproducibility Expectations

- Every activity should state expected inputs, outputs, and submission location.
- Every code activity should run in a fresh GitHub Codespace.
- Every dataset activity should record source, license, access date, and limitations.
- Every AI-assisted activity should include an audit note: what AI helped with, what changed, and how the result was verified.

## Review Cadence

- Before term: verify links, screenshots, Codespace setup, dependencies, and all assignment briefs.
- Weekly: update troubleshooting notes based on student issues.
- Before each milestone: test the milestone instructions from a fresh repository.
- After term: archive changes, refresh screenshots, and update the completion matrix.

# Definition-of-Done Checklists

## Course Page Done

- The page appears in `_quarto.yml` if it is student-facing.
- The title matches the week or reference purpose.
- Learning goals, activity steps, and expected outputs are clear.
- Links and image paths render from the Quarto project root.
- TODO blocks are either resolved or clearly scoped.

## Assignment Done

- The assignment has a student-facing brief.
- Required files and submission location are explicit.
- The work can be completed in GitHub Codespaces.
- Packages, data, and outputs are documented.
- The grading checklist matches the syllabus assessment logic.

## Milestone Done

- The milestone states the exact artifact students submit.
- The milestone maps to the final portfolio.
- The rubric includes reproducibility, auditability, data stewardship, and KT value.
- Peer or TA review instructions are actionable.

## Reproducibility Done

- A fresh Codespace can render or run the project.
- Paths are relative.
- Dependencies are declared with `renv`, `requirements.txt`, or a clearly documented equivalent.
- No required data exists only on one person's local computer.
- Randomness uses a seed when results depend on it.
- The README explains how to reproduce the output.

## AI Audit Done

- The AI-use note says what tool was used and for what task.
- Generated code was checked against real column names and expected outputs.
- Generated summaries are grounded in actual results.
- Security, privacy, and bias risks are considered.
- The final text does not cite sources invented by AI.

# NHANES Equity Case Study

## Teaching Notes

### Purpose

This note gives RAs and instructors the teaching plan for the recurring NHANES Health Equity Dashboard case study. The case study should function as a course spine: students revisit the same app, data, and workflow from different angles as their skills mature. It should not become a second term project.

### What Was Implemented

The case study now lives in one self-contained folder:

- `examples/nhanes-equity/app/app.R`
- `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- `examples/nhanes-equity/scripts/build_nhanes_equity.R`
- `examples/nhanes-equity/snippets/case-study-data-only.qmd`
- `examples/nhanes-equity/snippets/case-study-data-analysis.qmd`
- `examples/nhanes-equity/snippets/case-study-dashboard.qmd`
- `examples/nhanes-equity/README.md`

The app was adapted from `examples/bmi39.R`. The original task brief referred to `examples/bmi.39.R`, but that file was not present in this checkout.

The cached data are stored in `examples/nhanes-equity/data/` so the case-study app, script, README, and weekly snippets all use one consistent location. Legacy copies under `examples/data/` were left untouched to avoid breaking older references.

The weekly index pages for Weeks 1-11 now include phased case-study sections. Weeks 1-7 use data-first headings and snippets; Weeks 8-11 use the dashboard-forward snippet once app work becomes central.

## Pedagogical Design

The strongest use of this case study is incremental. Students should first learn to inspect and document a provided artifact, then audit and compare code, then critique visual and KT choices, and only later perform a fresh-run reproducibility stress test.

The progression is:

- Weeks 1-2: provenance, ethics, folder structure, relative paths, and rendering; no app run instructions.
- Weeks 3-5: data loading, missingness checks, AI audit comments, documentation-only Git changes, and R/Python parity checks using the CSV snapshot.
- Weeks 6-7: visualization critique, flawed-plot audit, EDA, Table 1, and planted-error audit.
- Week 8: first week where running the Shiny app is central.
- Weeks 9-10: slides, reports, source notes, limitations, and audience-specific communication.
- Week 11: fresh-session reproducibility testing and optional rebuild from CDC if internet access is available.

The cached RDS is the default classroom path. Rebuilding from CDC through `nhanesA` is valuable, but it is network-dependent and can be slow. Treat CDC rebuilds as an instructor demo or Week 11 challenge unless the course explicitly allocates time for them.

## Week-by-Week Instructor Notes

Week 1 introduces the case study through the Data Intake Card. Keep the focus on provenance, stewardship, and KT framing. Students do not need to run the app.

Week 2 uses the case study to make repo structure concrete. Students should locate the app, data, script, and snippet; confirm that paths are relative; and render the book.

Week 3 uses the CSV snapshot as a code-reading and data-inspection object. Students load, inspect, summarize, and ask AI to comment on code, then audit the comments against the actual operations.

Week 4 uses the case study for Git practice. The safest PR-sized change is documentation-only: label, caption, README wording, or a limitation note. Avoid asking students to add new analytic variables during the Git week.

Week 5 uses the CSV snapshot for R/Python parity. The goal is not Python mastery; it is learning how plausible outputs can diverge because of missing values, data types, or grouping choices.

Week 6 uses the worked ggplot example, flawed instructor plot, and visual-audit activity. Prefer instructor-provided flawed examples over asking students to invent misleading visuals from scratch.

Week 7 uses the worked Table 1 example and planted-error starter. Survey weights, strata, and PSU columns can be identified, but design-based standard errors should be instructor/demo material unless already taught.

Week 8 is the first app-forward week. Students should run the app, identify an audience, and improve one audience-facing label, help text, caption, or interpretation note. Deployment should remain optional unless explicitly assigned.

Week 9 uses one dashboard plot or table as the basis for a short slide explanation. Students should include a plain-language takeaway, a limitation, and a source note.

Week 10 turns the dashboard into a companion report. The report should document data, methods, limitations, and source notes rather than duplicating the dashboard interface.

Week 11 stress-tests reproducibility. Students should launch the app from a fresh session, confirm the cached RDS path,

and optionally rebuild from CDC if internet access and time allow.

## What RAs Should Watch For

- Students may confuse the cached case-study dataset with raw NHANES microdata.
- The app's Download CSV button exports the currently summarized table output, not raw NHANES microdata.
- The CDC rebuild path requires internet access and the `nhanesA` package.
- Week pages should include the shared snippet rather than duplicating instructions.
- Weeks 1-2 should not instruct students to run the app.
- Week 8 is the first week where app running is central.
- Student changes should stay small unless the instructor explicitly assigns a dashboard feature.
- Relative paths are part of the lesson; discourage absolute machine-specific paths.

## TODO Decisions

- TODO: Decide whether the legacy files in `examples/data/` should stay for backward compatibility or be retired after all references are updated.
- TODO: Decide whether `examples/bmi39.R` remains as a historical source file or becomes a short redirect note to `examples/nhanes-equity/app/app.R`.
- TODO: Decide whether CDC-rebuilt data should be committed each term or treated as generated output.
- TODO: Decide whether Week 5's R/Python parity check is optional practice or part of an assignment.
- TODO: Decide whether Week 8 includes only local app use or also demonstrates deployment.
- TODO: Decide whether the Week 6 flawed plot is distributed as code, image, or both.

- TODO: Decide whether the Week 7 planted-error starter is distributed from the instructor folder or copied into student assignment repos.
- TODO: Choose any additional required Table 1 variables for Week 7 beyond N, BMI mean (SD), and age mean (SD).

## Definition of Done for This Case Study

- The book renders from the repo root.
- Weeks 1-11 each include a short case-study touchpoint.
- The case-study folder contains runnable app code, cached data, rebuild script, shared snippet, and README.
- Weeks 6-7 include worked examples, student-facing activities, and instructor-only keys.
- Students can complete the default classroom path without internet.
- CDC rebuilds are documented as optional, network-dependent reproducibility work.



# 41 Syllabus-to-Content Coverage Audit

Source of truth: `syllabus/SPPH-381H-Course-Outline-v4.qmd` and `syllabus/syllabus-inventory-completion-matrix.md`.

Scope scanned: `_quarto.yml`, `weeks/**/*.qmd`, `references/**/*.qmd`, `assignments/**/*.qmd`, and `milestones/**/*.qmd`.

Decision rule: **Complete** requires both narrative coverage and a runnable example, activity, exercise, or concrete student output. **Needs Improvement** means content exists but is incomplete, scaffold-only, TODO-heavy, or missing the exercise/example needed by the syllabus. **Needs Review** means evidence is weak, ambiguous, or outside the primary week/module.

## 41.1 1) Executive Summary

### 41.1.1 Status Counts

- **Complete:** 3 topics
- **Needs Improvement:** 11 topics
- **Not Covered:** 0 topics
- **Needs Review:** 1 topic

### 41.1.2 Top 10 Highest-Priority Gaps

1. **Weeks 6-11 are scaffold-heavy.** Each has an overview, but most pages still contain `TODO Before Release / Content` `TODO` blocks rather than worked examples.

2. **Assignment briefs A1-A10 are stubs.** Every assignment README says `TODO: Build from assignments/assignment-template.qmd`.
3. **Milestone briefs M0-M5 and Final Portfolio are stubs and not in the Quarto TOC.** The milestone index is in the book, but the individual README briefs are orphaned.
4. **renv is promised in Week 3 but not taught with an activity.** Evidence is limited to assignment focus text and syllabus wording.
5. **Week 2 workflow mismatch: syllabus says fork a template, current content emphasizes GitHub Classroom.**
6. **Week 5 R deepening is weak.** Python/pandas coverage is substantial, but the syllabus promise of R mastery exercises on `dplyr` and functions is not clearly implemented in Week 5.
7. **Visualization lacks runnable ggplot2/plotly examples.** Week 6 explicitly says to add worked examples and flawed/corrected AI plots.
8. **Dashboard pathway is undecided.** Week 8 says to decide between Quarto dashboard, Shiny, or static HTML/JS.
9. **Reporting/publishing lacks concrete citation and GitHub Pages workflow.** Week 10 lists these as TO-DOs.
10. **Presentation weeks lack rubric/timing/submission details.** Weeks 12-13 contain overviews but still call for presentation timing rules, peer feedback forms, and rubric links.

## 41.2 2) Coverage Matrix

### 41.2.1 Week 0

- **Topic name (exact from syllabus):** Onboarding
- **Status:** Needs Improvement
- **Evidence:**

- syllabus/syllabus-inventory-completion-matrix.md / Extracted Schedule: “GitHub account, orientation video, pre-course survey.”
- weeks/week00-onboarding/index.qmd / Required Student Tasks: “Create a GitHub account”; “Watch the orientation video”; “Complete the pre-course survey”.
- weeks/week00-onboarding/index.qmd / Content TODO: “Add the actual Canvas orientation video link or placeholder.”

- **What’s missing:**

- Actual orientation video link or durable placeholder.
- Pre-course survey link or instructions.
- Clear confirmation/submission route.

- **Recommended next additions:**

- Add a Canvas-link placeholder with release instructions.
- Add a short completion checklist students can self-verify.
- Add expected support/contact path for account setup failures.

### 41.2.2 Week 1

- **Topic name (exact from syllabus):** Health Data, KT & Ethics
- **Status:** Complete
- **Evidence:**

- weeks/week01-health-data-ethics/index.qmd / Overview: “Knowledge Translation (KT)” and “ethical landscape of modern health data”.
- weeks/week01-health-data-ethics/index.qmd / The Open Data Landscape: “Explore 2-3 portals”.
- weeks/week01-health-data-ethics/index.qmd / The Data Intake Card: fields for dataset\_name, portal\_url, licence\_terms, and stewardship\_notes.

- weeks/week01-health-data-ethics/activity-data-intake-card.qmd  
/ Student Template: copyable YAML-style template for the activity.
- references/data-sources/nhis.qmd / Data Source: NHIS; references/data-sources/nhanes.qmd  
/ Data Source: NHANES; plus other data-source reference pages.
- **What’s missing:**
  - The activity page still has **TODO Before Release**.
  - A short rubric for provenance/licensing/stewardship is requested but not present.
- **Recommended next additions:**
  - Add a student-facing rubric to activity-data-intake-card.qmd.
  - Add one filled example Data Intake Card.
  - Link the Week 1 page directly to relevant reference data-source pages.

### 41.2.3 Week 2

- **Topic name (exact from syllabus):** Modern Workflows
- **Status:** Complete
- **Evidence:**
  - weeks/week02-modern-workflows/index.qmd  
/ Getting Started: GitHub Account and Classroom: account and repository setup steps.
  - weeks/week02-modern-workflows/index.qmd /  
Launching Your Codespace: “Create Codespace”.
  - weeks/week02-modern-workflows/index.qmd /  
Concepts: IDE, Notebook, Script: compares IDE, script, notebook.
  - weeks/week02-modern-workflows/index.qmd  
/ Concepts: Literate Programming: “Quarto documents mix Markdown text with code chunks.”
  - weeks/week02-modern-workflows/index.qmd /  
Tutorial: Stage, Commit, and Sync: staged commit workflow.

- weeks/week02-modern-workflows/index.qmd / Assignment 1: Your First Quarto Report: concrete student task.

- **What’s missing:**

- Syllabus says “Fork course template repo”; current content says GitHub Classroom/repository.
- Individual assignment folder is a stub even though the week page has a full assignment section.

- **Recommended next additions:**

- Decide whether the official workflow is fork-based or GitHub Classroom, then align Week 2 and syllabus language.
- Move or mirror the Week 2 assignment instructions into assignments/assignment01-workflows/README.md.

#### 41.2.4 Week 3

- **Topic name (exact from syllabus):** R with AI

- **Status:** Needs Improvement

- **Evidence:**

- weeks/week03-r-with-ai/index.qmd / Core Constructs: variables, functions, and pipes.
- weeks/week03-r-with-ai/index.qmd / Data Import and First Contact: read\_csv("data/patients.csv").
- weeks/week03-r-with-ai/index.qmd / Data Transformation: The Five dplyr Verbs: filter, select, mutate, summarise, group\_by.
- weeks/week03-r-with-ai/index.qmd / Writing a Simple Function: reusable summary function.
- weeks/week03-r-with-ai/index.qmd / Debugging: Reading the Red Text: “Fix It” loop.
- weeks/week03-r-with-ai/index.qmd / Assignment: The "Mad Libs" Report: step-by-step workflow and deliverable.
- assignments/assignment02-r-with-ai/README.md: “R basics, tidyverse transformations, custom functions, AI debugging, and renv awareness.”

- **What's missing:**
  - `renv` is promised in the syllabus as “intro to `renv` for dependency management” and “initialize `renv`”, but no worked `renv::init()` / restore activity was found.
  - Assignment 2 README is still a TODO stub.
- **Recommended next additions:**
  - Add a short `renv` tutorial/checkpoint to Week 3.
  - Add `renv.lock` expectations and a recovery note.
  - Expand `assignments/assignment02-r-with-ai/README.md` from the template.

#### 41.2.5 Week 4

- **Topic name (exact from syllabus):** Git & Collaboration
- **Status:** Complete
- **Evidence:**
  - `weeks/week04-git-collaboration/index.qmd` / Save Game of Reproducible Science: Git as provenance.
  - `weeks/week04-git-collaboration/index.qmd` / Repo Hygiene: `.gitignore` and relative paths.
  - `weeks/week04-git-collaboration/index.qmd` / Branching: Safe Experimentation: branch creation/merge tutorial.
  - `weeks/week04-git-collaboration/index.qmd` / Pull Requests: The Peer Review Gateway: PR concept and tutorial.
  - `weeks/week04-git-collaboration/index.qmd` / Setting Up Your Group Repository: term project group repository steps.
  - `weeks/week04-git-collaboration/index.qmd` / Practice Exercise: The Full Workflow Drill: concrete exercise.
- **What's missing:**
  - M0 brief exists but is a TODO stub.

- Assignment 3 README exists but is a TODO stub.
- **Recommended next additions:**
  - Expand `milestones/m0-group-formation/README.md` with submission evidence.
  - Expand `assignments/assignment03-git/README.md` with branch/PR screenshots or required proof.

### 41.2.6 Week 5

- **Topic name (exact from syllabus):** Polyglot Awareness & R Deepening
- **Status:** Needs Improvement
- **Evidence:**
  - `weeks/week05-polyglot-r-deepening/index.qmd`  
/ Python vs R--When and Why?: “R remains your primary analysis language.”
  - `weeks/week05-polyglot-r-deepening/index.qmd`  
/ Loading and Viewing Data: `pd.read_csv("data/patients.csv")`.
  - `weeks/week05-polyglot-r-deepening/index.qmd`  
/ Translation Traps: zero-index, indentation, assignment/equality.
  - `weeks/week05-polyglot-r-deepening/index.qmd`  
/ The AI Workflow: One Step, One Cell: parity audit rules.
  - `weeks/week05-polyglot-r-deepening/python13-translation-exercise.qmd`  
/ The Bilingual Translation Exercise: practice exercise.
  - `assignments/assignment04-polyglot/README.md`:  
“reading Python/pandas, translating a small R task, and verifying parity.”
- **What’s missing:**
  - Week 5 has strong Python coverage but weak explicit R-deepening exercises.
  - Assignment 4 README is a TODO stub.
  - Dependency documentation is covered for Python `requirements.txt`, but the syllabus framing of R deepening is not clearly tied to a student output.

- **Recommended next additions:**
  - Add one Week 5 exercise that repeats a task in R and Python and compares outputs.
  - Add a specific R `dplyr`/function refinement task.
  - Expand Assignment 4 into a concrete submission brief.

### 41.2.7 Week 6

- **Topic name (exact from syllabus):** Data Visualization
- **Status:** Needs Improvement
- **Evidence:**
  - `weeks/week06-visualization/index.qmd` / Overview: “publication-quality visualizations” and “ggplot2 or plotly”.
  - `weeks/week06-visualization/index.qmd` / Content TODO: “Add a worked ggplot2 example”; “Add one intentionally misleading AI-generated chart”.
  - `weeks/week06-visualization/activity-ai-visual-audit.qmd` / Goal: evaluate “correctness, clarity, bias, aggregation choices, and misleading design.”
  - `assignments/assignment05-visualization/README.md`: “original plot, revised plot, and audit note.”
- **What’s missing:**
  - Worked ggplot2 example.
  - Plotly or interactive extension.
  - Flawed AI chart plus corrected version.
  - Concrete starter data and prompt.
- **Recommended next additions:**
  - Add a runnable ggplot2 tutorial using shared data.
  - Add one intentionally flawed chart and student audit questions.
  - Add a small plotly extension as optional enrichment.



### 41.2.8 Week 7

- **Topic name (exact from syllabus):** EDA, Table 1 & AI Auditing
- **Status:** Needs Improvement
- **Evidence:**
  - weeks/week07-eda-ai-audit/index.qmd /  
Overview: EDA and “standard Table 1”.
  - weeks/week07-eda-ai-audit/index.qmd /  
Content TODO: “Build a worked EDA”; “Add Table 1 expectations”; “Add a planted-error AI audit exercise.”
  - weeks/week07-eda-ai-audit/activity-table1-audit.qmd /  
Goal: “create a descriptive summary and audit AI-generated code”.
  - assignments/assignment06-eda-ai-audit/README.md:  
“EDA, Table 1 or equivalent descriptive summary, planted-error audit, and provenance notes.”
- **What’s missing:**
  - Worked EDA with data path.
  - Table 1 variables and example output.
  - Planted-error code/summary/visualization.
  - Instructor key outside student page.
- **Recommended next additions:**
  - Build around examples/data/ or a week-local starter dataset.
  - Add a runnable Table 1 example.
  - Add AI audit prompts and expected fixes.

### 41.2.9 Week 8

- **Topic name (exact from syllabus):** Dashboard Prototypes for KT
- **Status:** Needs Improvement
- **Evidence:**
  - weeks/week08-dashboard-kt/index.qmd /  
Overview: Shiny apps, dashboards, and KT.

- `weeks/week08-dashboard-kt/index.qmd` /  
Content TODO: “Decide the required core pathway”; “Add a minimal dashboard starter”.
- `weeks/week08-dashboard-kt/starter-dashboard.qmd` /  
Required Elements: audience-specific question, control, visualization, interpretation, limitation.
- `assignments/assignment07-dashboard/README.md`:  
“minimal dashboard-style KT product from EDA outputs.”

- **What’s missing:**

- Actual dashboard code.
- Required pathway decision: Quarto dashboard, Shiny, or static HTML/JS.
- Codespaces testing instructions.
- Privacy-safe publishing guidance.

- **Recommended next additions:**

- Add a minimal Quarto dashboard starter that renders.
- Add a fallback static version.
- State which pathway is required and which are optional.

#### 41.2.10 Week 9

- **Topic name (exact from syllabus):** Communication of Scientific Findings
- **Status:** Needs Improvement
- **Evidence:**

- `weeks/week09-science-communication/index.qmd` /  
Overview: “Audience-Specific KT”, plain language, slide decks, peer review.
- `weeks/week09-science-communication/index.qmd` /  
Content TODO: “Add a revealjs starter deck”; “Add a plain-language summary exercise.”
- `weeks/week09-science-communication/revealjs-peer-feedback.qmd` /  
Peer Feedback Prompts: main message, evidence, overclaiming, reproducibility/audience fit.

- assignments/assignment08-communication/README.md:  
“revealjs slides, plain-language summary, citations,  
and peer feedback.”
- **What’s missing:**
  - Minimal format: revealjs example.
  - Three-slide starter structure.
  - Plain-language summary exercise.
  - Citation and accessibility expectations.
- **Recommended next additions:**
  - Add a renderable revealjs starter.
  - Add a before/after plain-language translation activity.
  - Convert peer prompts into a reusable feedback form.

#### 41.2.11 Week 10

- **Topic name (exact from syllabus):** Writing & Publishing Reports
- **Status:** Needs Improvement
- **Evidence:**
  - weeks/week10-reporting-publishing/index.qmd  
/ Overview: dynamic report and publishing workflows.
  - weeks/week10-reporting-publishing/index.qmd  
/ Content TODO: report skeleton, citation examples, GitHub Pages setup, grounding audit.
  - weeks/week10-reporting-publishing/report-template.qmd  
/ Suggested Sections: question, data, methods, results, limitations, plain-language summary, AI-use note.
  - assignments/assignment09-reporting/README.md:  
“Quarto report, citations, GitHub Pages or private equivalent, and grounding audit.”
- **What’s missing:**
  - Copyable report starter with real YAML.
  - Bibliography setup and citation example.

- GitHub Pages setup or private submission alternative.
- Interactive element example.
- **Recommended next additions:**
  - Add a minimal report template that renders with `references.bib`.
  - Add a GitHub Pages checklist with screenshots or text-only steps.
  - Add citation grounding audit prompts.

#### 41.2.12 Week 11

- **Topic name (exact from syllabus):** Portfolio Surgery & Optional Advanced Deployment
- **Status:** Needs Improvement
- **Evidence:**
  - `weeks/week11-portfolio-surgery/index.qmd` / Overview: dependency management, auditability, and peer review.
  - `weeks/week11-portfolio-surgery/index.qmd` / Content TODO: fresh Codespace test, broken-project exercise, Assignment 10, Milestone 4, final checklist.
  - `weeks/week11-portfolio-surgery/reproducibility-test.qmd` / Failure Log Template: checks for repository, packages, report, dashboard, README.
  - `assignments/assignment10-portfolio-surgery/README.md`: “fresh Codespace test, dependency/path repair, README clarity”.
  - `milestones/m4-peer-review/README.md`: peer review covering clarity, reproducibility, auditability, KT value.
- **What’s missing:**
  - Step-by-step fresh Codespace protocol.
  - Broken-project debugging exercise.
  - Final portfolio readiness checklist.
  - Optional Python-flavoured pathway from syllabus is not clearly present here.

- **Recommended next additions:**
  - Expand `reproducibility-test.qmd` into a run-book.
  - Add a sample failure log with fixes.
  - Add a final portfolio checklist linked from Week 11 and final portfolio README.

### 41.2.13 Week 12

- **Topic name (exact from syllabus):** Term Project Presentations, Part 1
- **Status:** Needs Improvement
- **Evidence:**
  - `weeks/week12-presentations/index.qmd` / Overview: first presentation block and peer review.
  - `weeks/week12-presentations/index.qmd` / Content TODO: timing rules, slide submission, peer feedback form, rubric.
  - `milestones/m5-presentation/README.md`: “presentation slides, live Q&A, and peer review feedback.”
- **What’s missing:**
  - Presentation timing rules.
  - Slide submission instructions.
  - Peer feedback form.
  - Presentation rubric.
- **Recommended next additions:**
  - Add a M5 rubric and link it from Week 12.
  - Add a slide checklist and submission timing.
  - Add a simple peer feedback form.

### 41.2.14 Week 13

- **Topic name (exact from syllabus):** Term Project Presentations, Part 2
- **Status:** Needs Improvement

- **Evidence:**
  - weeks/week13-presentations/index.qmd / Overview: completes M5 and feedback before final portfolio.
  - weeks/week13-presentations/index.qmd / Content TODO: group schedule, final reminders, closing reflection prompt.
  - milestones/m5-presentation/README.md: “slides render before class and claims are grounded”.
- **What’s missing:**
  - Group schedule or placeholder.
  - Final portfolio reminders.
  - Closing reflection prompt.
- **Recommended next additions:**
  - Add a repeatable Part 2 logistics block.
  - Link to final portfolio checklist.
  - Add a reflection prompt on reproducibility and KT.

#### 41.2.15 Final

- **Topic name (exact from syllabus):** Final Portfolio
- **Status:** Needs Review
- **Evidence:**
  - syllabus/syllabus-inventory-completion-matrix.md / Extracted Schedule: “Final portfolio due Dec 11, Fri 4 PM.”
  - milestones/final-portfolio/README.md: “complete repository, reproducible report, dashboard-style KT product, presentation materials, and AI-use audit note.”
  - milestones/final-portfolio/README.md: TODO: Build from milestones/milestone-template.qmd.
  - milestones/index.qmd / milestone table: “Complete reproducible portfolio.”
- **What’s missing:**
  - Detailed final portfolio checklist.

- Submission instructions.
- Explicit mapping to reproducibility contract.
- Public/private portfolio guidance from syllabus.

- **Recommended next additions:**

- Convert the syllabus reproducibility contract into a final checklist.
- Expand `milestones/final-portfolio/README.md`.
- Add a source page to `_quarto.yml` or link from Week 13.

## 41.3 3) Deliverables Alignment

### 41.3.1 Weekly Assignments A1-A10

| Deliverable | Syllabus expectation                                                    | Existing repo page(s)                                                                                                                                                  | Alignment                          | Mismatch / risk                                                                                              |
|-------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
| A1          | Assignment 1 due Week 2; fork/template, Codespace, Quarto, com-mit/sync | <del>assignments/index.qmd</del> ; <del>assignments/assignment01-workflows/README.md</del> ; <del>weeks/week02-modern-workflows/index.qmd</del><br><b>Assignment 1</b> | <del>Week 1</del><br><b>Week 2</b> | task detail, but assignment README is a TODO stub. Syllabus says fork; Week 2 content says GitHub Classroom. |

| Deliverable | Syllabus expectation                                                                | Existing repo page(s)                                            | Alignment                                                        | Mismatch / risk                                                                                                       |
|-------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| A2          | R basics, tidyverse, functions, debugging, <code>renv</code> , Quarto documentation | <code>assignments/week03-overview-ai/index.qmd</code>            | <code>assignments/week03-overview-ai/index.qmd</code>            | Week-3 with-ai/README.md; strong, but assignment README is a TODO stub and <code>renv</code> lacks a worked activity. |
| A3          | Git collaboration and repo hygiene                                                  | <code>assignments/week04-git-collaboration/index.qmd</code>      | <code>assignments/week04-git-collaboration/index.qmd</code>      | Week-4 Git/README.md; strong, but assignment README is a TODO stub.                                                   |
| A4          | Polyglot awareness and R deepening                                                  | <code>assignments/week05-polyglot-overdeepening/index.qmd</code> | <code>assignments/week05-polyglot-overdeepening/index.qmd</code> | Week-5 Polyglot/README.md; is strong; assignment README is a TODO stub; R deepening is not explicit enough.           |





| Deliverable | Syllabus expectation                       | Existing repo page(s)                                                                                                                                                                       | Alignment                                                                                                                                                                                   | Mismatch / risk                                                             |
|-------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| A8          | Scientific communication and peer feedback | <del>assignments/</del> assignmentsXassignment08-communication/README.md;<br>weeks/week09-science-communication/index.qmd;<br>weeks/week09-science-communication/revealjs-peer-feedback.qmd | <del>assignments/</del> assignmentsXassignment08-communication/README.md;<br>weeks/week09-science-communication/index.qmd;<br>weeks/week09-science-communication/revealjs-peer-feedback.qmd | revealjs starter, plain-language exercise, and assignment brief are TODO.   |
| A9          | Reporting, citations, publishing           | <del>assignments/</del> assignmentsXassignment09-reporting-publishing/README.md;<br>weeks/week10-reporting-publishing/index.qmd;<br>weeks/week10-reporting-publishing/report-template.qmd   | <del>assignments/</del> assignmentsXassignment09-reporting-publishing/README.md;<br>weeks/week10-reporting-publishing/index.qmd;<br>weeks/week10-reporting-publishing/report-template.qmd   | citations, GitHub Pages/private alternative, and assignment brief are TODO. |
| A10         | Portfolio surgery and reproducibility test | <del>assignments/</del> assignmentsXassignment10-portfolio-surgery/index.qmd;<br>weeks/week11-portfolio-surgery/reproducibility-test.qmd                                                    | <del>assignments/</del> assignmentsXassignment10-portfolio-surgery/index.qmd;<br>weeks/week11-portfolio-surgery/reproducibility-test.qmd                                                    | step protocol and assignment brief are TODO.                                |

### 41.3.2 Milestones M0-M5 and Final Portfolio

| Deliverable | Syllabus expectation                                                       | Existing repo page(s)                                                                           | Alignment | Mismatch / risk                                                                                                                |
|-------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| M0          | Group formation; one member forks template and adds team-mates             | milestones/No group for Week 4; weeks/week04-git-collaboration/Setting Up Your Group Repository |           | Week 4n/README.md; index.qmd<br>setup steps; milestone README is a TODO stub and not in _quarto.yml.                           |
| M1          | Proposal: research question, dataset, feasibility, prove-nance/stewardship | milestones/No proposal; weeks/week05-polyglot-in-eda/                                           |           | Week 5 README; python11-project-milestone.qmd<br>is a TODO stub.<br>Week 5 has a checklist, but syllabus places M1 due Week 6. |
| M2          | Preliminary analysis: cleaning, EDA, Table 1, runnable repo                | milestones/No preliminary; weeks/week07-redant-audit/                                           |           | Milestone analysis/README.md; READMEEx.qmd<br>and Week 7 are TODO-heavy; no worked EDA/Table 1 example.                        |

| Deliverable | Syllabus expectation                                            | Existing repo page(s)                                                                                                                            | Alignment                                                                                                   | Mismatch / risk                                                                                                                             |
|-------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| M3          | Project update: draft dashboard-style KT product + report draft | <del>milestones/NeedsProject-M3</del><br><del>weeks/week10-reporting-README</del><br><del>weeks/week08-dashboardskt/index.qmd</del>              | <del>NeedsProject-M3</del><br><del>reporting-README</del><br><del>index.qmd</del>                           | <del>Update/README.md;</del><br><del>index.qmd;</del><br><del>index.qmd</del><br>TODO stub;<br>Week 8/10 content is still scaffolded.       |
| M4          | Peer review with reproducibility and auditability checks        | <del>milestones/NeedsIm-review-M4</del><br><del>weeks/week11-portfolio-Survey</del>                                                              | <del>NeedsIm-review-M4</del><br><del>portfolio-Survey</del>                                                 | <del>Update/README.md;</del><br><del>index.qmd</del><br>is a<br>TODO stub;<br>Week 11 asks to add peer review instructions.                 |
| M5          | Presentations, live Q&A, peer review                            | <del>milestones/NeedsPresentation-M5</del><br><del>weeks/week12-presentations/index.qmd</del><br><del>weeks/week13-presentations/index.qmd</del> | <del>NeedsPresentation-M5</del><br><del>presentations/index.qmd</del><br><del>presentations/index.qmd</del> | <del>Update/README.md;</del><br><del>index.qmd;</del><br><del>index.qmd</del><br>is a<br>TODO stub;<br>rubric and timing rules are missing. |

| Deliverable        | Syllabus<br>expecta-<br>tion                                                   | Existing<br>repo<br>page(s)                                                              | Alignment                                                    | Mismatch<br>/ risk                                                                                               |
|--------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Final<br>Portfolio | Complete<br>report,<br>dashboard-<br>style KT<br>product,<br>auditable<br>repo | <a href="#">milestones/Week1-portfolio</a><br><a href="#">milestones/Week2-portfolio</a> | <a href="#">Find1-portfolio</a><br><a href="#">Index.qmd</a> | <a href="#">Find1-portfolio</a><br>README<br>is a<br>TODO<br>stub and<br>not<br>referenced<br>in<br>_quarto.yml. |

## 41.4 4) Consistency Checks

### 41.4.1 Topics Present in Repo but Not Explicitly in Syllabus

- `references/data-sources/*`: detailed standalone data-source tutorials for NHIS, NHANES, BRFSS, CCS, CADS, CTNS, CSADS, PHAC Infobase, CCDSS, GHO, Open Portal, and BC Data Catalogue. These support Week 1 but are more extensive than the syllabus schedule.
- `assistant-playbook/*`: present in `_quarto.yml` but outside the requested scan inputs and not part of the syllabus topic schedule.
- `syllabus/syllabus-inventory-completion-matrix.md`: included in `_quarto.yml` as Course Planning; useful internally, but not a student topic from the syllabus schedule.
- Several Week 2-5 tutorial fragments are more granular than the syllabus topic names, especially `workflows01-*` through `workflows11-*`, `r01-*` through `r10-*`, `git01-*` through `git10-*`, and `python01-*` through `python14-*`.

### 41.4.2 Syllabus Topics in TOC but Missing or Weak Content Pages

- Week 6 through Week 13 all have TOC pages, but each primary week page still contains `TODO Before Release` and `Content TODO`.
- Assignment and milestone index/template pages are in the TOC, but individual A1-A10 and M0-M5 README briefs are not.
- `syllabus/SPPH-381H-Course-Outline-v4.qmd` is a source-of-truth file but is not referenced in `_quarto.yml`; only `syllabus/syllabus-inventory-completion-matrix.md` is in the book.
- Final Portfolio appears in `milestones/index.qmd` and `milestones/final-portfolio/README.md`, but there is no dedicated TOC entry for the final portfolio brief.

### 41.4.3 Orphan Pages Not Referenced in `_quarto.yml`

- `assignments/assignment01-workflows/README.md`
- `assignments/assignment02-r-with-ai/README.md`
- `assignments/assignment03-git/README.md`
- `assignments/assignment04-polyglot/README.md`
- `assignments/assignment05-visualization/README.md`
- `assignments/assignment06-eda-ai-audit/README.md`
- `assignments/assignment07-dashboard/README.md`
- `assignments/assignment08-communication/README.md`
- `assignments/assignment09-reporting/README.md`
- `assignments/assignment10-portfolio-surgery/README.md`
- `milestones/final-portfolio/README.md`
- `milestones/m0-group-formation/README.md`
- `milestones/m1-proposal/README.md`
- `milestones/m2-preliminary-analysis/README.md`
- `milestones/m3-project-update/README.md`
- `milestones/m4-peer-review/README.md`
- `milestones/m5-presentation/README.md`
- `syllabus/SPPH-381H-Course-Outline-v4.qmd`

## 41.5 Appendix A: Files Scanned

- `_quarto.yml`
- `syllabus/SPPH-381H-Course-Outline-v4.qmd`
- `syllabus/syllabus-inventory-completion-matrix.md`
- `weeks/week00-onboarding/index.qmd`
- `weeks/week01-health-data-ethics/index.qmd`
- `weeks/week01-health-data-ethics/activity-data-intake-card.qmd`
- `weeks/week02-modern-workflows/index.qmd`
- `weeks/week02-modern-workflows/workflows01-cloud-drive.qmd`
- `weeks/week02-modern-workflows/workflows02-repository-tour.qmd`
- `weeks/week02-modern-workflows/workflows03-open-codespace.qmd`
- `weeks/week02-modern-workflows/workflows04-reopen-codespace.qmd`
- `weeks/week02-modern-workflows/workflows05-vscode-orientation.qmd`
- `weeks/week02-modern-workflows/workflows06-files-folders.qmd`
- `weeks/week02-modern-workflows/workflows07-render-quarto.qmd`
- `weeks/week02-modern-workflows/workflows08-commit-work.qmd`
- `weeks/week02-modern-workflows/workflows09-troubleshooting.qmd`
- `weeks/week02-modern-workflows/workflows10-stop-codespace.qmd`
- `weeks/week02-modern-workflows/workflows11-checklist.qmd`
- `weeks/week03-r-with-ai/index.qmd`
- `weeks/week03-r-with-ai/r01-setup.qmd`
- `weeks/week03-r-with-ai/r02-packages.qmd`
- `weeks/week03-r-with-ai/r03-basics.qmd`
- `weeks/week03-r-with-ai/r04-control-flow.qmd`
- `weeks/week03-r-with-ai/r05-data-import.qmd`
- `weeks/week03-r-with-ai/r06-debugging.qmd`
- `weeks/week03-r-with-ai/r07-ai-audit-checklist.qmd`
- `weeks/week03-r-with-ai/r08-comments.qmd`
- `weeks/week03-r-with-ai/r09-assignment-mad-libs.qmd`
- `weeks/week03-r-with-ai/r10-final-checklist.qmd`
- `weeks/week04-git-collaboration/index.qmd`
- `weeks/week04-git-collaboration/git01-concepts.qmd`
- `weeks/week04-git-collaboration/git02-tutorial.qmd`
- `weeks/week04-git-collaboration/git03-repo-hygiene.qmd`
- `weeks/week04-git-collaboration/git04-branching.qmd`
- `weeks/week04-git-collaboration/git05-pull-requests.qmd`
- `weeks/week04-git-collaboration/git06-ai-auditing.qmd`
- `weeks/week04-git-collaboration/git07-group-repository.qmd`
- `weeks/week04-git-collaboration/git08-knowledge-check.qmd`

- weeks/week04-git-collaboration/git09-practice-exercise.qmd
- weeks/week04-git-collaboration/git10-quick-reference.qmd
- weeks/week05-polyglot-r-deepening/index.qmd
- weeks/week05-polyglot-r-deepening/python01-when-and-why.qmd
- weeks/week05-polyglot-r-deepening/python02-environment.qmd
- weeks/week05-polyglot-r-deepening/python03-packages-reproducibility.qmd
- weeks/week05-polyglot-r-deepening/python04-translation-traps.qmd
- weeks/week05-polyglot-r-deepening/python05-load-inspect-data.qmd
- weeks/week05-polyglot-r-deepening/python06-exceptions.qmd
- weeks/week05-polyglot-r-deepening/python07-one-step-one-cell.qmd
- weeks/week05-polyglot-r-deepening/python08-mixing-r-python.qmd
- weeks/week05-polyglot-r-deepening/python09-reusable-scripts.qmd
- weeks/week05-polyglot-r-deepening/python10-git-progress.qmd
- weeks/week05-polyglot-r-deepening/python11-project-milestone.qmd
- weeks/week05-polyglot-r-deepening/python12-knowledge-check.qmd
- weeks/week05-polyglot-r-deepening/python13-translation-exercise.qmd
- weeks/week05-polyglot-r-deepening/python14-cheat-sheet.qmd
- weeks/week06-visualization/index.qmd
- weeks/week06-visualization/activity-ai-visual-audit.qmd
- weeks/week07-eda-ai-audit/index.qmd
- weeks/week07-eda-ai-audit/activity-table1-audit.qmd
- weeks/week08-dashboard-kt/index.qmd
- weeks/week08-dashboard-kt/starter-dashboard.qmd
- weeks/week09-science-communication/index.qmd
- weeks/week09-science-communication/revealjs-peer-feedback.qmd
- weeks/week10-reporting-publishing/index.qmd
- weeks/week10-reporting-publishing/report-template.qmd
- weeks/week11-portfolio-surgery/index.qmd
- weeks/week11-portfolio-surgery/reproducibility-test.qmd
- weeks/week12-presentations/index.qmd
- weeks/week13-presentations/index.qmd
- references/data-sources/nhis.qmd
- references/data-sources/nhanes.qmd
- references/data-sources/brfss.qmd
- references/data-sources/ccs.qmd
- references/data-sources/cads.qmd
- references/data-sources/ctns.qmd
- references/data-sources/csads.qmd
- references/data-sources/phac-infobase.qmd
- references/data-sources/ccdss.qmd
- references/data-sources/gho.qmd



- `references/data-sources/open-portal.qmd`
- `references/data-sources/bc-data-catalogue.qmd`
- `assignments/index.qmd`
- `assignments/assignment-template.qmd`
- `assignments/assignment01-workflows/README.md`
- `assignments/assignment02-r-with-ai/README.md`
- `assignments/assignment03-git/README.md`
- `assignments/assignment04-polyglot/README.md`
- `assignments/assignment05-visualization/README.md`
- `assignments/assignment06-eda-ai-audit/README.md`
- `assignments/assignment07-dashboard/README.md`
- `assignments/assignment08-communication/README.md`
- `assignments/assignment09-reporting/README.md`
- `assignments/assignment10-portfolio-surgery/README.md`
- `milestones/index.qmd`
- `milestones/milestone-template.qmd`
- `milestones/m0-group-formation/README.md`
- `milestones/m1-proposal/README.md`
- `milestones/m2-preliminary-analysis/README.md`
- `milestones/m3-project-update/README.md`
- `milestones/m4-peer-review/README.md`
- `milestones/m5-presentation/README.md`
- `milestones/final-portfolio/README.md`

## 41.6 Appendix B: Keyword Patterns Used

- Week/module labels: Week 0, Week 1, Week 2, Week 3, Week 4, Week 5, Week 6, Week 7, Week 8, Week 9, Week 10, Week 11, Week 12, Week 13, Final Portfolio
- Workflow terms: Codespaces, VS Code, Quarto, render, commit, sync, fork, GitHub Classroom, stage, stop Codespace
- R terms: R, tidyverse, dplyr, read\_csv, filter, select, mutate, summarise, group\_by, function, debugging, renv
- Python terms: Python, pandas, pd.read\_csv, Jupyter, requirements.txt, reticulate, translation, parity
- Ethics/data terms: health data, open data, portal, metadata, provenance, stewardship, privacy,

- security, OCAP, CARE, AI hallucination, bias
- Visualization terms: ggplot2, plotly, visual audit, misleading, clarity, aggregation, caption, bias
- EDA terms: EDA, Exploratory Data Analysis, Table 1, descriptive summary, cleaning, planted-error, audit
- Dashboard terms: dashboard, KT, Knowledge Translation, Shiny, Shinylive, static HTML, GitHub Pages, filter, interactive
- Communication/reporting terms: plain language, revealjs, slides, peer feedback, citations, references, BibTeX, Zotero, report, publishing, grounding audit
- Portfolio terms: fresh Codespace, reproducibility, README, dependency, path, peer review, presentation, AI-use note

Centers for Disease Control and Prevention, National Center for Health Statistics. 2026. “National Health and Nutrition Examination Survey.” <https://wwwn.cdc.gov/nchs/nhanes/>.